



# **Shark Oracle: a Comprehensive Living Database of Elasmobranch Research for Literature Reviews, Research Geography, and Conservation Prioritisation**

**Simon Dedman, Guuske Tiktak, Nicholas K Dulvy, Elena Fernández Corredor, & David Ruiz García**

**Florida International University (FIU), MacAlister Elliott and Partners (MEP), Nederlandse Elasmobranchen Vereniging (NEV), Simon Fraser University (SFU), Institut de Ciències del Mar (CSIC), & MARBEC (IFREMER)**

with David Shiffman, Ulrich Bösing, Chris Mull, Alex McInturf, Sophia Pelletier, Deven Guerrero, Ryan McMullen, Carylanne Maier, Tobi-Dawne Smith, Kate Wing, Cat Gordon, Ali Hood, Brit Finucci, David Green, Andrew Temple, Lola Riesgo, Emily Warren, Chiara Gambardella, & Nathan Perisic

Slides at [https://github.com/SimonDedman/elasmo\\_analyses](https://github.com/SimonDedman/elasmo_analyses)



# The team



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AES abstracts



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**Nicolas Straube**  
Shark References



**Ulrich Boesing**  
RIS library



**David Green**  
JMIH abstracts



**Brit Finucci**  
OCS abstracts



AES



EEA



OCS

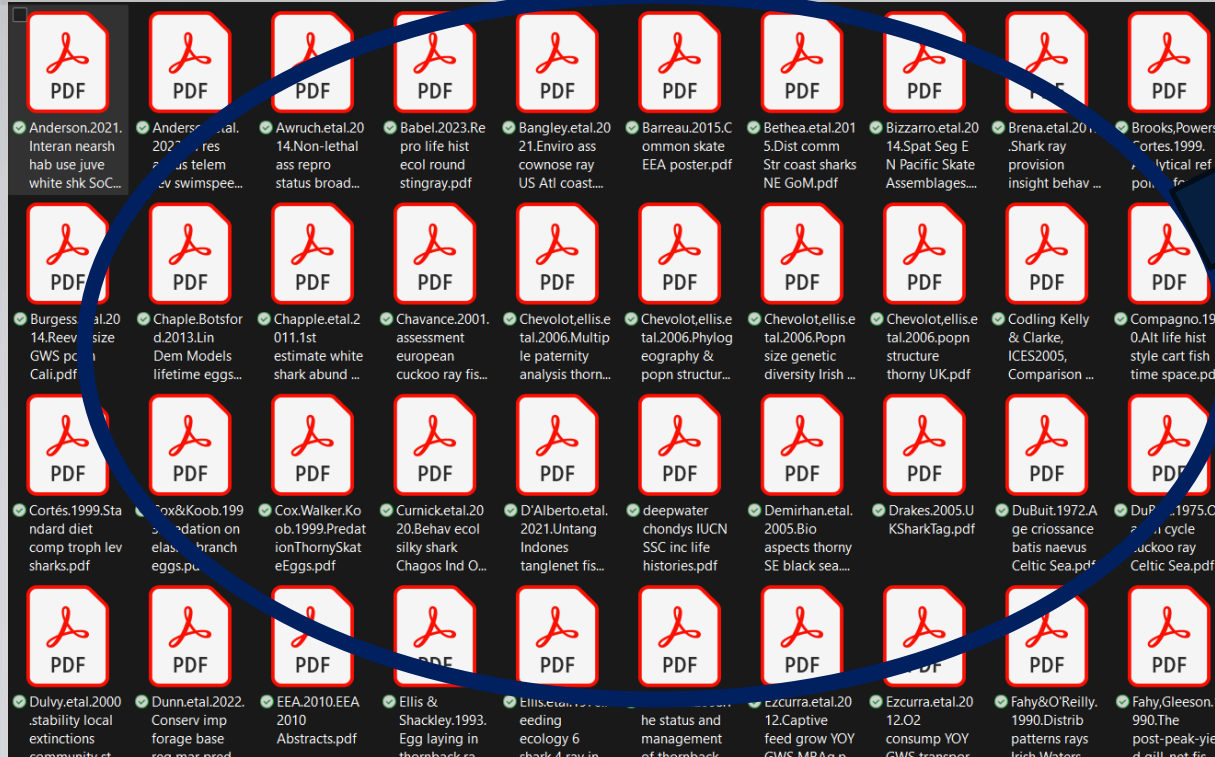


Sharks International





# Publish > SharkRefs > Download > Rules > Extract



 SharkReferences.com



## Rules



## Extraction



- Keyword presence
- Keyword frequency & thresholds
- Keyword presence/frequency + local anchors, e.g. needs elasmowords in nearby sentences

[https://github.com/SimonDedman/elasmowords\\_analyses/tree/main/docs/schema\\_proposals](https://github.com/SimonDedman/elasmowords_analyses/tree/main/docs/schema_proposals)

# Shark Oracle: data pipeline

34,425 papers | 21,907 PDFs | 19,592 in the Oracle index | 23,491 abstracts from 124 meetings | 22 topics

## 1 Sources

Shark-References  
monthly sync  
(34,425 records)

Coauthor libraries +  
drop folders

Conference abstract  
books (124 meetings)

OpenAlex  
(authors, citations)

NamSor  
(gender, origin)

Sharkipedia traits +  
IUCN Red List

Unpaywall, SciMago  
(OA, journal rank)

## 2 Acquire, Extract

PDF library  
(21,907 PDFs)

Download hub:  
team push to fill gaps

PDF id map  
(literature\_id to file)

Text extraction +  
keyword scoring

1,748 schema columns  
in 8 groups

Evidence audit trail  
(431,000 rows)

## 3 Enrich

30,316  
unique authors

Gender: NamSor  
(Genderize retired)

Institution country,  
North / South

Study country,  
ocean basin

215 analytical  
techniques

Species, bycatch,  
life-history links

## 4 Validate

190 expert  
(gold) labels

48,306 Fable  
(silver) labels

Rule-improvement  
loop: score, propose,  
retest

Per-discipline  
review pages

Per-author  
validation pages

## 5 Outputs

Shark Oracle: cited  
answers, 19,592  
papers indexed

Abstracts browser  
(23,491 abstracts)

22 topic expert  
review pages

Researcher  
network atlas

Download hub +  
remaining papers

Enriched parquet,  
118+ figures

# PDF download helpers: publisher & journal index



## Download helper: publishers & journals

Every publisher and journal with papers still missing from the corpus. Each row opens the download helper already filtered to that slice, so you can work through one source you have institutional access to without wading through the full queue.

[Full download helper](#)
[Closed-access helpers !\[\]\(830769b31eeeaca920791081939ff8ba\_img.jpg\)](#)
[Results index](#)
[Slide deck](#)
**9,424**

Papers remaining

**193**

Publishers

**5,009**

Journals

**533**

Open access available

Counts are read live from the same `papers_data.json` that drives the helper, so this page never goes stale: when papers are ingested and the queue is regenerated, these numbers follow. **Open access rows are the quickest wins** — those PDFs are free to fetch, they just haven't been collected yet.

For the paywalled tail, the **closed-access helpers** give four routes per paper (publisher DOI, JSTOR, your library, Google Scholar) across the 1,492 papers known to be closed. Those pages now record progress to the same shared record as the helper, so it doesn't matter which you work from. If you aren't at FIU, set your own library search URL once at the top of any of those pages.

[By publisher](#)
[By journal](#)


193 publishers.

1,005 conference abstracts are excluded from these counts.

| Name                          | Remaining | OA available | Open                   |
|-------------------------------|-----------|--------------|------------------------|
| Unknown publisher             | 6,857     | —            | <a href="#">Open →</a> |
| Elsevier                      | 434       | 100          | <a href="#">Open →</a> |
| JSTOR                         | 390       | 55           | <a href="#">Open →</a> |
| Wiley                         | 206       | 27           | <a href="#">Open →</a> |
| Springer                      | 107       | 4            | <a href="#">Open →</a> |
| Biodiversity Heritage Library | 106       | 86           | <a href="#">Open →</a> |
| Company of Biologists         | 80        | 10           | <a href="#">Open →</a> |
| Taylor & Francis              | 68        | 2            | <a href="#">Open →</a> |
| Schweizerbart                 | 63        | —            | <a href="#">Open →</a> |
| Oxford University Press       | 55        | 32           | <a href="#">Open →</a> |
| CSIRO Publishing              | 49        | 23           | <a href="#">Open →</a> |
| Magnolia Press (Zootaxa)      | 47        | 14           | <a href="#">Open →</a> |
| Inter-Research                | 40        | 30           | <a href="#">Open →</a> |

# PDF download helpers: what's still missing

**SHARK ORACLE**

Remaining Papers

YOU  
--

JOURNAL

All journals

PUBLISHER

All

YEAR

All

OA

All

PRIORITY

All

Clear

☐ HIDE CLICKED

☐ INCLUDE CONFERENCE ABSTRACTS (1004)

Synched (1270)

9279  
REMAINING

31  
CLICKED

502  
OPEN ACCESS

8777  
CLOSED

Showing 1–100 of 9,279 remaining

Show 100 entries

Search:

CSV

[Help / OA legend](#)

Remaining = 9,279 papers to get. The [dashboard](#) = 10,283 outstanding = these + 1,004 conference abstracts (hidden here unless ticked). 136 more are already downloaded and await filing, so they are not listed.

[Download helpers](#): every remaining paper as cards with search buttons, by publisher or journal. Pick a journal or publisher above to jump to its page.

# Helpers: closed-access publisher guides

## Download helpers

Every paper still to get, 9,279 in all, laid out for working through: one card per paper with the right search buttons, and a **Got it** button that tells the team it is done. The [hub](#) tracks progress across the whole list; these pages are for doing the downloading. Conference abstracts are left out (the abstracts project covers them).

**Generated:** 2026-10-08 03:21

### Where to start

1. **Papers with a DOI** (2,251): pick a publisher your institution subscribes to.
2. **Papers without a DOI** (7,028): pick a journal you know, or one in your language or region. These need searching rather than a login, and many turn up on Scholar, BHL, or a society website.
3. Save the PDFs to your named folder on Simon's NAS (any filename), and click **Got it** on each.

## Papers with a DOI, by publisher

- [Elsevier](#): 431 papers
- [JSTOR](#): 380 papers
- [Wiley](#): 205 papers
- [Springer](#): 99 papers
- [Company of Biologists](#): 80 papers
- [Biodiversity Heritage Library](#): 78 papers
- [Taylor & Francis](#): 68 papers
- [Schweizerbart](#): 63 papers
- [Oxford University Press](#): 53 papers
- [CSIRO Publishing](#): 49 papers
- [Magnolia Press \(Zootaxa\)](#): 47 papers
- [Inter-Research](#): 40 papers
- [American Physiological Society](#): 39 papers
- [Taylor & Francis \(CRC\)](#): 34 papers
- [University of Chicago Press](#): 30 papers
- [Cambridge University Press](#): 30 papers
- [AAAS \(Science\)](#): 27 papers
- [Geological Society of London](#): 23 papers
- [ASIH \(Copeia\)](#): 20 papers
- [Smaller publishers \(fewer than 15 each\)](#): 455 papers

## Papers without a DOI, by journal

- [Biologia Marina Mediterranea](#): 77 papers
- [Report of Japanese Society for Elasmobranch Studies](#): 71 papers
- [Unknown](#): 68 papers
- [Journal of Ichthyology](#): 56 papers
- [Het Zeepaard](#): 50 papers
- [Annales](#): 46 papers
- [Tertiary Research](#): 41 papers
- [Palaeontographica](#): 41 papers
- [Journal of Paleontology](#): 38 papers
- [Thesis](#): 37 papers
- [California Fish and Game](#): 36 papers
- [Bulletin of Marine Science](#): 36 papers
- [Uyeno](#): 36 papers
- [CSIRO Marine and Atmospheric Research Paper](#): 35 papers
- [P.W. Gilbert](#): 35 papers
- [Program and Abstracts for the](#): 31 papers
- [Rivista di Parassitologia](#): 30 papers
- [FAO Fisheries Technical Paper](#): 27 papers
- [Journal of the Elisha Mitchell Scientific Society](#): 24 papers
- [Bulletin de l'Institut Royal des Sciences Naturelles de Belgique](#): 23 papers
- [Shark News](#): 23 papers
- [Cahiers de Biologie Marine](#): 22 papers
- [Indian Journal of Fisheries](#): 22 papers
- [Proceedings of the](#): 22 papers
- [Comptes Rendus des séances de la Société de Biologie et de ses Filiales](#): 20 papers
- [Other journals: \(no journal\) ... Anales Museo Historia Natura](#): 300 papers
- [Other journals: Anancho Educational Committe ... Arquipélago. Life and Marine](#): 295 papers
- [Other journals: Arquivos de Ciências do Mar ... Biologist](#): 292 papers
- [Other journals: Biologiya Morya \(Vladivostok ... Bulletin de la Société de Ch](#): 300 papers
- [Other journals: Bulletin de la Société de l' ... Bulletins of American Paleon](#): 299 papers
- [Other journals: Bundesamt für Landestopograf ... Cuadernos de Geología Ibéric](#): 300 papers
- [Other journals: Current Science ... Falco](#): 292 papers
- [Other journals: FAO ... Geological Society of New Zea](#): 299 papers

# Helpers: one publisher guide (Elsevier)

[← All download helpers](#) | [Downloading hub](#)

## Download helper: Elsevier

**Publisher:** Elsevier. Papers with a DOI; 97 are flagged open access but our scripts failed to fetch them.  
**431** papers across 132 journal(s). Generated 2026-10-08 03:21; papers filed since then still appear until the next rebuild.

### How to use this page

1. Pick your name under **You** and, if you like, set your own library's search link.
2. For each paper try the **Publisher page (DOI)** first; if your login doesn't reach it, try JSTOR, then your library (WorldCat by default; set your own below), then Scholar.
3. If you get the PDF, save it and click **Got it**. That greys it out for everyone, here and on the downloading hub. Clicking a search button records nothing, so a dead end costs nobody anything.
4. If you can't get it, click **Can't get it**. That only marks it on your own browser, so you don't try it twice; someone with different access may still succeed.
5. Put the PDFs in your named folder on Simon's NAS. Any filename is fine. No NAS link yet? Email Simon and he'll send one.

**You:** -- ▼ Synced with the shared record

**Library link:**

*Paste your own library's search URL with {TITLE} where the title goes; this browser remembers it.*

**Got: 4 / 431** · **Can't get (you): 0** ☐ hide papers already got

### Journals on this page:

[A.P. Klimley & D. Ainley \(41\)](#) [Academic Press \(1\)](#) [Acta Biomaterialia \(1\)](#) [Advances in Marine Biology \(1\)](#) [Ameghiniana \(1\)](#)

[American Journal of Emergency Medicine \(1\)](#) [Animal Behaviour \(6\)](#) [Annales de Paleontologie \(1\)](#) [Annales de Paléontologie \(1\)](#) [Aquatic Toxicology \(1\)](#)

[Archiv für Fischereiwissenschaft \(1\)](#) [Association Géologique Aubeise \(1\)](#) [Biochemical and Biophysical Research Communications \(2\)](#)

A.P. Klimley & D. Ainley (41)

**1** ID 11461 10.1016/b978-012415031-7/50029-x

### White Shark-Inflicted Wounds of Sea Otters in California, 1968?1992

Ames, J.A. & Geibel, J.J. & Wendell, F.E. & Pattison, C.A. (1996)

Year: 1996

In A.P. Klimley & D. Ainley, (Eds.), Great White Sharks. The biology of Carcharodon carcharias (pp. 309–316). Academic Press.

**2** ID 83 10.1016/b978-012415031-7/50005-7

### The Fossil History of Carcharodon and Its Possible Ancestor, Cretolamna A Study in Tooth Identification

Applegate, S.P. & Espinosa-Arrubarrena, L. (1996)

Year: 1996

In A.P. Klimley & D. Ainley, (Eds.), Great White Sharks. The biology of Carcharodon carcharias (pp. 19–36). Academic Press.

Marked as got by Andrew on 2026-09-28; not filed yet. If you have it too, drop it in anyway.

**3** ID 11456 10.1016/b978-012415031-7/50024-0

### Behavior of the White Shark An Emerging Picture

Barlow, G.W. (1996)

Year: 1996

In A.P. Klimley & D. Ainley, (Eds.), Great White Sharks. The biology of Carcharodon carcharias (pp. 257–260). Academic Press.

**4** ID 284 10.1016/b978-012415031-7/50043-4

### Worldwide Patterns of White Shark Attacks on Humans

Burgess, G.H. & Callahan, M. (1996)

Year: 1996

In A.P. Klimley & D. Ainley, (Eds.), Great White Sharks. The biology of Carcharodon carcharias (pp. 457–469). Academic Press.

**5** ID 11472 10.1016/b978-012415031-7/50039-2

### An Evaluation of Methodologies to Study the Population Biology of White Sharks

Cailliet, G.M. (1996)

Year: 1996

In A.P. Klimley & D. Ainley, (Eds.), Great White Sharks. The biology of Carcharodon carcharias (pp. 415–416). Academic Press.

**6** ID 497 10.1016/b978-012415031-7/50033-1



# Keyword rules > 8 Schema Groups, 1,748 Columns



Ecosystem (20 cols)

Habitat, depth zones, ocean features

Species (1,308 cols)



Every known elasmobranch species matched by binomial name

Discipline (19) + Techniques (215 cols)



Biology, genetics, movement, etc. Analytical methods: telemetry, stable isotopes, eDNA, etc.;



Ocean Basin (9 cols)

Global geographic coverage



Pressure (26) + Gear (29) + Impact (25)



Bycatch, pollution, climate; Gear types, mitigation; abundance, mortality, growth, genetic, etc



Your keywords, your rules, your results.

*Shagreen?  
Social aggregations?  
Shenanigans?  
Mislabelling?  
Data used?*

# Validate your data: find your name



## Elasmobranch Analytical Methods Review

Validate the automated extractions for your papers

Search by name...

Search for your name above to find your author page. Accents, hyphens, and name order don't matter ("Fernandez Corredor" finds "Elena Fernández-Corredor"). Names in our database are matched here; other researchers are looked up on [OpenAlex](#). If your page is not yet in our database, please locate your [OpenAlex author ID](#) and submit your papers via [Shark References](#).

**Multiple versions of your name?** OpenAlex sometimes creates several IDs for the same researcher (e.g. "Nicholas K. Dulvy", "N. Dulvy", "N.K. Dulvy"). The one with the most works — usually marked "In our database" — is typically your main profile. The others are usually stub profiles with 1–15 works. We automatically merge obvious duplicates within our corpus, but external OpenAlex ghost profiles may still appear.

## OpenAlex authors: search "Chris Mull"

[api.openalex.org/authors?search=Chris%20Mull](https://api.openalex.org/authors?search=Chris%20Mull) (rows whose name is not a Mull spelling omitted)

| display_name        | works_count | last known institution | OpenAlex id                                                                     |
|---------------------|-------------|------------------------|---------------------------------------------------------------------------------|
| Christopher G. Mull | 35          | Dalhousie University   | <a href="https://openalex.org/A5025880572">https://openalex.org/A5025880572</a> |
| Chris Mull          | 3           | Dalhousie University   | <a href="https://openalex.org/A5072645810">https://openalex.org/A5072645810</a> |

**FIU**

# Validate your data: an author page

## David Ruiz-García

Parc Científic de la Universitat de València · OpenAlex: A5009396914 · 6 papers in database

- 2025 Ruiz-García et al. Key drivers of at-vessel mortality in demersal sharks. Conservation Biology Pending
- 2024 Figuerola et al. Adapting to a pollution hotspot? Catsharks shift to plastic substrates for ovip... Science of The Total Environment Pending
- 2024 Carrasco-Puig et al. Heavy metal concentrations in sharks, rays and chimaeras from the western Medit... Marine Pollution Bulletin Pending
- 2023 Ruiz-García et al. Spatial distribution of the demersal chondrichthyan community from the western ... Frontiers in Marine Science Pending
- 2023 Ruiz-García et al. New record of Hexanchus griseus in the northwestern Mediterranean Sea with insi... Mediterranean Marine Science Pending
- 2020 Ruiz-García et al. Determining Stingray Movement Patterns in a Wave-Swept Coastal Zone Using a Bli... Fishes Pending

0 of 6 reviewed

[Download JSON](#)

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**FIU**

| Institution                      | Gender*    | Origin country* | Ethnicity / diaspora* |
|----------------------------------|------------|-----------------|-----------------------|
| Florida International University | — select — | GB              | British               |

Corrections welcome. Changes saved automatically. \*NamSor inference from name & country only.

# Validate your data

▼ 2025 Guttridge et al. Stay or go? Space and resource use of the great hammerhead (Sphyrna mokarran) o... Frontiers In Marine Science Pending

Year: 2025 Journal: Frontiers In Marine Science DOI: 10.3389/fmars.2025.1544482 Study type: empirical Epoch: Recent OA: gold (cc-by) Altmetric: held locally ID: 34390 SR habitat: marine pelagic

Authors: Guttridge, T.L. & Heim, V. & Dedman, S. & Guttridge, A.E. & Bain, S.A. & Keller, B.A. & Matich, P. (2025)

## Ocean Basin

|                                       |                                               |                                       |                                        |                                                    |                                        |                                         |                                                   |                                         |
|---------------------------------------|-----------------------------------------------|---------------------------------------|----------------------------------------|----------------------------------------------------|----------------------------------------|-----------------------------------------|---------------------------------------------------|-----------------------------------------|
| <input type="checkbox"/> arctic ocean | <input checked="" type="checkbox"/> caribbean | <input type="checkbox"/> indian ocean | <input type="checkbox"/> mediterranean | <input checked="" type="checkbox"/> north atlantic | <input type="checkbox"/> north pacific | <input type="checkbox"/> south atlantic | <input checked="" type="checkbox"/> south pacific | <input type="checkbox"/> southern ocean |
|---------------------------------------|-----------------------------------------------|---------------------------------------|----------------------------------------|----------------------------------------------------|----------------------------------------|-----------------------------------------|---------------------------------------------------|-----------------------------------------|

► Extraction rules — Ocean basin — 9 binary columns indicating which ocean basin(s) the study covers. (9 rules)

Rating: Correct Partially correct Incorrect

Notes:

Rule feedback: Threshold: current Reset Add term: e.g. demersal Remove term: e.g. pelagic

## Ocean Sub-basin

|                                                    |                                         |                                             |                                                   |                                       |                                             |                                             |                                          |                                         |                                             |                                         |                                        |
|----------------------------------------------------|-----------------------------------------|---------------------------------------------|---------------------------------------------------|---------------------------------------|---------------------------------------------|---------------------------------------------|------------------------------------------|-----------------------------------------|---------------------------------------------|-----------------------------------------|----------------------------------------|
| <input type="checkbox"/> adriatic sea              | <input type="checkbox"/> aegean sea     | <input type="checkbox"/> alboran sea        | <input type="checkbox"/> andaman sea              | <input type="checkbox"/> arabian sea  | <input type="checkbox"/> azores             | <input checked="" type="checkbox"/> bahamas | <input type="checkbox"/> baltic sea      | <input type="checkbox"/> barents sea    | <input type="checkbox"/> bay of bengal      | <input type="checkbox"/> bay of biscay  | <input type="checkbox"/> benguela      |
| <input type="checkbox"/> bering sea                | <input type="checkbox"/> black sea      | <input type="checkbox"/> california current | <input checked="" type="checkbox"/> caribbean sea | <input type="checkbox"/> celtic sea   | <input type="checkbox"/> coral sea          | <input type="checkbox"/> east china sea     | <input type="checkbox"/> english channel | <input type="checkbox"/> gulf of alaska | <input type="checkbox"/> gulf of california | <input type="checkbox"/> gulf of guinea | <input type="checkbox"/> gulf of maine |
| <input checked="" type="checkbox"/> gulf of mexico | <input type="checkbox"/> hawaii         | <input type="checkbox"/> ionian sea         | <input type="checkbox"/> irish sea                | <input type="checkbox"/> ligurian sea | <input type="checkbox"/> mozambique channel | <input type="checkbox"/> north sea          | <input type="checkbox"/> norwegian sea   | <input type="checkbox"/> persian gulf   | <input type="checkbox"/> philippine sea     | <input type="checkbox"/> red sea        | <input type="checkbox"/> sargasso sea  |
| <input type="checkbox"/> sea of japan              | <input type="checkbox"/> sea of marmara | <input type="checkbox"/> sea of okhotsk     | <input type="checkbox"/> south china sea          | <input type="checkbox"/> tasman sea   | <input type="checkbox"/> tyrrhenian sea     | <input type="checkbox"/> yellow sea         |                                          |                                         |                                             |                                         |                                        |

► Extraction rules — Ocean sub-basin — 66 binary columns covering finer geographic regions (Mediterranean, Gulf of Mexico, Sea of Japan, California Current, ...). (2 rules)

Rating: Correct Partially correct Incorrect

Notes:

Rule feedback: Threshold: current Reset Add term: e.g. demersal Remove term: e.g. pelagic

## Discipline

|                                               |                                                |                                                  |                                                  |                                                  |                                     |                                             |                                              |                                           |                                    |                                     |                                              |
|-----------------------------------------------|------------------------------------------------|--------------------------------------------------|--------------------------------------------------|--------------------------------------------------|-------------------------------------|---------------------------------------------|----------------------------------------------|-------------------------------------------|------------------------------------|-------------------------------------|----------------------------------------------|
| <input checked="" type="checkbox"/> behaviour | <input checked="" type="checkbox"/> biology    | <input type="checkbox"/> biomechanics            | <input checked="" type="checkbox"/> conservation | <input checked="" type="checkbox"/> data science | <input type="checkbox"/> ecotourism | <input type="checkbox"/> fisheries          | <input checked="" type="checkbox"/> genetics | <input type="checkbox"/> human dimensions | <input type="checkbox"/> husbandry | <input type="checkbox"/> immunology | <input checked="" type="checkbox"/> movement |
| <input type="checkbox"/> paleontology         | <input checked="" type="checkbox"/> physiology | <input checked="" type="checkbox"/> reproductive | <input type="checkbox"/> sensory                 | <input type="checkbox"/> taxonomy                | <input type="checkbox"/> toxicology | <input checked="" type="checkbox"/> trophic |                                              |                                           |                                    |                                     |                                              |

► Extraction rules — Discipline — 19 binary columns classifying the research area (biology, behaviour, trophic, genetics, movement, fisheries, data science, taxonomy...). (19 rules)

[https://simondedman.github.io/elasmo\\_analyses/validate/](https://simondedman.github.io/elasmo_analyses/validate/)

[https://github.com/SimonDedman/elasmo\\_analyses/issues](https://github.com/SimonDedman/elasmo_analyses/issues)



# Topic review: 22 topics, one champion

## Topic review

Each card is a topic hub: open it for what the build found, the review page, and the tour. One question, asked one discipline at a time: of every paper the corpus holds, which are genuinely about *this*? Each topic bundles a discipline's extraction column with the pressure, impact and gear columns that belong to it. A subject expert judges the papers near the line, and the rules are then fitted to those judgements.

First time here? [Take the tour](#) (four minutes, then a sandbox where nothing counts).

Looking for the keywords themselves? [Every rule in every topic](#), one sortable table: keywords, anchors, thresholds, counts, agreement with Fable, and a link to the line that defines each one.

### Ecology

#### Trophic ecology

d\_trophic · 2 live columns · champion: (unassigned)

**4,348** **2,042** **1,941**  
mention a keyword   qualify today   Fable-read

Open the hub: what the build found, the review page, and the tour →

#### Sub-basins and seas

43 columns, no discipline column · · champion: (unassigned)

**814** **303** **1,941**  
mention a keyword   qualify today   Fable-read

Open the hub: what the build found, the review page, and the tour →

### Behavioural biology

#### Behaviour

d\_behaviour · 6 live columns · champion: (unassigned)

**5,654** **1,868** **1,941**  
mention a keyword   qualify today   Fable-read

Open the hub: what the build found, the review page, and the tour →

#### Sensory biology

d\_sensory · 5 live columns · champion: (unassigned)

**1,965** **1,368** **1,941**  
mention a keyword   qualify today   Fable-read

Open the hub: what the build found, the review page, and the tour →

#### Movement and spatial ecology

d\_movement · 2 live columns · champion: (unassigned)

**5,035** **1,828** **1,941**  
mention a keyword   qualify today   Fable-read

Open the hub: what the build found, the review page, and the tour →

#### Habitat and ecosystems

20 columns, no discipline column · · champion: (unassigned)

**9,022** **3,920** **1,941**  
mention a keyword   qualify today   Fable-read

Open the hub: what the build found, the review page, and the tour →

#### Ocean basins

9 columns, no discipline column · · champion: (unassigned)

**4,896** **2,335** **1,941**  
mention a keyword   qualify today   Fable-read

Open the hub: what the build found, the review page, and the tour →

### Biology

#### Taxonomy and systematics

d\_taxonomy · 1 live column · champion: (unassigned)

**5,175** **3,418** **1,941**  
mention a keyword   qualify today   Fable-read

Open the hub: what the build found, the review page, and the tour →

#### Biomechanics

d\_biomechanics · 1 live column · champion: (unassigned)

**1,132** **747** **1,941**  
mention a keyword   qualify today   Fable-read

Open the hub: what the build found, the review page, and the tour →

### Human interactions

#### Conservation and management

d\_conservation · 14 live columns · champion: (unassigned)

**8,085** **3,716** **1,941**  
mention a keyword   qualify today   Fable-read

Open the hub: what the build found, the review page, and the tour →

#### Human dimensions

d\_human\_dimensions · 6 live columns · champion: (unassigned)

**1,534** **537** **1,941**  
mention a keyword   qualify today   Fable-read

Open the hub: what the build found, the review page, and the tour →

### Methods

#### Data science and methods

d\_data\_science · 1 live column · champion: (unassigned)

**806** **312** **1,941**  
mention a keyword   qualify today   Fable-read

Open the hub: what the build found, the review page, and the tour →

#### Palaeontology

d\_paleontology · 1 live column · champion: (unassigned)

**3,597** **2,784** **1,941**  
mention a keyword   qualify today   Fable-read

Open the hub: what the build found, the review page, and the tour →

#### Reproductive biology

d\_reproductive · 2 live columns · champion: (unassigned)

**6,052** **3,010** **1,941**  
mention a keyword   qualify today   Fable-read

Open the hub: what the build found, the review page, and the tour →

#### Fisheries

d\_fisheries · 40 live columns · 66 proposed · champion: **David Ruiz-García (proposed)**

**5,064** **2,096** **1,941**  
mention a keyword   qualify today   Fable-read

Open the hub: what the build found, the review page, and the tour →

#### Toxicology and pollution

d\_toxicology · 4 live columns · champion: (unassigned)

**1,291** **611** **1,941**  
mention a keyword   qualify today   Fable-read

Open the hub: what the build found, the review page, and the tour →

#### Biology and life history

d\_biology · 3 live columns · champion: (unassigned)

**7,836** **3,970** **1,941**  
mention a keyword   qualify today   Fable-read

Open the hub: what the build found, the review page, and the tour →

#### Genetics and genomics

d\_genetics · 2 live columns · champion: (unassigned)

**6,498** **3,347** **1,941**  
mention a keyword   qualify today   Fable-read

Open the hub: what the build found, the review page, and the tour →

#### Husbandry and aquaria

d\_husbandry · 2 live columns · champion: (unassigned)

**2,151** **798** **1,941**  
mention a keyword   qualify today   Fable-read

Open the hub: what the build found, the review page, and the tour →

#### Physiology

d\_physiology · 9 live columns · champion: (unassigned)

**4,614** **2,143** **1,941**  
mention a keyword   qualify today   Fable-read

Open the hub: what the build found, the review page, and the tour →

#### Immunology and disease

d\_immunology · 2 live columns · champion: (unassigned)

**1,960** **1,145** **1,941**  
mention a keyword   qualify today   Fable-read

Open the hub: what the build found, the review page, and the tour →

#### Ecotourism

d\_ecotourism · 3 live columns · champion: (unassigned)

**851** **564** **1,941**  
mention a keyword   qualify today   Fable-read

Open the hub: what the build found, the review page, and the tour →

FIU

## How to read this dashboard

It is a busy page, so here it is one piece at a time. Click a numbered pin, or walk through in order with **Next**. Nothing here is live: it is a photograph of the real thing. When you have had enough, there is a sandbox at the bottom with ten papers where you can press every button and none of it counts.

The screenshot shows the 'Topic review › Fisheries' dashboard. It features a top navigation bar with buttons like 'Ladder', 'Margin queue', 'Reviewer', 'Simon', 'Export', 'Import', and 'Submit'. A central panel displays a list of papers with columns for 'above the line', 'Table', 'IN', 'OUT', 'bycatch', 'discards', 'retention', 'stock assessment', 'catch data', 'fishing mortality', 'fishery management', 'fishery-independent', 'catch data', 'discards', 'retention', 'stock assessment', 'catch data', 'fishing mortality', 'fishery management', 'fishery-independent'. A right sidebar contains a 'Rule' section with a 'Threshold' of 2 and a 'Reset to live rule' button. Below this is a 'Section weights' table with columns for 'title', 'keywords', 'abstract', 'introduction', 'method', 'results', 'discussion', 'conclusions', 'other'. At the bottom is a 'Keywords' table with columns for 'term', 'weight', 'papers', 'IN share', and 'decisive'. Numbered callouts 1-13 highlight various UI elements: 1 points to the 'CONCEPT' dropdown, 2 to the 'Judge the concept' text, 3 to the 'Ladder' button, 4 to the 'Unsure' button, 5 to the 'verwrite' button, 6 to the 'Rule' section, 7 to the 'fisheries live column' section, 8 to the 'Table' button, 9 to the 'Paper' button, 10 to the 'Threshold' input, 11 to the 'Section weights' table, and 12 to the 'Keywords' table.

STEP 1 OF 13

### The concept you are judging

Each option in this selector is one extraction column: a yes/no column in the database that records whether a paper is about this concept. The fisheries topic has 106 of them, 40 live columns the pipeline extracts today and 66 David has proposed that have never been run. The line under the selector gives that count; the two lines below it say what the chosen concept means, in the champion's words, with examples of what counts and what does not.

Judge the concept, not the keywords. Picking one of David's proposed columns scores it across the whole corpus on the spot, so you can see what his wording would pick up before anybody commits to it.

Back

Next



### Now try it, with nothing at stake

The sandbox gives you ten papers from this concept: a few that clearly qualify, a few that clearly do not, and several on the line. Label them, drag the threshold about, drop a keyword, run the search. None of it is saved, sent, or counted, and closing the tab forgets

everything.

[Open the sandbox \(10 papers\)](#)

[Go to the real thing](#)

CONCEPT (extraction column)

Topic review › Fisheries

d\_fisheries

106 concepts in the Fisheries topic: 40 live extraction columns and 66 proposed, never run.

d\_fisheries is a live extraction column.

Judge the **concept**, not the keywords: the champion writes a two-line definition with inclusion and exclusion examples here before anyone labels.

5,064 papers mention a keyword; 2,096 qualify; topic champion: **David Ruiz-García (proposed)**.

IN I

OUT 0

Unsure U

Clear X

All above → IN A

All below → OUT B

☐ overwrite

Undo Ctrl Z

Show

all papers with a hit

Order

closest to the line first

Margin ±

0.5

title, author, journal, id

☐ suspect text

7,309 shown, 2,095 above the line

1

**2 A survey of ciguatera at Enewetak and Bikini, Marshall Islands, with notes on the systematics and food habits of ciguatoxic fishes.**

1980 · Randall · Fishery Bulletin

above the line

discard\* ×3

retention ×1

...(1975a), and Tester (see footnote 7), have added to the list of the great variety ofmarine animals (mainly fishes) that this species will take as food, as well as sundry item...

2

**2 New record of a great white shark, Carcharodon carcharias (Lamnidae) from the eastern Adriatic Sea.**

2005 · Soldo et al. · Cybium

above the line

discard\* ×2

...Due to its great size, it was not possible to preserve the specimen in the vessel's fish hold, so it was discarded to sea....

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**2 Distribution, abundance, and habits of pelagic sharks in the central Pacific Ocean.**

1958 · Strasburg · Fishery Bulletin

above the line

catch data ×3

...The catch data for the mackerel shark consist of 28 records, 26 of which are from north of 35° ;N. latitude indicating that this shark is a temperate- or cold-water form (fig. 4)....

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**2 Distribution, biology and fisheries importance of the pencil shark, Hypogaleus hyugaensis (Elasmobranchii: Tria...**

2002 · Simpfendorfer et al. · Marine and Freshwater Research

above the line

catch data ×1

discard\* ×4

stock assessment ×1

fisheries management ×1

...Map of study area and fisheries management zones....

5

**2 Reproductive biology and abundance of the white-dotted skate, Bathyraja albomaculata, in the Southwest Atlantic**

2006 · Ruocco et al. · ICES Journal of Marine Science

above the line

catch data ×1

bycatch ×3

discard\* ×2

...The catch data and samples of skates used in this paper were obtained from eight research cruises conducted in the Southwest Atlantic between 34(S an...

6

**2 The role of protected areas in elasmobranch fisheries management and conservation.**

2002 · Stevens · In Elasmobranch Biodiversity

above the line

fisheries management ×1

...The role of protected areas in elasmobranch fisheries management and conservation....

7

**2 Movements and swimming behaviour of white sharks (Carcharodon carcharias) in Australian waters.**

2006 · Bruce et al. · Marine Biology

above the line

bycatch ×6

...Free-swimming Game-Wshing Commercial bycatch Total...

8

**2 Lipid, fatty acid and squalene composition of liver oil from six species of deepwater sharks collected in so...**

1995 · Bakes et al. · Comparative Biochemistry and Physiology – Part B

above the line

bycatch ×3

discard\* ×1

retention ×2

...Peaks were quantified with software from DAPA Scientific Software Individual components were identified using mass spectral data and by comparing rete...

9

**2 Long-term changes in ray populations in the North Sea.**

1996 · Walker et al. · ICES Journal of Marine Science

above the line

fishing mortality ×2

catch data ×1

...Landings statistics Nominal catch data on rays and skates have been collected by ICES member countries since 1903 and published in Bulletins Statistiques....

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fishing mortality ×2

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**2 Size composition, monthly condition factor and morphometrics for fishery-dependent samples of Rioraja agassizi (Chondrich...**

2007 · Oddone et al. · Neotropical Ichthyology

above the line

fishery-dependent ×2

discard\* ×1

...Size composition, monthly condition factor and morphometrics for fishery-dependent samples of Rioraja agassizi (Chondrichthyes: Rajidae), off Santos, Southe...

12

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1992 · Cailliet et al. · Australian Journal of Marine and Freshwater Research

above the line

fishing mortality ×7

...To simulate what would happen to the demography of the Pacific angel shark population under a range of natural and fishing mortalities, the survivorship function (L<sub>t</sub>) was m...

14

**2 Spatial, seasonal and biological analysis of sharks caught in the New South Wales protective beach meshin...**

1994 · Krogh · Australian Journal of Marine and Freshwater Research

above the line

bycatch ×2

...A total of 753 angel sharks (including 100 recorded as bycatch) was caught in the NSW beach meshing programme between October 1972 and...

Paper

Rule

Impact

Optimiser

Keys

Paper.

Everything the rule saw in the paper under the cursor: the verdict and why, the quotations, and every keyword the rule looks for, including the ones it did not find. Move the cursor with the arrow keys.

**A survey of ciguatera at Enewetak and Bikini, Marshall Islands, with notes on the systematics and food habits of ciguatoxic fishes.**

Randall 1980 · Fishery Bulletin · id 1995

ABSTRACT

No abstract held for this paper. The DOI link above opens the publisher page.

Qualifies:

score 2 reaches the threshold of 2.

score

0

1

2

3

4

5+

n

threshold

n

this paper

(1975a), and Tester (see footnote 7), have added to the list of the great variety ofmarine animals (mainly fishes) that this species will take as food, as well as sundry items of garbage and refuse discarded into the sea by man. results

(1966) to demonstrate the long periods of retention of ciguatoxin in the tissues of poisonous fishes when removed from the source of the toxin. results

Every keyword in this rule, found or not

A letter marks the section a keyword was found in (T=title, K=keywords, A=abstract, I=introduction, M=methods, R=results, RD=results and discussion, D=discussion, C=conclusions), with the count after it when there is more than one. Greyed rows are keywords this rule looks for and did **not** find. -n means mentions discarded for having no shark or ray term within a sentence. The last column is what the keyword contributed: mentions x that section's weight, shown small beside each letter. A section at weight 0 (usually text the section labeller could not place) contributes nothing.

| keyword                   | T <sub>2</sub> | K <sub>2</sub> | A <sub>0.5</sub> | I <sub>0.5</sub> | M <sub>1</sub> | R <sub>1</sub> | RD <sub>1</sub> | D <sub>0.5</sub> | C <sub>0.5</sub> | score     |
|---------------------------|----------------|----------------|------------------|------------------|----------------|----------------|-----------------|------------------|------------------|-----------|
| stock assessment          |                |                |                  |                  |                |                |                 |                  |                  | not found |
| fisheries management      |                |                |                  |                  |                |                |                 |                  |                  | not found |
| catch data                |                |                |                  |                  |                |                |                 |                  |                  | not found |
| fishing mortality         |                |                |                  |                  |                |                |                 |                  |                  | not found |
| maximum sustainable yield |                |                |                  |                  |                |                |                 |                  |                  | not found |
| MSY Aa                    |                |                |                  |                  |                |                |                 |                  |                  | not found |
| fishery-dependent         |                |                |                  |                  |                |                |                 |                  |                  | not found |
| fishery-independent       |                |                |                  |                  |                |                |                 |                  |                  | not found |
| bycatch                   |                |                |                  |                  |                |                |                 |                  |                  | not found |
| discard*                  |                |                | -1               |                  |                | R2             |                 |                  |                  | 2         |
| retention                 |                |                |                  |                  |                | -1             |                 |                  |                  | 0         |
| non-target species        |                |                |                  |                  |                |                |                 |                  |                  | not found |
| non-target catch          |                |                |                  |                  |                |                |                 |                  |                  | not found |
| total                     |                |                |                  |                  |                |                |                 |                  |                  | 2         |

CONCEPT (extraction column)

Topic review > Fisheries

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IN I

OUT O

Unsure U

Clear X

All above → IN A

All below → OUT B

☐ overwrite

Undo Ctrl Z

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all papers with a hit

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closest to the line first

Margin ±

0.5

title, author, journal, id

☐ suspect text

7,309 shown, 2,095 above the line

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above the line

catch data ×1

discard\* ×4

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...Map of study area and fisheries management zones....

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above the line

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above the line

fisheries management ×1

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above the line

fishing mortality ×2

catch data ×1

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bycatch ×2

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Paper

Rule

Impact

Optimiser

Keys

Rule. The rule itself: which keywords count, how much a mention is worth in each section of a paper, the score a paper must reach, and any anchor word that must be present. Change anything here and every paper in the list is re-scored at once, so the threshold bar moves as you type. Nothing you do here touches the corpus.

d\_fisheries

live column

Every change re-scores all 7,313 papers here in the browser. Nothing is written to the corpus until a proposal is accepted.

Threshold

2

☒ elasmobranch proximity

☐ round the score (retired)

Reset to live rule

Section weights

title

keywords

abstract

introduction

methods

2

2

0.5

0.5

1

results

results and discussion

discussion

conclusions

other

1

1

0.5

0.5

0

anchors

This rule has none.

Keywords

|                                     | term                      | weight | papers | IN share | decisive    |
|-------------------------------------|---------------------------|--------|--------|----------|-------------|
| <input checked="" type="checkbox"/> | bycatch                   | 1      | 2,665  | 67%      | 643 (77/38) |
| <input checked="" type="checkbox"/> | discard*                  | 1      | 1,636  | 69%      | 279 (24/16) |
| <input checked="" type="checkbox"/> | retention                 | 1      | 1,090  | 54%      | 192 (5/17)  |
| <input checked="" type="checkbox"/> | fisheries management      | 1      | 1,018  | 89%      | 136 (21/7)  |
| <input checked="" type="checkbox"/> | fishing mortality         | 1      | 699    | 86%      | 107 (5/4)   |
| <input checked="" type="checkbox"/> | catch data                | 1      | 656    | 87%      | 107 (8/3)   |
| <input checked="" type="checkbox"/> | stock assessment          | 1      | 599    | 87%      | 87 (9/1)    |
| <input checked="" type="checkbox"/> | fishery-independent       | 1      | 271    | 81%      | 36 (5/1)    |
| <input checked="" type="checkbox"/> | non-target species        | 1      | 204    | 63%      | 23 (1/3)    |
| <input checked="" type="checkbox"/> | fishery-dependent         | 1      | 169    | 86%      | 23 (2/2)    |
| <input checked="" type="checkbox"/> | maximum sustainable yield | 1      | 108    | 93%      | 11 (1/0)    |
| <input checked="" type="checkbox"/> | MSY <div>Aa</div>         | 1      | 108    | 90%      | 10 (0/1)    |

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IN I

OUT O

Unsure U

Clear X

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☐ overwrite

Undo Ctrl Z

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Paper

Rule

Impact

Optimiser

Keys

Impact. What your rule edits and your labels add up to across the whole corpus: how many papers qualify now against the live rule, where the scores sit, how much of the margin you have judged, and how well the rule agrees with each source of labels.

Corpus effect

2,095

qualify now

2,095

under the live rule

0

newly IN

0

newly OUT

No rule edits: this is the live rule.

Score distribution

1,350

675

0

0 2 4 6 8 10 12 14

line at 2

Not drawn: 2,737 papers that mention a keyword but score 0, because no mention sits near a shark or ray term or in a weighted section.

labelled IN

labelled OUT

unsure

unlabelled

Review progress

958

in the margin band

0

of those judged

0

inspected labels

0

assumed labels

Accuracy against each label source

| labels                           | n        | precision | recall | F1    | live F1 | false + | false -  |
|----------------------------------|----------|-----------|--------|-------|---------|---------|----------|
| My labels, inspected only        | none yet |           |        |       |         |         |          |
| My labels, including assumed     | none yet |           |        |       |         |         |          |
| Earlier author validations       | 3        | 66.7%     | 100.0% | 80.0% | 80.0%   | 1       | 0        |
| Fable silver labels (not expert) | 1,941    | 76.5%     | 44.8%  | 56.5% | 56.5%   | 80      | 322 (93) |

Accuracy from labels made only on papers the keywords already found cannot see papers the keywords miss. The Fable row covers every paper Fable read, including those with no keyword hit, which is why its false-negative count includes unreachable papers.

CONCEPT (extraction column)

Topic review › Fisheries

d\_fisheries

106 concepts in the Fisheries topic: 40 live extraction columns and 66 proposed, never run.

d\_fisheries is a live extraction column. Judge the concept, not the keywords: the champion writes a two-line definition with inclusion and exclusion examples here before anyone labels. 5,064 papers mention a keyword; 2,096 qualify; topic champion: David Ruiz-García (proposed).

IN I

OUT O

Unsure U

Clear X

All above → IN A

All below → OUT B

☐ overwrite

Undo Ctrl Z

Show

all papers with a hit

Order

closest to the line first

Margin ± 0.5

title, author, journal, id

☐ suspect text

7,309 shown, 2,095 above the line

1

2 A survey of ciguatera at Enewetak and Bikini, Marshall Islands, with notes on the systematics and food habits of ciguatoxic fishes.

1980 · Randall · Fishery Bulletin

above the line

discard\* ×3

retention ×1

...(1975a), and Tester (see footnote 7), have added to the list of the great variety ofmarine animals (mainly fishes) that this species will take as food, as well as sundry item...

2

2 New record of a great white shark, Carcharodon carcharias (Lamnidae) from the eastern Adriatic Sea.

2005 · Soldo et al. · Cybium

above the line

discard\* ×2

...Due to its great size, it was not possible to preserve the specimen in the vessel's fish hold, so it was discarded to sea....

3

2 Distribution, abundance, and habits of pelagic sharks in the central Pacific Ocean.

1958 · Strasburg · Fishery Bulletin

above the line

catch data ×3

...The catch data for the mackerel shark consist of 28 records, 26 of which are from north of 35° ;N. latitude indicating that this shark is a temperate- or cold-water form (fig. 4)...

4

2 Distribution, biology and fisheries importance of the pencil shark, Hypogaleus hyugaensis (Elasmobranchii: Tria...

2002 · Simpfendorfer et al. · Marine and Freshwater Research

above the line

catch data ×1

discard\* ×4

stock assessment ×1

fisheries management ×1

...Map of study area and fisheries management zones....

5

2 Reproductive biology and abundance of the white-dotted skate, Bathyraja albomaculata, in the Southwest Atlantic

2006 · Ruocco et al. · ICES Journal of Marine Science

above the line

catch data ×1

bycatch ×3

discard\* ×2

...The catch data and samples of skates used in this paper were obtained from eight research cruises conducted in the Southwest Atlantic between 34(S an...

6

2 The role of protected areas in elasmobranch fisheries management and conservation.

2002 · Stevens · In Elasmobranch Biodiversity

above the line

fisheries management ×1

...The role of protected areas in elasmobranch fisheries management and conservation....

7

2 Movements and swimming behaviour of white sharks (Carcharodon carcharias) in Australian waters.

2006 · Bruce et al. · Marine Biology

above the line

bycatch ×6

...Free-swimming Game-Wshing Commercial bycatch Total...

8

2 Lipid, fatty acid and squalene composition of liver oil from six species of deepwater sharks collected in so...

1995 · Bakes et al. · Comparative Biochemistry and Physiology – Part B

above the line

bycatch ×3

discard\* ×1

retention ×2

...Peaks were quantified with software from DAPA Scientific Software Individual components were identified using mass spectral data and by comparing rete...

9

2 Long-term changes in ray populations in the North Sea.

1996 · Walker et al. · ICES Journal of Marine Science

above the line

fishing mortality ×2

catch data ×1

...Landings statistics Nominal catch data on rays and skates have been collected by ICES member countries since 1903 and published in Bulletins Statistiques...

10

2 Age determination and growth of the smalltail shark, Carcharhinus porosus, from northern Brazil.

1998 · Lessa et al. · Marine and Freshwater Research

above the line

fishing mortality ×2

...Such biases and distortions referred to as the 'phenomenon of apparent change of growth rate' can be caused by length-selective fishing mortality (Lee 1912; Ricker 1969)....

11

2 Size composition, monthly condition factor and morphometrics for fishery-dependent samples of Rioraja agassizi (Chondrich...

2007 · Oddone et al. · Neotropical Ichthyology

above the line

fishery-dependent ×2

discard\* ×1

...Size composition, monthly condition factor and morphometrics for fishery-dependent samples of Rioraja agassizi (Chondrichthyes: Rajidae), off Santos, Southe...

12

2 Vertebral size-at-age heterogeneity in gummy shark harvested off southern Australia.

1999 · Troynikov et al. · Journal of Fish Biology

above the line

fishing mortality ×6

...Finally we examined whether our results are consistent with the hypothesis of the phenomenon of apparent change in growth rate caused by length-selective fishing mortalit...

13

2 Growth and demography of the Pacific Angle Shark (Squatina californica), based upon tag returns off ...

1992 · Cailliet et al. · Australian Journal of Marine and Freshwater Research

above the line

fishing mortality ×7

...To simulate what would happen to the demography of the Pacific angel shark population under a range of natural and fishing mortalities, the survivorship function (L<sub>t</sub>) was m...

14

2 Spatial, seasonal and biological analysis of sharks caught in the New South Wales protective beach meshin...

1994 · Krogh · Australian Journal of Marine and Freshwater Research

above the line

bycatch ×2

...A total of 753 angel sharks (including 100 recorded as bycatch) was caught in the NSW beach meshing programme between October 1972 and...

Paper

Rule

Impact

Optimiser

Keys

Optimiser. You judge papers; this searches for the rule that best reproduces your judgements, instead of anyone tuning numbers by hand. It reports accuracy on labels held back from the fitting, which is the figure that means anything.

Search for the rule that best reproduces the labels

The judge labels papers; the search proposes the keyword weights, section weights and threshold. It tries one setting at a time, keeps whatever raises the score, repeats until nothing improves, and charges a small penalty per change so that it prefers rules close to the current one.

Labels

Fable silver (1,941 papers read)

Optimise

F1 (balance)

☒ threshold

☒ section weights

☒ drop keywords

☐ keyword weights

☒ proximity switch

☐ rounding

Run search

CONCEPT (extraction column)

Topic review › Fisheries

d\_fisheries

106 concepts in the Fisheries topic: 40 live extraction columns and 66 proposed, never run.

d\_fisheries is a live extraction column.

Judge the **concept**, not the keywords: the champion writes a two-line definition with inclusion and exclusion examples here before anyone labels.

5,064 papers mention a keyword; 2,096 qualify; topic champion: **David Ruiz-García (proposed)**.

IN I

OUT O

Unsure U

Clear X

All above → IN A

All below → OUT B

☐ overwrite

Undo Ctrl Z

Show

all papers with a hit

Order

closest to the line first

Margin ±

0.5

title, author, journal, id

☐ suspect text

7,309 shown, 2,095 above the line

|                                                                                                                                                                                                                                                                          |                                                                                                                                              |                                                                               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| 1                                                                                                                                                                                                                                                                        | <b>2 A survey of ciguatera at Enewetak and Bikini, Marshall Islands, with notes on the systematics and food habits of ciguatoxic fishes.</b> | 1980 · Randall · Fishery Bulletin                                             |
| <div>above the line</div> <div>discard* ×3</div> <div>retention ×1</div> <div>...(1975a), and Tester (see footnote 7), have added to the list of the great variety ofmarine animals (mainly fishes) that this species will take as food, as well as sundry item...</div> |                                                                                                                                              |                                                                               |
| 2                                                                                                                                                                                                                                                                        | <b>2 New record of a great white shark, Carcharodon carcharias (Lamnidae) from the eastern Adriatic Sea.</b>                                 | 2005 · Soldo et al. · Cybium                                                  |
| <div>above the line</div> <div>discard* ×2</div> <div>...Due to its great size, it was not possible to preserve the specimen in the vessel's fish hold, so it was discarded to sea....</div>                                                                             |                                                                                                                                              |                                                                               |
| 3                                                                                                                                                                                                                                                                        | <b>2 Distribution, abundance, and habits of pelagic sharks in the central Pacific Ocean.</b>                                                 | 1958 · Strasburg · Fishery Bulletin                                           |
| <div>above the line</div> <div>catch data ×3</div> <div>...The catch data for the mackerel shark consist of 28 records, 26 of which are from north of 35° ;N. latitude indicating that this shark is a temperate- or cold-water form (fig. 4)...</div>                   |                                                                                                                                              |                                                                               |
| 4                                                                                                                                                                                                                                                                        | <b>2 Distribution, biology and fisheries importance of the pencil shark, Hypogaleus hyugaensis (Elasmobranchii: Tria...</b>                  | 2002 · Simpfendorfer et al. · Marine and Freshwater Research                  |
| <div>above the line</div> <div>catch data ×1</div> <div>discard* ×4</div> <div>stock assessment ×1</div> <div>fisheries management ×1</div> <div>...Map of study area and fisheries management zones....</div>                                                           |                                                                                                                                              |                                                                               |
| 5                                                                                                                                                                                                                                                                        | <b>2 Reproductive biology and abundance of the white-dotted skate, Bathyraja albomaculata, in the Southwest Atlantic</b>                     | 2006 · Ruocco et al. · ICES Journal of Marine Science                         |
| <div>above the line</div> <div>catch data ×1</div> <div>bycatch ×3</div> <div>discard* ×2</div> <div>...The catch data and samples of skates used in this paper were obtained from eight research cruises conducted in the Southwest Atlantic between 34(S an...</div>   |                                                                                                                                              |                                                                               |
| 6                                                                                                                                                                                                                                                                        | <b>2 The role of protected areas in elasmobranch fisheries management and conservation.</b>                                                  | 2002 · Stevens · In Elasmobranch Biodiversity                                 |
| <div>above the line</div> <div>fisheries management ×1</div> <div>...The role of protected areas in elasmobranch fisheries management and conservation....</div>                                                                                                         |                                                                                                                                              |                                                                               |
| 7                                                                                                                                                                                                                                                                        | <b>2 Movements and swimming behaviour of white sharks (Carcharodon carcharias) in Australian waters.</b>                                     | 2006 · Bruce et al. · Marine Biology                                          |
| <div>above the line</div> <div>bycatch ×6</div> <div>...Free-swimming Game-Wshing Commercial bycatch Total...</div>                                                                                                                                                      |                                                                                                                                              |                                                                               |
| 8                                                                                                                                                                                                                                                                        | <b>2 Lipid, fatty acid and squalene composition of liver oil from six species of deepwater sharks collected in so...</b>                     | 1995 · Bakes et al. · Comparative Biochemistry and Physiology – Part B        |
| <div>above the line</div> <div>bycatch ×3</div> <div>discard* ×1</div> <div>retention ×2</div> <div>...Peaks were quantified with software from DAPA Scientific Software Individual components were identified using mass spectral data and by comparing rete...</div>   |                                                                                                                                              |                                                                               |
| 9                                                                                                                                                                                                                                                                        | <b>2 Long-term changes in ray populations in the North Sea.</b>                                                                              | 1996 · Walker et al. · ICES Journal of Marine Science                         |
| <div>above the line</div> <div>fishing mortality ×2</div> <div>catch data ×1</div> <div>...Landings statistics Nominal catch data on rays and skates have been collected by ICES member countries since 1903 and published in Bulletins Statistiques....</div>           |                                                                                                                                              |                                                                               |
| 10                                                                                                                                                                                                                                                                       | <b>2 Age determination and growth of the smalltail shark, Carcharhinus porosus, from northern Brazil.</b>                                    | 1998 · Lessa et al. · Marine and Freshwater Research                          |
| <div>above the line</div> <div>fishing mortality ×2</div> <div>...Such biases and distortions referred to as the 'phenomenon of apparent change of growth rate' can be caused by length-selective fishing mortality (Lee 1912; Ricker 1969)....</div>                    |                                                                                                                                              |                                                                               |
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| <div>above the line</div> <div>fishery-dependent ×2</div> <div>discard* ×1</div> <div>...Size composition, monthly condition factor and morphometrics for fishery-dependent samples of Rioraja agassizi (Chondrichthyes: Rajidae), off Santos, Southe...</div>           |                                                                                                                                              |                                                                               |
| 12                                                                                                                                                                                                                                                                       | <b>2 Vertebral size-at-age heterogeneity in gummy shark harvested off southern Australia.</b>                                                | 1999 · Troynikov et al. · Journal of Fish Biology                             |
| <div>above the line</div> <div>fishing mortality ×6</div> <div>...Finally we examined whether our results are consistent with the hypothesis of the phenomenon of apparent change in growth rate caused by length-selective fishing mortalit...</div>                    |                                                                                                                                              |                                                                               |
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| 14                                                                                                                                                                                                                                                                       | <b>2 Spatial, seasonal and biological analysis of sharks caught in the New South Wales protective beach meshin...</b>                        | 1994 · Krogh · Australian Journal of Marine and Freshwater Research           |
| <div>above the line</div> <div>bycatch ×2</div> <div>...A total of 753 angel sharks (including 100 recorded as bycatch) was caught in the NSW beach meshing programme between October 1972 and...</div>                                                                  |                                                                                                                                              |                                                                               |

Paper

Rule

Impact

Optimiser

Keys

Selecting

Click a row to move the cursor. Shift-click or drag selects a range, Ctrl-click toggles one paper.

↑ I (or ↓ K) move, with Shift to extend. Space toggles the cursor paper in the selection.

Esc clears it.

Labelling

I IN, O OUT, U unsure, X clear. With nothing selected they act on the cursor paper and move down one.

A labels every unlabelled paper above the cursor IN; B labels every unlabelled paper below it

OUT. These are stored as assumed labels (dashed pill), separately from papers you looked at, and the accuracy figures can be shown with or without them.

Ctrl Z undoes the last labelling action, however large.

What the numbers mean

The score is the extractor's own: each keyword mention counts once, multiplied by the weight of the section it sits in, and only if a shark or ray term appears within one sentence either side (when the proximity filter is on). A paper is IN when the score reaches the threshold and, if the rule has anchors, at least one anchor word occurs.

Topic review › Fisheries

CONCEPT (extraction column)

Live extraction columns — 40 concepts

- pr\_fishing\_commercial
- pr\_fishing\_artisanal
- pr\_fishing\_recreational
- pr\_fishing\_iuu
- pr\_bycatch
- pr\_shark\_finning
- pr\_targeted\_fishing
- pr\_discarding
- gear\_longline
- gear\_gillnet
- gear\_trawl
- gear\_purse\_seine
- gear\_seine
- gear\_hook\_line
- gear\_trap
- gear\_net\_other
- gear\_harpoon
- gear\_dredge
- gear\_trawl\_beam
- gear\_trawl\_otter
- gear\_survey
- gear\_pelagic
- gear\_demersal
- gear\_artisanal
- gear\_mit\_circle\_hook
- gear\_mit\_brd
- gear\_mit\_deterrent
- gear\_mit\_time\_area
- gear\_mit\_handling

106 concepts in the Fisheries topic: 40 live extraction columns and 66 proposed, never run.

Ladder

Margin queue

Reviewer

Simon

Export

Import

Submit

d\_fisheries is a live extraction column. Judge the concept, not the keywords: the champion writes a two-line definition with inclusion and exclusion examples here before anyone labels. 5,064 papers mention a keyword; 2,096 qualify; topic champion: David Ruiz-García (proposed).

IN I OUT O Unsure U Clear X All above → IN A All below → OUT B overwrite Undo Ctrl Z

Show all papers with a hit Order closest to the line first Margin ± 0.5 title, author, journal, id 7,313 shown, 2,096 above the line

|   |                                                                                                                                                                                                                                |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>2 A survey of ciguatera at Enewetak and Bikini, Marshall Islands, with notes on the systematics and food habits of ciguatoxic fishes.</b> 1980 · Randall · Fishery Bulletin                                                 |
|   | above the line discard* ×3 retention ×1 ... (1975a), and Tester (see footnote 7), have added to the list of the great variety of marine animals (mainly fishes) that this species will take as food, as well as sundry item... |
| 2 | <b>2 New record of a great white shark, Carcharodon carcharias (Lamnidae) from the eastern Adriatic Sea.</b> 2005 · Soldo et al. · Cybium                                                                                      |
|   | above the line discard* ×2 ... Due to its great size, it was not possible to preserve the specimen in the vessel's fish hold, so it was discarded to sea....                                                                   |
| 3 | <b>2 Distribution, abundance, and habits of pelagic sharks in the central Pacific Ocean.</b> 1958 · Strasburg · Fishery Bulletin                                                                                               |
|   | above the line catch data ×3 ... The catch data for the mackerel shark consist of 28 records, 26 of which are from north of 35° N. latitude indicating that this shark is a temperate- or cold-water form (fig. 4)....         |
| 4 | <b>2 Distribution, biology and fisheries importance of the pencil shark, Hypogaleus hyugaensis (Elasmobranchii: Triacanthidae).</b> 2002 · Simpfendorfer et al. · Marine and Freshwater Research                               |
|   | above the line catch data ×1 discard* ×4 stock assessment ×1 fisheries management ×1 ... Map of study area and fisheries management zones....                                                                                  |
| 5 | <b>2 Reproductive biology and abundance of the white-dotted skate, Bathyraja albomaculata, in the Southwest Atlantic</b> 2006 · Ruocco et al. · ICES Journal of Marine Science                                                 |

Paper Rule Impact Optimiser Keys

Rule. The rule itself: which keywords count, how much a mention is worth in each section of a paper, the score a paper must reach, and any anchor word that must be present. Change anything here and every paper in the list is re-scored at once, so the threshold bar moves as you type. Nothing you do here touches the corpus.

d\_fisheries live column

Every change re-scores all 7,313 papers here in the browser. Nothing is written to the corpus until a proposal is accepted.

Threshold 2 elasmobranch proximity round the score (retired)

Reset to live rule

Section weights

|         |                        |            |              |         |
|---------|------------------------|------------|--------------|---------|
| title   | keywords               | abstract   | introduction | methods |
| 2       | 2                      | 0.5        | 0.5          | 1       |
| results | results and discussion | discussion | conclusions  | other   |

Every rule in every topic

The live extraction columns and the champions' proposed rules, with the keywords and anchors each one counts, its threshold, how many papers it touches, and how far it agrees with Fable's reading of the corpus sample (silver labels, not expert). A column that belongs to several topics is listed under each; the counts are the same. The **Source** column opens the line of the extractor that defines a live column, or the workbook a proposed rule came from.

filter: topic, rule, keyword, anchor...

topic

Fisheries (106)

▼

kind

all

▼

schema

all

▼

group by topic

☒

each column once

☐

106 rows of 244; 106 distinct rules

all 22 topics (244)

Trophic ecology 2

Movement and spatial ecology 2

Habitat and ecosystems 20

Ocean basins 9

Sub-basins and seas 43

Behaviour 6

Sensory biology 5

Taxonomy and systematics 1

Palaeontology 1

Biology and life history 3

Physiology 9

Biomechanics 1

Reproductive biology 2

Genetics and genomics 2

Immunology and disease 2

Conservation and management 14

Fisheries 106

Husbandry and aquaria 2

Ecotourism 3

Human dimensions 6

Toxicology and pollution 4

Data science and methods 1

| rule                                                                    | keywords (anchors in blue; case-sensitive in bold)    | thr                                                                                                                                                                                                                       | prox | papers with a mention | qualify | margin | Fable IN | precision | recall | F1   | unreachable | source                   |
|-------------------------------------------------------------------------|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----------------------|---------|--------|----------|-----------|--------|------|-------------|--------------------------|
| Fisheries (Human interactions) · champion: David Ruiz-García (proposed) |                                                       |                                                                                                                                                                                                                           |      |                       |         |        |          |           |        |      |             |                          |
| d_fisheries                                                             | <a href="#">discipline</a> <a href="#">quotations</a> | stock assessment, fisheries management, catch data, fishing mortality, maximum sustainable yield, <b>MSY</b> , fishery-dependent, fishery-independent, bycatch, discard*, retention, non-target species, non-target catch | 2    | on                    | 5,064   | 2,096  | 958      | 623       | 77%    | 45%  | 56%         | 160 <a href="#">L258</a> |
| gear_artisanal                                                          |                                                       | artisanal, traditional gear, small-scale, hand-operated                                                                                                                                                                   | 2    |                       | 2,363   | 700    | 346      | 236       | 76%    | 49%  | 60%         | 32 <a href="#">L188</a>  |
| gear_demersal                                                           |                                                       | demersal longline, bottom trawl, demersal trawl, bottom longline                                                                                                                                                          | 1    |                       | 1,508   | 984    | 651      | 327       | 89%    | 31%  | 46%         | 186 <a href="#">L187</a> |
| gear_dredge                                                             |                                                       | dredge, towed dredge, scallop dredge, clam dredge, oyster dredge, hydraulic dredge                                                                                                                                        | 1    |                       | 235     | 132    | 127      | 11        | 6%     | 13%  | 8%          | 5 <a href="#">L182</a>   |
| gear_ghost                                                              |                                                       | ghost gear, ghost net, ghost fishing, <b>ALDFG</b> , abandoned gear, lost gear, derelict gear, derelict fishing                                                                                                           | 1    |                       | 60      | 30     | 22       | 13        | 33%    | 9%   | 14%         | 7 <a href="#">L204</a>   |
| gear_gillnet                                                            |                                                       | gillnet, gill net, trammel net, entangling net, drift net, driftnet                                                                                                                                                       | 2    |                       | 1,798   | 556    | 255      | 319       | 89%    | 21%  | 33%         | 140 <a href="#">L174</a> |
| gear_harpoon                                                            |                                                       | harpoon, spearfish*, spear gun                                                                                                                                                                                            | 1    |                       | 413     | 222    | 207      | 59        | 37%    | 22%  | 28%         | 28 <a href="#">L181</a>  |
| gear_hook_line                                                          |                                                       | hook and line, handline, rod and reel, trolling, jigging, pole and line                                                                                                                                                   | 2    |                       | 1,017   | 201    | 159      | 221       | 96%    | 11%  | 20%         | 107 <a href="#">L178</a> |
| gear_longline                                                           | <a href="#">quotations</a>                            | longline, long-line, pelagic longline, demersal longline, bottom longline                                                                                                                                                 | 2    |                       | 2,802   | 1,261  | 447      | 328       | 86%    | 43%  | 57%         | 72 <a href="#">L173</a>  |
| gear_mit_brd                                                            |                                                       | bycatch reduction device, <b>BRD</b> , turtle excluder, <b>TED</b>                                                                                                                                                        | 1    |                       | 156     | 74     | 56       | 31        | 77%    | 37%  | 50%         | 13 <a href="#">L191</a>  |
| gear_mit_circle_hook                                                    |                                                       | circle hook, non-offset hook                                                                                                                                                                                              | 1    |                       | 203     | 177    | 132      | 67        | 75%    | 20%  | 31%         | 47 <a href="#">L190</a>  |
| gear_mit_deterrent                                                      |                                                       | shark deterrent, SharkGuard, shark guard, electropositive, Rare Earth, <b>EPM</b> , LED deterrent, magnetic deterrent                                                                                                     | 1    |                       | 179     | 99     | 74       | 15        | 46%    | 40%  | 43%         | 8 <a href="#">L192</a>   |
| gear_mit_handling                                                       |                                                       | safe release, handling practice*, live release, post-release mortality, <b>PRM</b>                                                                                                                                        | 1    |                       | 524     | 227    | 191      | 96        | 56%    | 38%  | 46%         | 32 <a href="#">L194</a>  |
| gear_mit_illumination                                                   |                                                       | illuminat* net, illuminat* gillnet, LED net, net light*, lightstick*, light attract*                                                                                                                                      | 1    |                       | 32      | 19     | 15       | 7         | 100%   | 17%  | 29%         | 4 <a href="#">L200</a>   |
| gear_mit_line_weight                                                    |                                                       | line weight*, weighted branchline, leaded swivel, sliding lead, lumo lead, sink rate                                                                                                                                      | 1    |                       | 14      | 12     | 7        | 3         | 50%    | 33%  | 40%         | 0 <a href="#">L197</a>   |
| gear_mit_pinger                                                         |                                                       | pinger, acoustic alarm, acoustic deterrent, porpoise alerting device, <b>PAL</b>                                                                                                                                          | 1    |                       | 209     | 113    | 80       | 2         | 15%    | 100% | 27%         | 0 <a href="#">L199</a>   |
| gear_mit_setting                                                        |                                                       | night set*, deep set*, deep-set buoy gear, side-set*, underwater set*                                                                                                                                                     | 1    |                       | 64      | 37     | 35       | 11        | 33%    | 33%  | 33%         | 4 <a href="#">L198</a>   |
| gear_mit_time_area                                                      |                                                       | time-area closure, spatial closure, fishing closure, seasonal closure, <b>MPA</b>                                                                                                                                         | 2    |                       | 720     | 316    | 100      | 160       | 42%    | 20%  | 27%         | 82 <a href="#">L193</a>  |
| gear_mit_weak_hook                                                      |                                                       | weak hook, corrodible hook, designed to straighten                                                                                                                                                                        | 1    |                       | 1       | 1      | 0        | 1         | –      | 0%   | –           | 1 <a href="#">L196</a>   |



Topic review › All rules

Every rule in every topic

The live extraction columns and the champions' proposed rules, with the keywords and anchors each one counts, its threshold, how many papers it touches, and how far it agrees with Fable's reading of the corpus sample (silver labels, not expert). A column that belongs to several topics is listed under each; the counts are the same. The **Source** column opens the line of the extractor that defines a live column, or the workbook a proposed rule came from.

filter: topic, rule, keyword, anchor...

topic **Genetics and genomics (2)**

kind **all**

schema **all**

☒ group by topic

☐ each column once

2 rows of 244; 2 distinct rules

all 22 topics (244)

Trophic ecology 2

Movement and spatial ecology 2

Habitat and ecosystems 20

Ocean basins 9

Sub-basins and seas 43

Behaviour 6

Sensory biology 5

Taxonomy and systematics 1

Palaeontology 1

Biology and life history 3

Physiology 9

Biomechanics 1

Reproductive biology 2

**Genetics and genomics 2**

Immunology and disease 2

Conservation and management 14

Fisheries 106

Husbandry and aquaria 2

Ecotourism 3

Human dimensions 6

Toxicology and pollution 4

Data science and methods 1

| rule                                                     | keywords (anchors in blue; case-sensitive in bold)                                                                                                                                                         | thr | prox | papers with a mention | qualify | margin | Fable IN | precision | recall | F1  | unreachable | source               |
|----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|------|-----------------------|---------|--------|----------|-----------|--------|-----|-------------|----------------------|
| Genetics and genomics (Biology) · champion: (unassigned) |                                                                                                                                                                                                            |     |      |                       |         |        |          |           |        |     |             |                      |
| d_genetics                                               | <a href="#">discipline</a> <a href="#">quotations</a> genetic*, genomic*, <b>eDNA</b> , environmental DNA, microsatellite, mitochondrial, phylogenet*, haplotype, <b>SNP</b> , RADseq, population genetics | 2   |      | 6,498                 | 3,347   | 966    | 339      | 60%       | 80%    | 69% | 23          | <a href="#">L256</a> |
| imp_genetic                                              | <a href="#">quotations</a> genetic diversity, effective population size, bottleneck, inbreeding                                                                                                            | 2   |      | 867                   | 380     | 115    | 46       | 70%       | 70%    | 70% | 4           | <a href="#">L222</a> |

Built 2026-10-08 03:08 PDT from 22 topics (232 distinct rules); live-column source lines refer to extractor commit f52d48ec4d. Precision, recall and F1 compare the live rule with Fable's labels on the papers Fable has read; "unreachable" counts Fable-IN papers that mention no keyword of the rule at all. The full keyword reference by schema, with notes, is [extraction review reference](#).

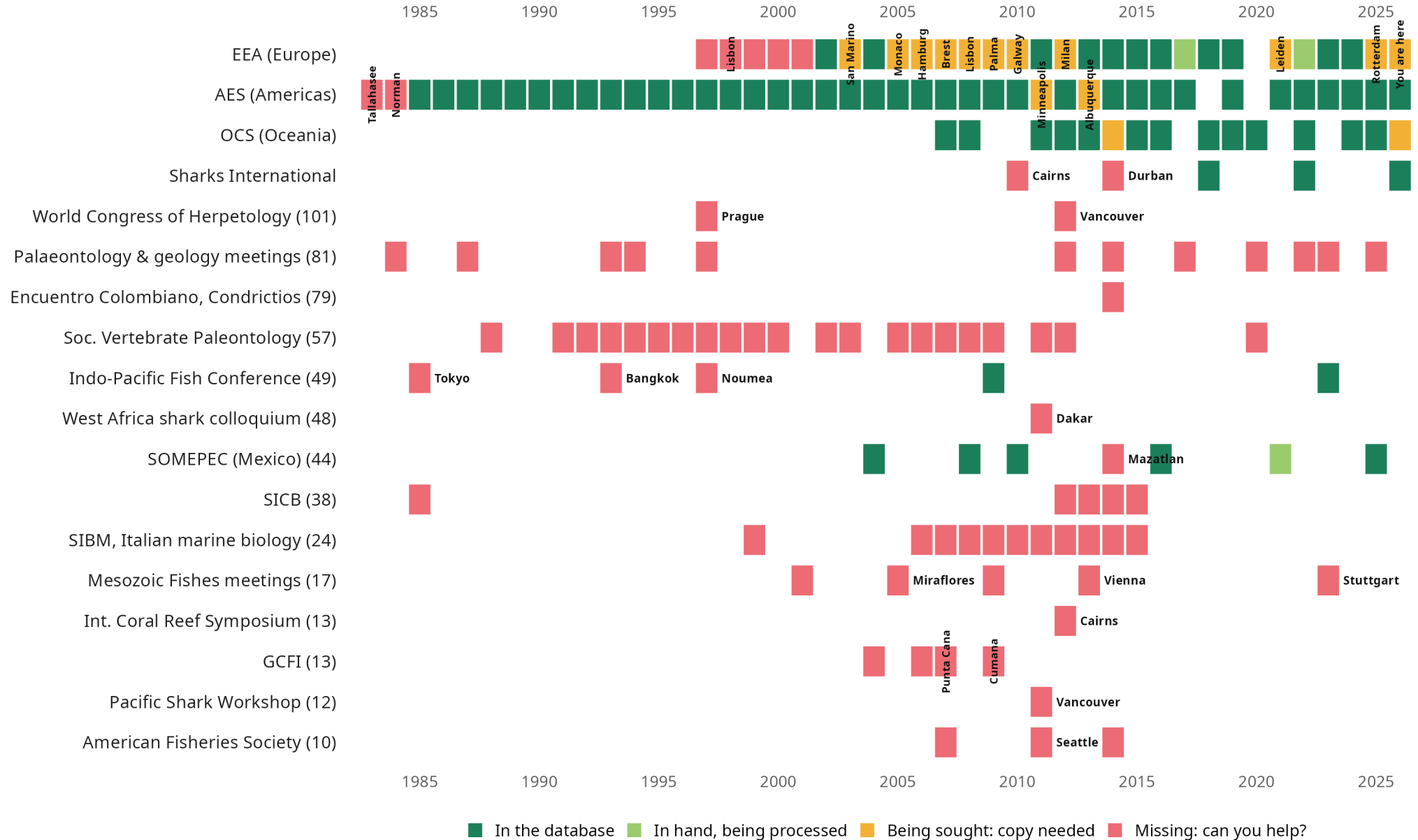
# Topic champions wanted: 1 of 22 topics has one

| Discipline / topic           | Champion          |
|------------------------------|-------------------|
| Fisheries                    | David Ruiz-García |
| Habitat and ecosystems       | you?              |
| Ocean basins                 | you?              |
| Sub-basins and seas          | you?              |
| Biology and life history     | you?              |
| Behaviour                    | you?              |
| Trophic ecology              | you?              |
| Genetics and genomics        | you?              |
| Movement and spatial ecology | you?              |
| Conservation and management  | you?              |
| Data science and methods     | you?              |

| Discipline / topic       | Champion |
|--------------------------|----------|
| Husbandry and aquaria    | you?     |
| Palaeontology            | you?     |
| Taxonomy and systematics | you?     |
| Physiology               | you?     |
| Reproductive biology     | you?     |
| Biomechanics             | you?     |
| Sensory biology          | you?     |
| Ecotourism               | you?     |
| Human dimensions         | you?     |
| Immunology and disease   | you?     |
| Toxicology and pollution | you?     |



72 of 170 meetings since 1983 are in the database. Blank = no meeting that year.



# Conference abstracts: browse what we hold

FIU

## Conference abstracts

Every abstract we hold from shark, ray, and chimaera meetings (and the general fish meetings that carry them), searchable by title, author, or any word in the abstract. Missing a book from your own shelves? [See what we lack.](#)



|                 |                     |                                 |                                 |
|-----------------|---------------------|---------------------------------|---------------------------------|
| 124<br>meetings | 23,491<br>abstracts | 7,867<br>elasmobranch abstracts | 38,463<br>distinct author names |
|-----------------|---------------------|---------------------------------|---------------------------------|

Author

Title

Abstract text

Meeting series

Presentation type

Year from

to

Sort by

All

All

1985

2026

Newest first ▾

☒ Elasmobranch abstracts only

SI meeting

EEA meeting

AES meeting

OCS meeting

All

All

All

All

Leave a list empty to include everything. Several boxes narrow the results together.

7,867 abstracts (elasmobranch only; untick to include all fish)

- Post-release survival of silky sharks in tropical tuna purse seine fishery under best handling practices: insights from tagging and predictive modelling** elasm

EEA 2026 · Leire Lopetegui Eguren + 13 more talk
- Estimates of elasmobranch bycatch in the Atlantic by tuna fisheries to guide mitigation priorities** elasm

EEA 2026 · Gloria Marino Briceno + 4 more talk
- A dedicated survey of the spinetail devil ray (Mobula mobular) along the west coast of Corsica** elasm

EEA 2026 · William Travers + 2 more talk
- Spatio-temporal patterns and biological characteristics of silky shark (Carcharhinus falciformis) and scalloped hammerhead (Sphyrna lewini) landings at Tanjung Luar, Indonesia** elasm

EEA 2026 · Baiq Lista Azkia Sulhana + 4 more lightning
- Evaluating ICCAT Conservation and Management Measures for Pelagic Sharks and Rays: Progress in Reporting, Mitigation, and Research** elasm

EEA 2026 · Lola Lacambra + 4 more talk
- Turning Regulation into Practice: Advancing the Shark and Ray Conservation Measures and Tuna Industry Engagement** elasm

EEA 2026 · Efin Muttaqin + 5 more talk
- Porbeagle aggregation in Brittany: fine-scale monitoring and perspectives on North Atlantic connectivity** elasm

EEA 2026 · Lucas Zaccagnini + 4 more talk
- Environmental Constraints of the Reproduction of Myliobatiform Rays within the Pelagic Zone** elasm

EEA 2026 · Nicolas Ziani + 9 more lightning
- Hidden interactions in the North-East Atlantic: sea lamprey parasitism of porbeagle sharks (Lamna nasus)** elasm

EEA 2026 · Cassiopé Muller + 3 more lightning
- Unlocking key information for Mediterranean sharks conservation through Fishers' Ecological Knowledge** elasm

EEA 2026 · Malo Rio talk
- Catching tope (Galeorhinus galeus): using recreational angling competitions to drive species recovery.** elasm

EEA 2026 · Gordon Watson + 2 more talk

# Conference abstracts: the gaps, year by year

## Conference abstracts

Every abstract we hold from shark, ray, and chimaera meetings (and the general fish meetings that carry them), searchable by title, author, or any word in the abstract. Missing a book from your own shelves? [See what we lack](#).



**124** meetings  
**23,491** abstracts  
**7,867** elasmobranch abstracts  
**38,463** distinct author names

Search: zzzzzq

Author: e.g. Heithaus Title: e.g. juvenile white shark Abstract text: e.g. acoustic telemetry

Meeting series: All Presentation type: All Year from: 1985 to: 2026 Sort by: Newest first ☒ Elasmobranch abstracts only

SI meeting: All EEA meeting: All AES meeting: All OCS meeting: All

Leave a list empty to include everything. Several boxes narrow the results together.

0 abstracts (elasmobranch only; untick to include all fish) matching "zzzzzq"

Nothing matches. Try fewer words, clear a filter, or untick "Elasmobranch only".

## Coverage: what we hold

Each cell is one meeting. Colours run from nothing known (red) to fully ingested (green). Blank means no meeting that year for that series.

Series with no abstracts yet appear in the grid but not in the search filters.

Missing Schedule Hardcopy Programme Pending OCR Digital Extracted Partial Ingested

**Help us fill the gaps.** Do you have an abstract book, programme, or newsletter from one of these meetings, on paper or as a file? We would like a scan or a photo of the abstract pages. Years with no abstract book in our hands (1983 onward):

**ASIH/JMIH:** 1983, 1984, 1985, 1986, 1987, 1988, 1989, 1990, 1991, 1992, 1993, 1994, 1995, 1996, 2002, 2004, 2011, 2013

**AES:** 1983, 1984, 2011, 2013

**EEA:** 1997, 1998, 1999, 2000, 2001, 2003, 2005, 2006, 2007, 2008, 2009, 2010, 2012, 2017, 2021, 2025

**OCS:** 2014, 2026

**SI:** 2010, 2014

| Year | ASIH/JMIH                | AES                      | EEA                           | OCS                        | SI                  | ECC | SBEEL | SOMEPEC              | W Africa | GCFI | Pac Shark Wksp | I |
|------|--------------------------|--------------------------|-------------------------------|----------------------------|---------------------|-----|-------|----------------------|----------|------|----------------|---|
| 2026 | New Orleans<br>Extracted | New Orleans<br>Extracted | online<br>Ingested            | Pending                    | Colombo<br>Ingested |     |       |                      |          |      |                |   |
| 2025 | St. Paul<br>Ingested     | St. Paul<br>Ingested     | Rotterdam<br>Programme        | Sunshine Coast<br>Ingested |                     |     |       | Ensenada<br>Ingested |          |      |                |   |
| 2024 | Pittsburgh<br>Ingested   | Pittsburgh<br>Ingested   | Thessaloniki<br>Digital       | Geelong<br>Ingested        |                     |     |       |                      |          |      |                |   |
| 2023 | Norfolk<br>Ingested      | Norfolk<br>Ingested      | Brighton<br>Ingested          |                            |                     |     |       |                      |          |      |                |   |
| 2022 | Spokane<br>Ingested      | Spokane<br>Ingested      | Valencia (=SI2022)<br>Digital | Virtual<br>Ingested        | Valencia<br>Digital |     |       |                      |          |      |                |   |
| 2021 | Phoenix<br>Ingested      | Phoenix<br>Ingested      | Leiden<br>Programme           |                            |                     |     |       | Puebla<br>Digital    |          |      |                |   |

|      |                              |                              |                             |                                  |                         |         |  |  |  |  |                                  |                       |
|------|------------------------------|------------------------------|-----------------------------|----------------------------------|-------------------------|---------|--|--|--|--|----------------------------------|-----------------------|
| 2020 |                              |                              |                             | Virtual<br>Ingested              |                         |         |  |  |  |  |                                  |                       |
| 2019 | Snowbird<br>Ingested         | Snowbird<br>Ingested         | Rende<br>Ingested           | Dunedin<br>Ingested              |                         |         |  |  |  |  |                                  |                       |
| 2018 | Rochester<br>Ingested        |                              | Peniche<br>Ingested         | North Stradbroke Is.<br>Ingested | Joao Pessoa<br>Ingested |         |  |  |  |  |                                  |                       |
| 2017 | Austin<br>Ingested           | Austin<br>Ingested           | Amsterdam<br>Schedule       |                                  |                         |         |  |  |  |  |                                  |                       |
| 2016 | New Orleans<br>Ingested      | New Orleans<br>Ingested      | Bristol<br>Ingested         | Wrest Point<br>Ingested          |                         |         |  |  |  |  | Puerto Vallarta<br>Ingested      |                       |
| 2015 | Reno<br>Ingested             | Reno<br>Ingested             | Peniche<br>Digital          | Auckland<br>Ingested             |                         |         |  |  |  |  |                                  |                       |
| 2014 | Chattanooga<br>Ingested      | Chattanooga<br>Ingested      | Leeuwarden<br>Ingested      | Pending                          | Durban<br>Missing       | Missing |  |  |  |  | Mazatlan<br>Missing              |                       |
| 2013 | Albuquerque<br>Hardcopy      | Albuquerque<br>Hardcopy      | Plymouth<br>Ingested        | Brisbane<br>Ingested             |                         |         |  |  |  |  |                                  |                       |
| 2012 | Vancouver<br>Ingested        | Vancouver<br>Ingested        | Milan<br>Hardcopy           | Adelaide<br>Ingested             |                         |         |  |  |  |  |                                  |                       |
| 2011 | Minneapolis<br>Hardcopy      | Minneapolis<br>Hardcopy      | Berlin<br>Ingested          | Gold Coast<br>Ingested           |                         |         |  |  |  |  | Dakar<br>Missing                 | Vancouver<br>Missing  |
| 2010 | Providence<br>Ingested       | Providence<br>Ingested       | Galway<br>Hardcopy          |                                  | Cairns<br>Missing       |         |  |  |  |  | Ciudad Universitaria<br>Ingested |                       |
| 2009 | Portland<br>Ingested         | Portland<br>Ingested         | Palma de Majorca<br>Pending |                                  |                         |         |  |  |  |  |                                  | Cumana<br>Missing     |
| 2008 | Montreal<br>Ingested         | Montreal<br>Ingested         | Lisbon<br>Pending           | Sydney<br>Ingested               |                         |         |  |  |  |  | Ciudad Universitaria<br>Ingested |                       |
| 2007 | St. Louis<br>Ingested        | St. Louis<br>Ingested        | Brest<br>Pending            | Queenscliff<br>Ingested          |                         |         |  |  |  |  |                                  | Punta Cana<br>Missing |
| 2006 | New Orleans<br>Ingested      | New Orleans<br>Ingested      | Hamburg<br>Pending          |                                  |                         |         |  |  |  |  |                                  | Missing               |
| 2005 | Tampa<br>Ingested            | Tampa<br>Ingested            | Monaco<br>Pending           |                                  |                         |         |  |  |  |  |                                  |                       |
| 2004 | Norman<br>Hardcopy           | Norman<br>Ingested           | London<br>Ingested          |                                  |                         |         |  |  |  |  | Ciudad Universitaria<br>Ingested | Missing               |
| 2003 | Manaus<br>OCR                | Manaus<br>Ingested           | San Marino<br>Pending       |                                  |                         |         |  |  |  |  |                                  |                       |
| 2002 | Kansas City<br>Hardcopy      | Kansas City<br>Ingested      | Cardiff<br>Digital          |                                  |                         |         |  |  |  |  | Missing                          |                       |
| 2001 | University Park<br>OCR       | University Park<br>Ingested  | Missing                     |                                  |                         |         |  |  |  |  |                                  |                       |
| 2000 | La Paz<br>OCR                | La Paz<br>Ingested           | Missing                     |                                  |                         |         |  |  |  |  |                                  |                       |
| 1999 | University Park<br>OCR       | University Park<br>Ingested  | Missing                     |                                  |                         |         |  |  |  |  |                                  |                       |
| 1998 | Guelph<br>Ingested           | Guelph<br>Ingested           | Lisbon<br>Missing           |                                  |                         |         |  |  |  |  |                                  |                       |
| 1997 | Seattle<br>OCR               | Seattle<br>Ingested          | Missing                     |                                  |                         |         |  |  |  |  |                                  |                       |
| 1996 | New Orleans<br>Hardcopy      | New Orleans<br>Ingested      |                             |                                  |                         |         |  |  |  |  |                                  |                       |
| 1995 | Edmonton<br>Hardcopy         | Edmonton<br>Ingested         |                             |                                  |                         |         |  |  |  |  |                                  |                       |
| 1994 | Los Angeles<br>Hardcopy      | Los Angeles<br>Ingested      |                             |                                  |                         |         |  |  |  |  |                                  |                       |
| 1993 | Austin<br>Hardcopy           | Austin<br>Ingested           |                             |                                  |                         |         |  |  |  |  |                                  |                       |
| 1992 | Champaign-Urbana<br>Hardcopy | Champaign-Urbana<br>Ingested |                             |                                  |                         |         |  |  |  |  |                                  |                       |
| 1991 | New York<br>Missing          | New York<br>Ingested         |                             |                                  |                         |         |  |  |  |  |                                  |                       |
| 1990 | Charleston<br>Missing        | Charleston<br>Ingested       |                             |                                  |                         |         |  |  |  |  |                                  |                       |
| 1989 | San Francisco<br>Missing     | San Francisco<br>Ingested    |                             |                                  |                         |         |  |  |  |  |                                  |                       |
| 1988 | Ann Arbor<br>Missing         | Ann Arbor<br>Ingested        |                             |                                  |                         |         |  |  |  |  |                                  |                       |
| 1987 | Albany<br>Missing            | Albany<br>Ingested           |                             |                                  |                         |         |  |  |  |  |                                  |                       |
| 1986 | Victoria<br>Missing          | Victoria<br>Ingested         |                             |                                  |                         |         |  |  |  |  |                                  |                       |
| 1985 | Knoxville<br>Missing         | Knoxville<br>Ingested        |                             |                                  |                         |         |  |  |  |  |                                  |                       |
| 1984 | Norman<br>Missing            | Norman<br>Missing            |                             |                                  |                         |         |  |  |  |  |                                  |                       |
| 1983 | Tallahassee<br>Missing       | Tallahassee<br>Missing       |                             |                                  |                         |         |  |  |  |  |                                  |                       |

ASIH, JMIH, UL, EEP, and NIA share one course book and appear as one column. Leave a cell for the host city and status.

## ▼ Author

Reaches papers with OpenAlex author records (~half the corpus); papers without them won't match an author.

## ► Discipline

## ► Technique

## ► Pressure

## ► Gear

## ► Impact

## ► Ecosystem

## ► Ocean basin (text-mined)

## ► Sub-basin (text-mined)

## ► Study country

## ► Study ocean basin

## ► First-author country

## ► First-author region

## ► Country (record field)

## ► Epoch

## ► Journal

## ► Data source

## ► Open-access (OA) status

## ► OA and geography flags

## ► Publication year

How is age determined in sharks from vertebrae?

[Ask](#)[List papers](#)

What methods are used to determine age and growth in sharks?

How does bycatch in commercial fisheries affect elasmobranch populations?

How is environmental DNA (eDNA) used to detect and monitor sharks?

What do stable isotope analyses reveal about shark trophic ecology?

How are acoustic telemetry and satellite tags used to track shark movement?

What are the effects of climate change on the distribution of rays and skates?

► Recent questions

mode: generated (anthropic:claude-fable-5-1)

Age is estimated by counting growth-band pairs on the corpus calcareum (the outer anterior and posterior calcified structures of the centrum), with bands on the intermedialis used only as a guide when those on the corpus calcareum are indistinct [26051]. – Vertebrae may be read whole (e.g. with digitally enhanced images) or as sagittal sections, and in young blue sharks the two approaches gave no systematic age difference to about age 8 (CV <8%), though sections of larger or older animals typically require increased magnification at the centrum margin to resolve the outermost band pairs [5394, 12594]. – Band visibility is often enhanced by staining, such as alizarin red for concentric growth increments in gummy shark [6459]. – Reads are made independently by two or more readers blind to sex and length, with disagreements re-read together and precision/bias assessed via percent agreement and related metrics following Cailliet and Goldman (2004) [5169]. – Because band periodicity is not guaranteed, counts must be validated, for example by oxytetracycline marking of captive or recaptured individuals, tag-recapture, or bomb radiocarbon assays; the latter have shown that band counts can underestimate age by 5–11 years in adult sandbar sharks and by >15 years in large white sharks, where vertebral growth may cease, while confirming annual banding in whale sharks [12594, 14019, 21125, 28332]. – The underlying vertebral development remains poorly understood, which helps explain poor band readability in some species (notably deep-water sharks) and discrepancies between validated and unvalidated ages [26051]. ✓

● Sentence check (experimental): LIMITED — 1 paper(s) entail an answer sentence; 1/6 cited sentences entailed ● Topic coverage: WELL-SUPPORTED — 8 papers retrieved, 8 strongly relevant

8 source papers

Export

BibTeX



Copy to clipboard

Download

Copy answer with references

✓ [26051] Localized zinc distribution in shark vertebrae suggests differential deposition during ontogeny and across vertebral structures.

cos 0.848 · rel 6.546

Raoult, V., Howell, N., Zahra, D., Peddemors, V.M., Howard, D.L., de Jonge, M.D., Buchan, B.L., Williamson, J.E. (2018) · 2018 · PLoS ONE

Determining age at maturity and longevity are two components that are critical for effective fisheries management [7, 8]. In elasmobranchs, age is determined by counting bands on the outer anterior and posterior calcified structures of the vertebrae (corpus ca...

► Why this paper?

✓ [5394] Comparison of whole and sectioned vertebrae for determining the age of young blue shark (Prionace glauca).

cos 0.861 · rel 5.134

Macneil, M.A., Campana, S.E. (2003) · 2003 · Journal of Northwest Atlantic Fishery Science

KEYWORDS age-bias, age determination, precision, Prionace glauca vertebrae OTHER Abstract Age determinations using whole and sectioned vertebrae from 185 blue shark (Prionace glauca) showed no systematic between-method difference in ages to a minimum age of 8 ...

► Why this paper?

✓ [21125] Validated age estimates for large white sharks of the northeastern Pacific Ocean: altered perceptions of vertebral growth shed light on complicated bomb  $\Delta 14C$  results.

cos 0.84 · rel 5.084

Andrews, A.H., Kerr, L.A. (2015) · 2015 · Environmental Biology of Fishes

Discussion New perspectives on the growth of shark vertebrae have shed light on previously analyzed bomb  $14C$  data and enabled age validation of white sharks from the northeastern Pacific Ocean. The prior interpretation of white shark  $14C$  data assumed that samp...

► Why this paper?

✓ [28332] Annual Bands in Vertebrae Validated by Bomb Radiocarbon Assays Provide Estimates of Age and Growth of Whale Sharks.

cos 0.839 · rel 4.862

Ong, J.J.L., Meekan, M.G., Hsu, H.H., Fanning, L.P., Campana, S.E. (2020) · 2020 · Frontiers in Marine Science



# Author & journal info enrichment



**Authors**



**Journal**

**SJR**

Scimago Journal & Country Rank



 **Altmetric**  
(held locally, no longer  
shown on public pages)



# Researcher network: the author atlas

## OCEAN BASIN

## MIN EDGE WEIGHT: 2



## YEAR RANGE: 1950–2026



## COLOUR BY

Main discipline ▼

## SHAPE BY

None (all circles) ▼

Best shape-by attrs in current filter (fewest categories first):

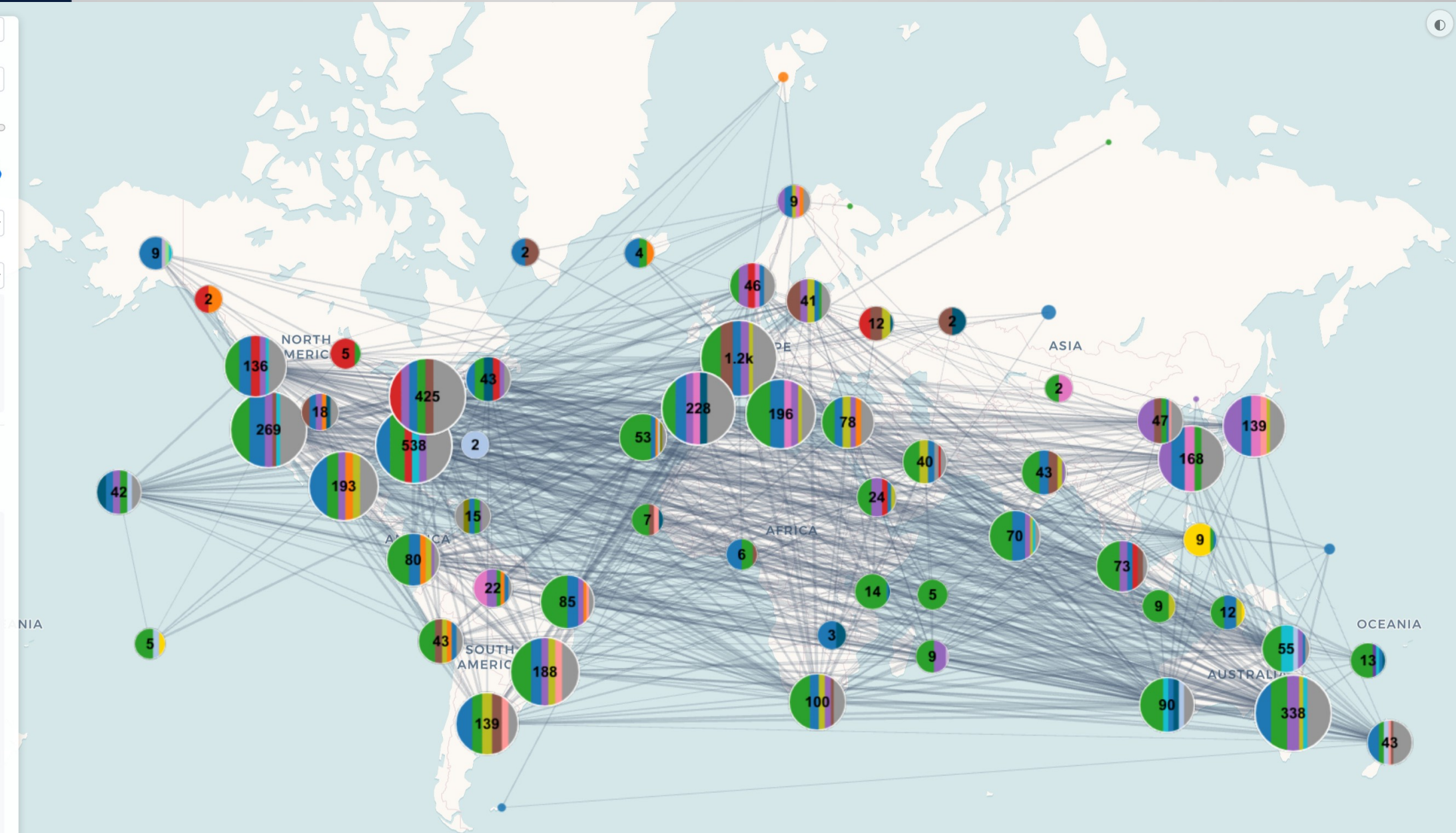
- **gender**: 3 categories — top: M(3651), F(1782), Unknown(2)
- **disc\_main**: 20 categories — top: Conservation(1303), Biology(878), Genetics(599)
- **country**: 125 categories — top: US(1317), AU(489), GB(343)

- ☒ Show edges
- ☒ Cluster at low zoom
- ☒ Group lines per institution

## MAIN DISCIPLINE

- Conservation
- Biology
- Genetics
- Paleontology
- Taxonomy
- Physiology
- Immunology
- Reproductive
- Movement
- Trophic
- Behaviour
- Fisheries
- Sensory
- Toxicology
- Biomechanics
- Husbandry
- Ecotourism
- Human Dimensions
- Data Science
- Unclassified
- other / unknown

Reset filters





# Progress, & what's next

- ✓ **Living Database:** monthly Shark References sync, running
- ✓ **Rule validation:** 190 expert labels (15 papers, 3 reviewers) + 48,306 AI silver labels; 10 rule-improvement rounds
- ✓ **Dashboards:** validation, remaining papers, 22 per-topic expert-review pages
- ✓ **AI chatbot:** Shark Oracle: hybrid search, cited answers from 19,592 papers; first evaluation harness (85 questions)
- ✓ **Conference abstracts:** 23,491 from 124 meetings, 1985-2026 (EEA 2026 ingested 7 Oct)
- **More downloading:** 21,895 PDFs on disk (+72% since EEA25, +18% SI, +6% AES); ~10,300 left
- **Sharkipedia & IUCN Red List:** in discussion
- **Gap analysis:** timelines, maps; researcher network atlas live (5,278 authors, 2,014 institutions)



**SHARKIPEDIA**  
elasmobranch traits & trends

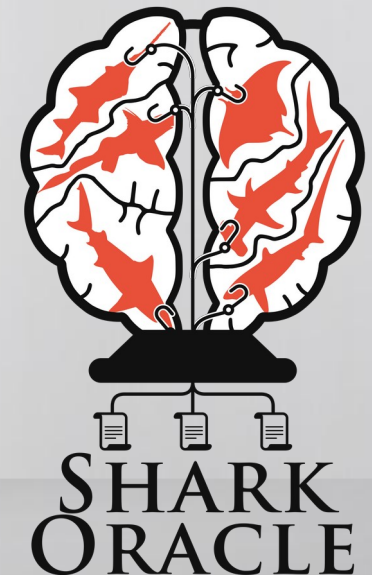
THE IUCN RED LIST  
OF THREATENED SPECIES™

## HELP US! (to help you)

Submit your papers to [SharkReferences.com](https://sharkreferences.com)

Validate your papers on our GitHub [https://simondedman.github.io/elasmo\\_analyses/validate/](https://simondedman.github.io/elasmo_analyses/validate/)

Talk to us about lit reviews



# Planned studies & project leads

Simon Dedman

**1. Meta-paper:** the full systematic review & pipeline

Guuske Tiktak

**2. Geographic bias:** research deserts, global north & south

*existing analyses: first-author institution maps (Global North 74%), ocean-basin effort, collaboration network*

Elena Fernández Corredor,  
Carylanne Maier, Tobin  
Dawne Smith

**3. Gender, Diversity, Disability:** representation across disciplines & regions

*Carylanne's hand-coded AES abstracts 1992-2024 (gender, conservation, taxa) as ground truth; NamSor + OpenAlex for the corpus*

David Shiffman

**4. Social attention:** what drives public engagement?

*Shiffman's 22-category coding of 2,701 AES abstracts; Altmetric scores across the corpus*

David Ruiz García

**5. Fishing pressure:** overlap → mortality → mitigation framework

Chris Mull

**6. Species/Family/Genus, & Habitat:** what are we studying, where?

*feeds Sharkipedia and IUCN Red List assessments: species, family, genus and habitat coverage gaps*

Alex McInturf

**7. Social aggregations:** sharks together

Carylanne Maier, AES,  
EEA, OCS

**8. Conference abstracts:** 23,408 talks & posters, 1985-2026

Kate Wing

**9. Data sources:** which databases hold what, and how they connect

*connects to MarSpatAuto: automated marine spatial analysis fed by the same database*

Emily Warren, Others

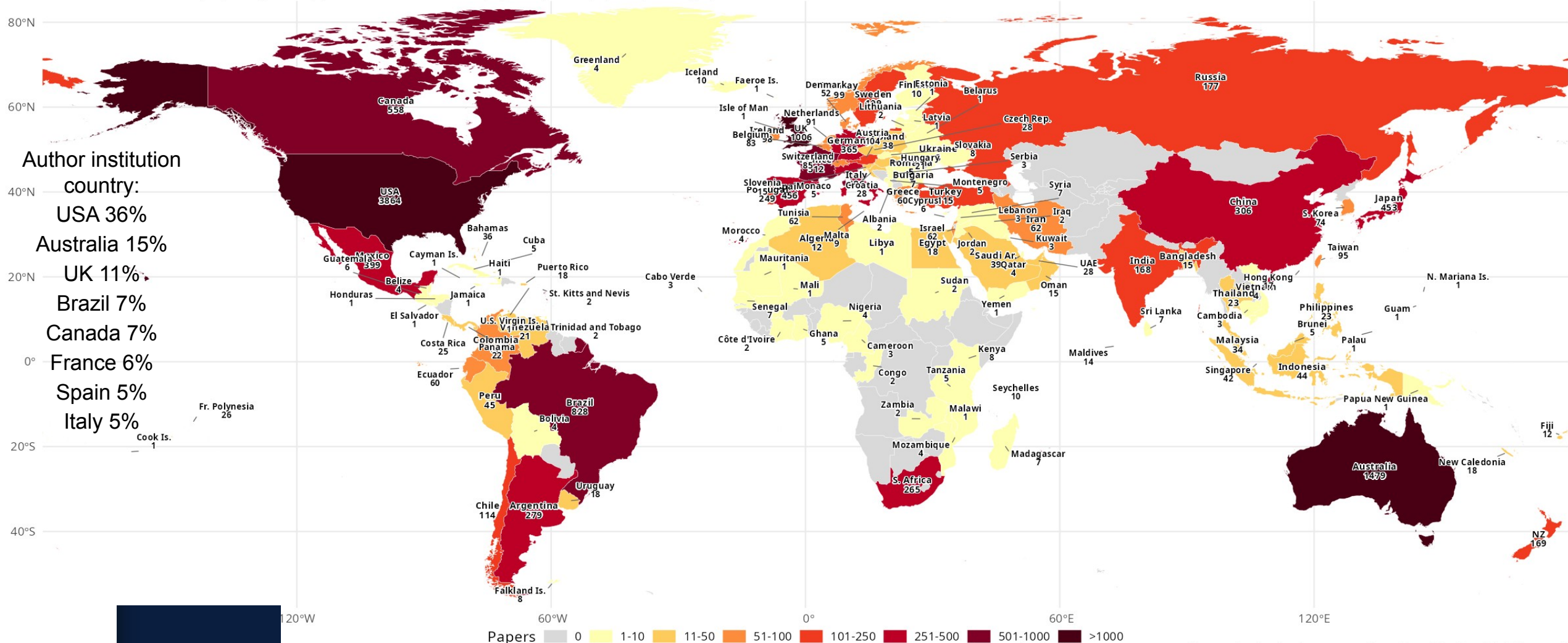
**10. Others:** ecological roles, swim speed, journal quality, techniques



# Global North dominates chondrichthyan research

## Global Distribution of Shark Research Papers

Global North: 10738 papers (74.4%) | Global South: 3696 papers (25.6%) | Grey = no papers

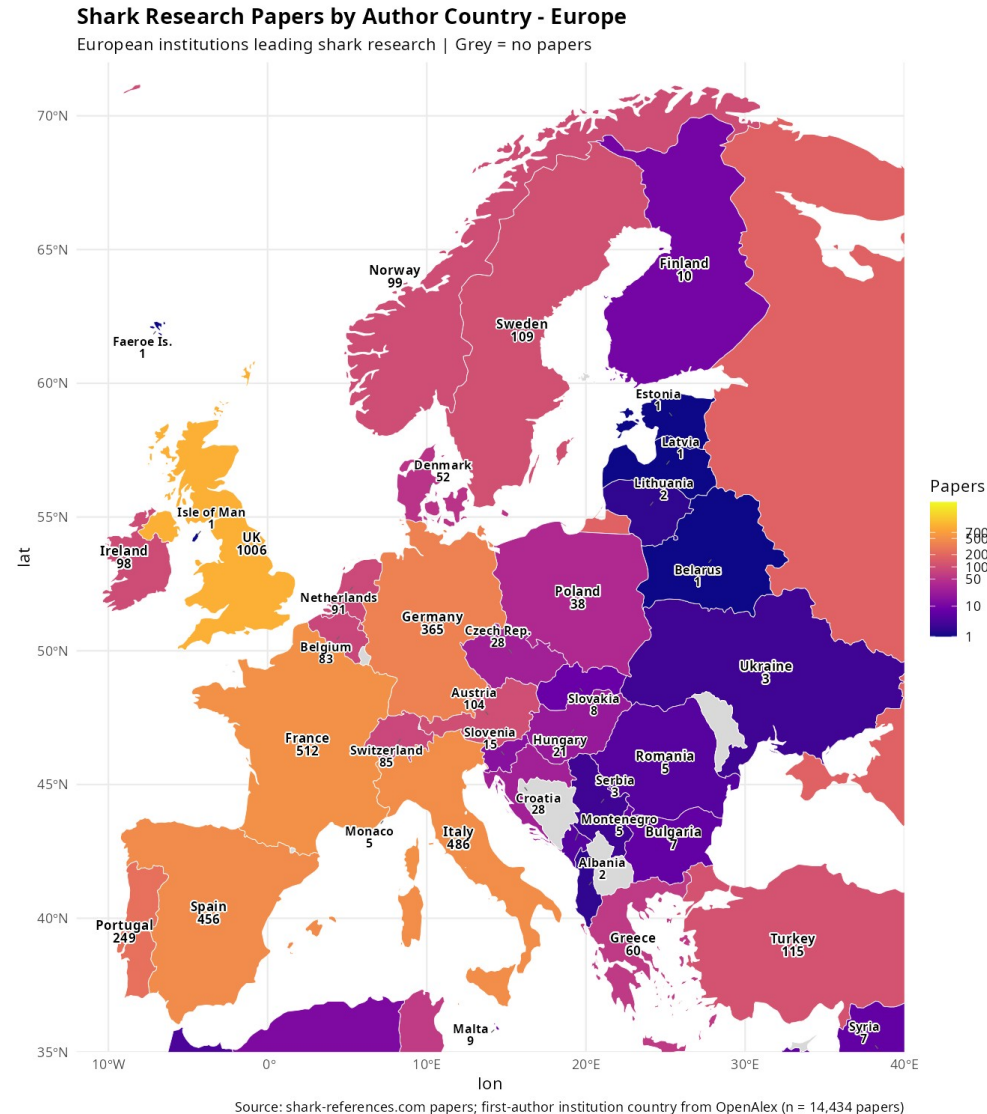




# Within Europe: the UK, France, Spain, & Italy lead

European author  
institution country:

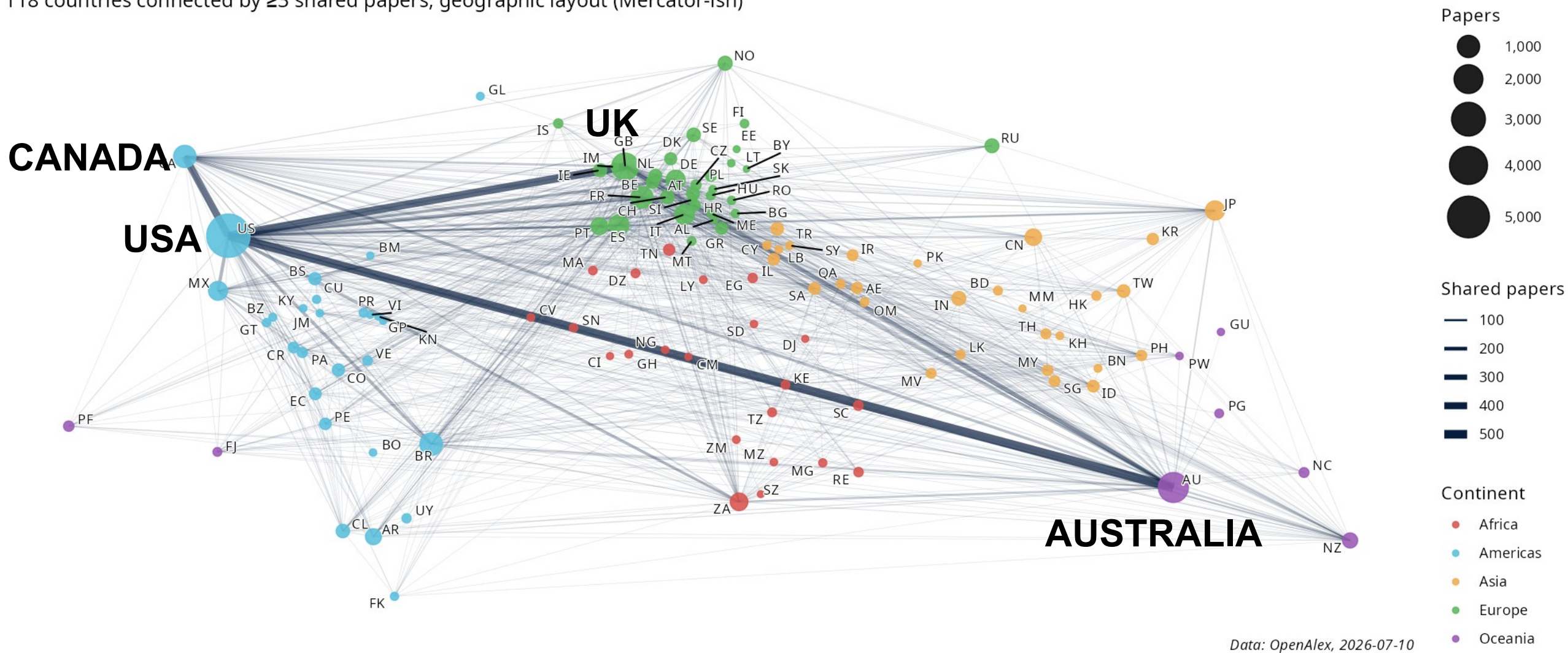
UK 29%  
France 17%  
Spain 13%  
Italy 13%  
Germany 11%  
Portugal 8%  
Norway 4%  
Russia 4%





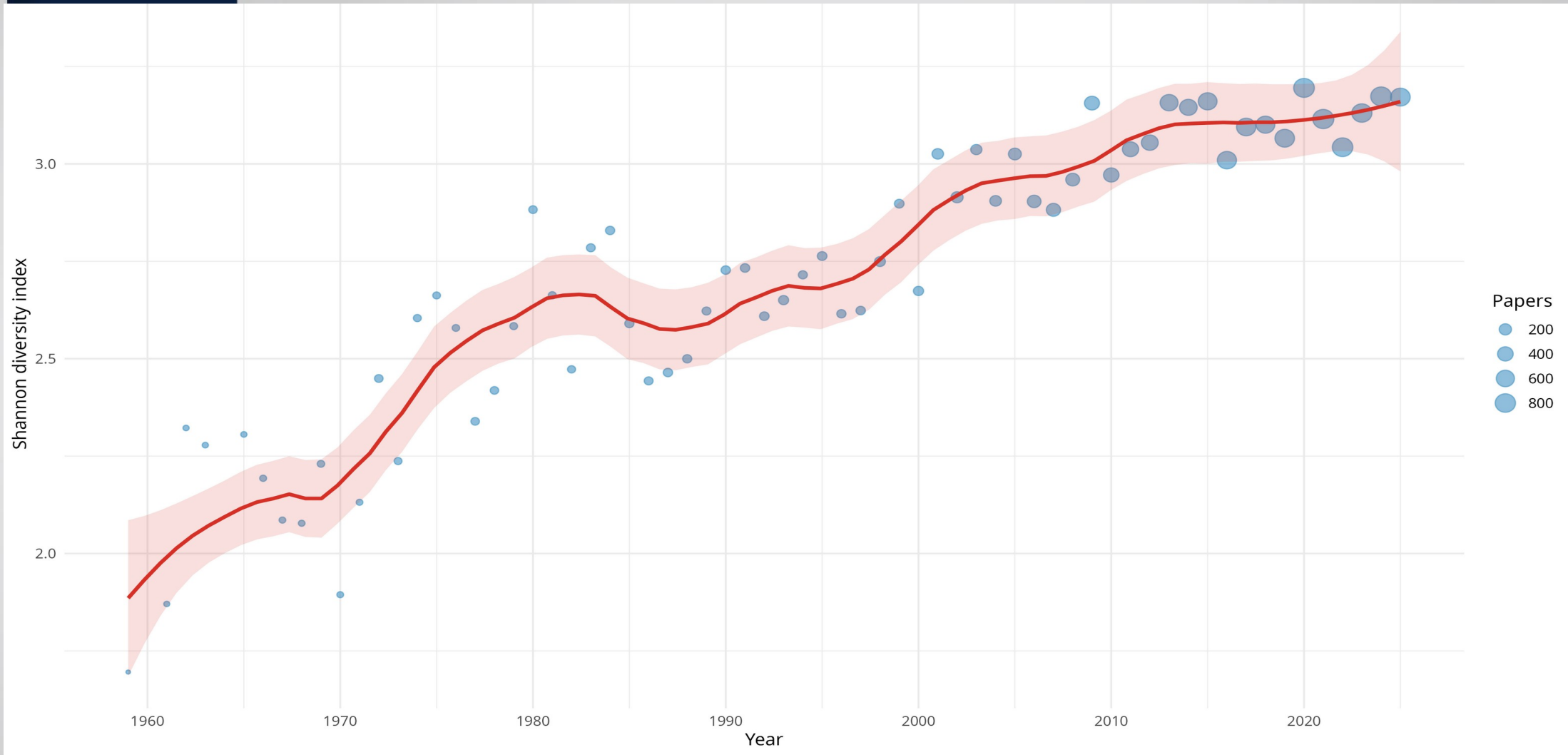
# Key collaborations between Global North

118 countries connected by  $\geq 3$  shared papers; geographic layout (Mercator-ish)





# Ethnic diversity of 1st authors is increasing!



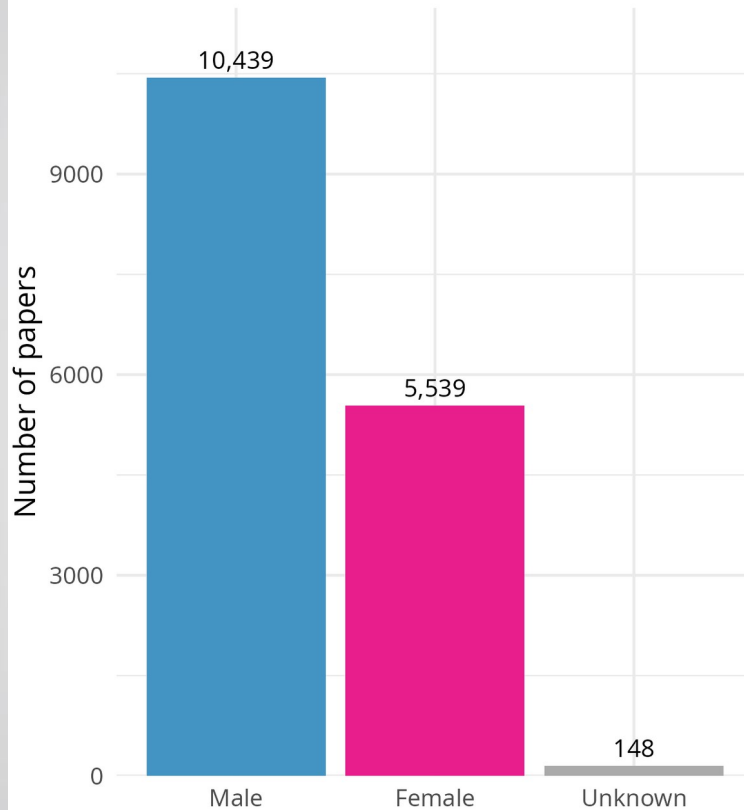


# Gender gap is closing!

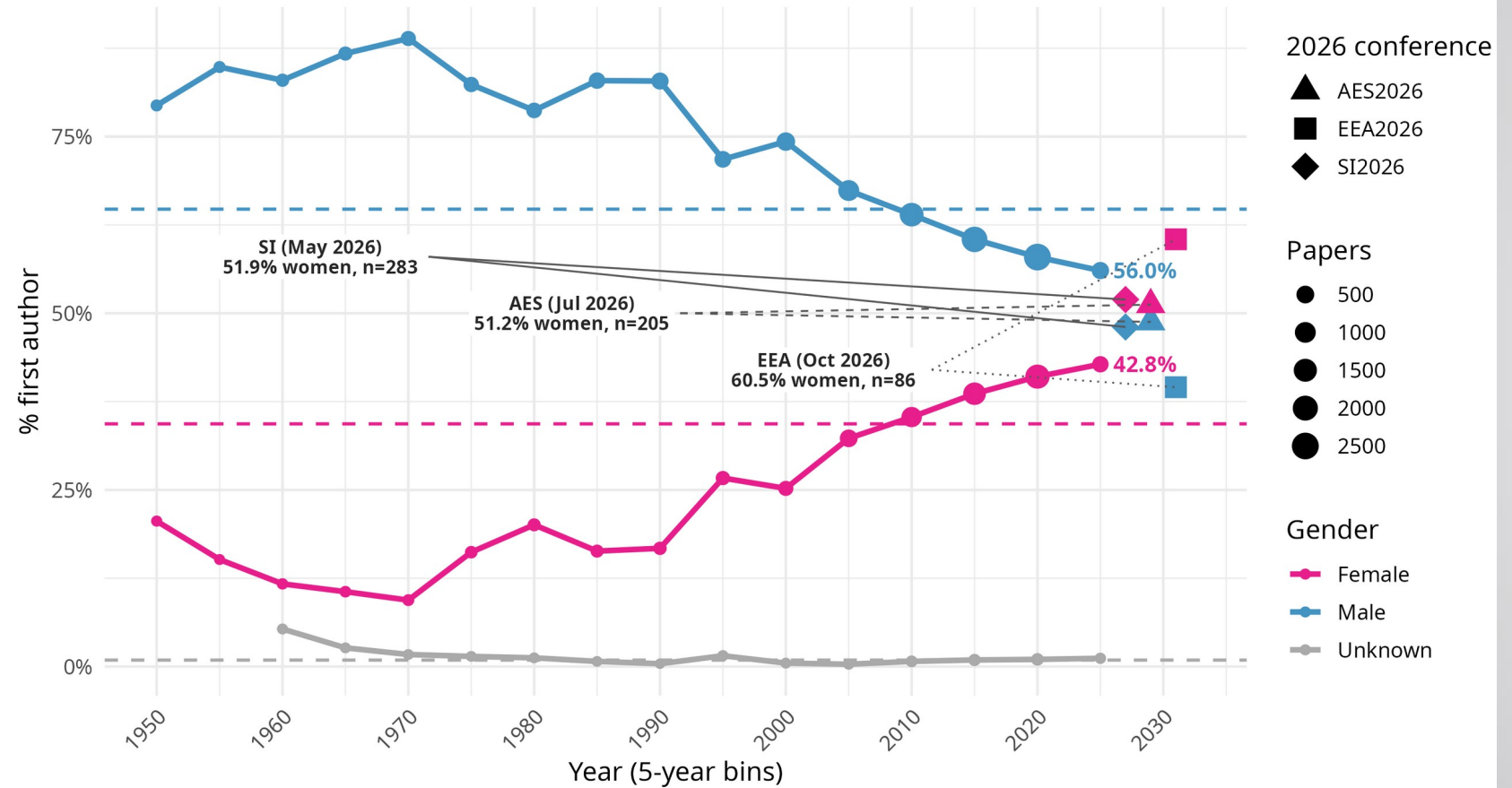
First author gender in elasmobranch research

Left: total counts | Right: % by 5-year period (NamSor-enriched) + 2026 conference presenters

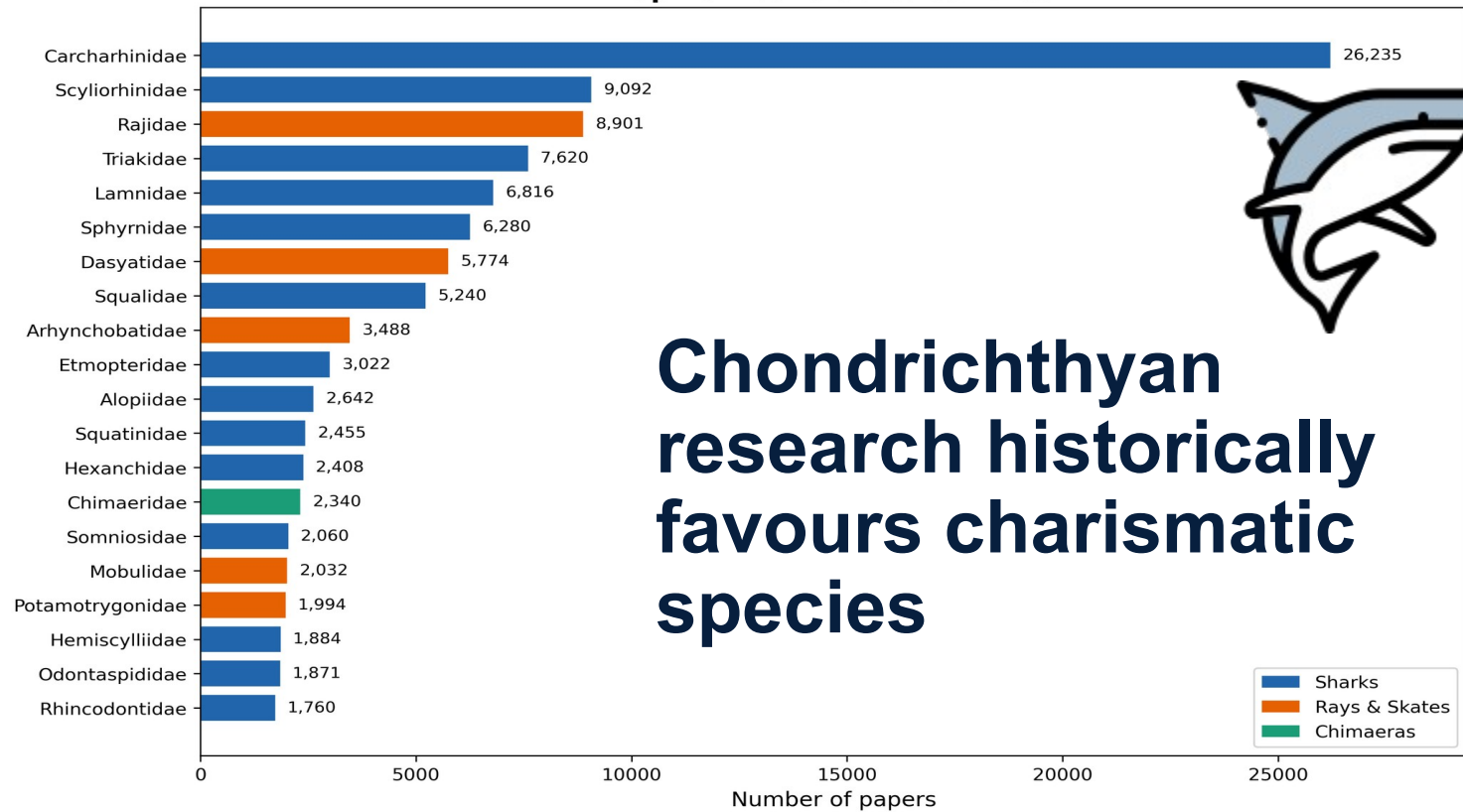
## Gender counts



## Trend over time

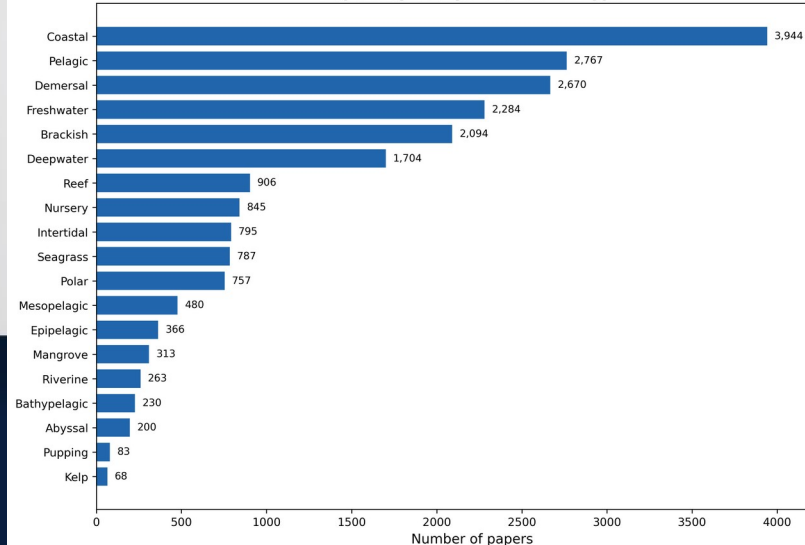


Top 20 most-studied families



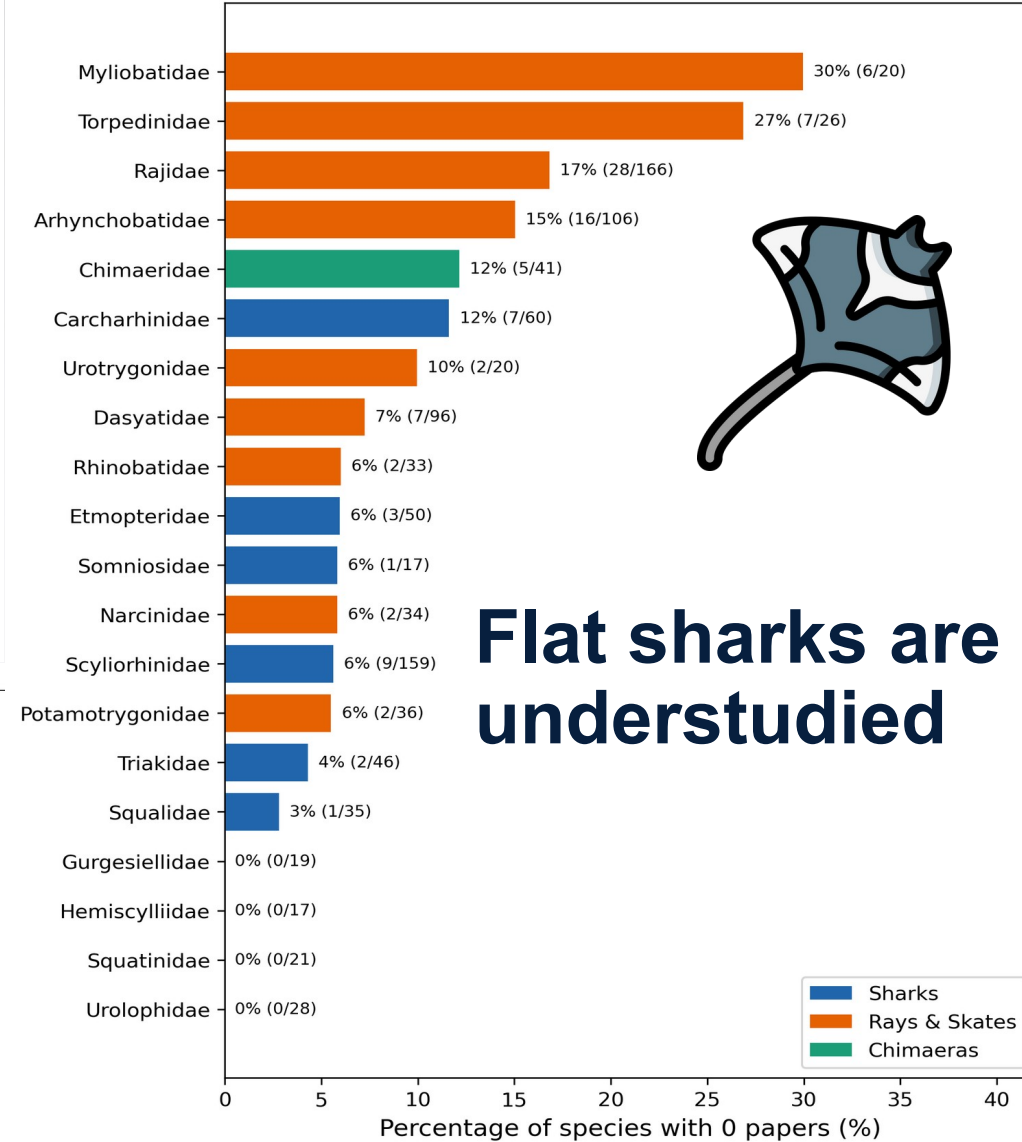
Chondrichthyan research historically favours charismatic species

Papers by ecosystem/habitat type



# Species & Habitat Bias

Research gaps: unstudied species in the 20 largest families



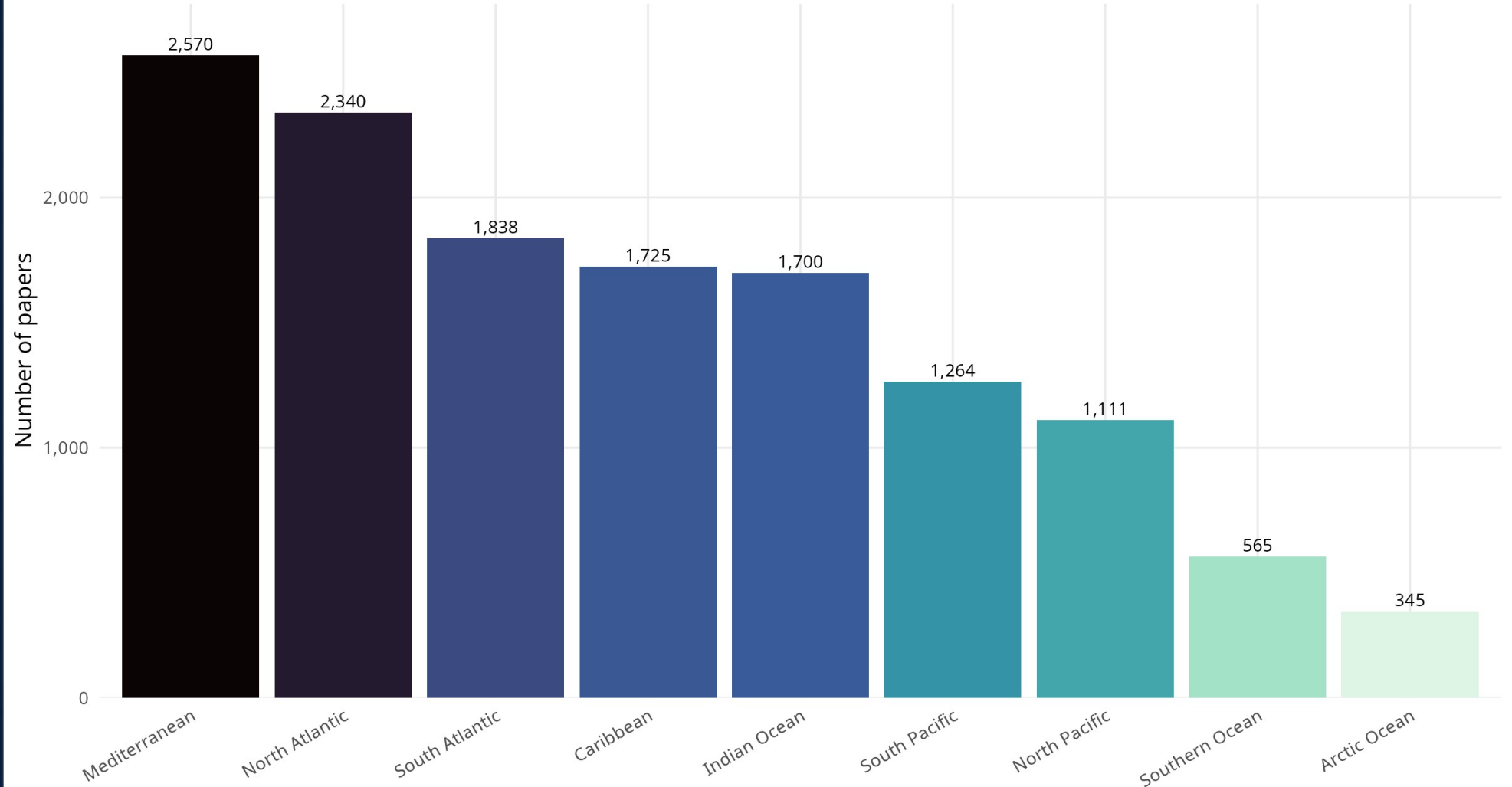
Flat sharks are understudied



# More research effort in the North Atlantic & Med

## Elasmobranch research by ocean basin

Papers classified per basin (multi-label, papers with PDFs)

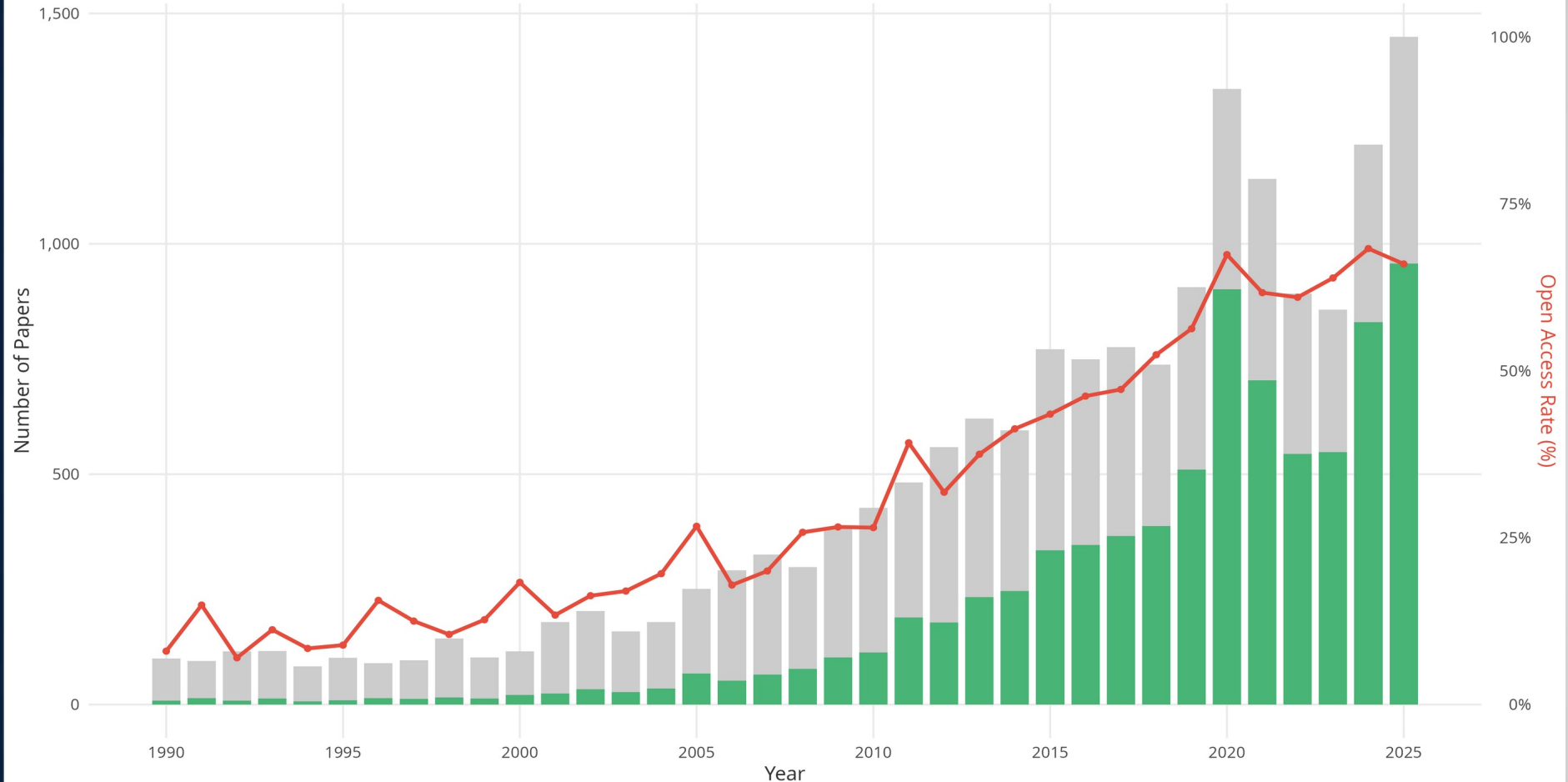




# Open Access has increased: win for data transparency and global reach... But!

## Open Access in Elasmobranch Research

Grey = total papers; green = open access; red line = OA percentage



## The problem

Field labs now build their own detectors. 26 of the 53 shark and marine-megaafauna computer-vision projects inventoried in October 2026 have built a detector and 14 a tracker, almost all on the same Ultralytics YOLO stack, each trained on one region and failing in the next. Only 8 publish data; 8 have automated individual ID, while the long-running photo-ID catalogues presented at this meeting are matched by hand. The largest video archives hold no models at all. Nobody connects the layers.

## What we propose

One open platform, six modules with stable interfaces: capture and ingest; detection and tracking; species and attribute classification; individual ID, morphometrics and behaviour; human validation and active learning; open data, models and attribution. Each module is owned by a group that already builds it. Data stay with their owners under licences they choose; code is MIT or Apache; credit is automatic.

## Next steps

► **Gather the team.** A standing consortium of module owners and data holders, now, while the inventory is fresh. A monthly call and a shared GitHub organisation.

► **Release the benchmark.** A licensed shark and ray BRUV dataset with held-out test regions, hosted free on AWS Open Data: the precondition for everything else.

► **Write the grant.** NSF 26-512 AI Datasets (software line) is due 4 November 2026; one joint application per quarter after that.

► **Propose the hackathons.** A three-day build sprint with a corporate host in February or March 2027, ahead of NVIDIA GTC: parallel modules, an integration contract, one merge, no prize. Then repeat with another host.

► **Open the validation layer.** The sprint's review-and-active-learning tooling behind a public Zooniverse front end.

### Join

Bring a module you build, a catalogue or video archive you hold, or a region we lack. No member is asked for raw video to join.

**Project page:** [github.com/SimonDedman/ocean-ID](https://github.com/SimonDedman/ocean-ID)

**Contact:** Simon Dedman, [simondedman@gmail.com](mailto:simondedman@gmail.com)

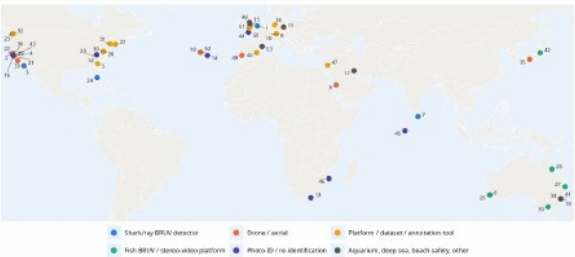


## Who builds which layer: 53 shark and marine-megaafauna computer-vision efforts

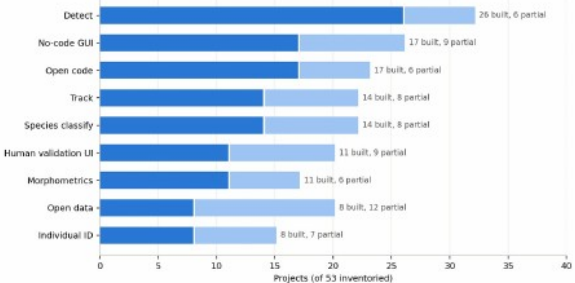
|                                                 | Detect | Track | Species classify | Individual ID | Morpho-metrics | Human validation UI | Open code | Open data | No-code GUI | Layers covered |
|-------------------------------------------------|--------|-------|------------------|---------------|----------------|---------------------|-----------|-----------|-------------|----------------|
| SharkTrack fork (Dedman)                        |        |       |                  |               |                |                     |           |           |             | 6.5            |
| SharkTrack (Varini)                             |        |       |                  |               |                |                     |           |           |             | 5              |
| AFID stereo sharks (Barker/Marrable)            |        |       |                  |               |                |                     |           |           |             | 4              |
| CICESE rays (López-Posada)                      |        |       |                  |               |                |                     |           |           |             | 3.5            |
| Global FinPrint (Chapman, Heithaus)             |        |       |                  |               |                |                     |           |           |             | 3              |
| Sri Lanka CNN (Jayasinghe)                      |        |       |                  |               |                |                     |           |           |             | 2              |
| FishID (Connolly, Griffith)                     |        |       |                  |               |                |                     |           |           |             | 5.5            |
| Tohoku stereo fish (Ishikawa)                   |        |       |                  |               |                |                     |           |           |             | 4.5            |
| AIMS/CSIRO OzFish, COTS (Cure, Harvey)          |        |       |                  |               |                |                     |           |           |             | 3.5            |
| EventMeasure (SeaGIS)                           |        |       |                  |               |                |                     |           |           |             | 3              |
| UWA pelagic BRUVs (Meeuwig)                     |        |       |                  |               |                |                     |           |           |             | 1.5            |
| MPI-AB Maldives drone (Albi, Couzin)            |        |       |                  |               |                |                     |           |           |             | 3.5            |
| Drone U-Net (DiGiacomo)                         |        |       |                  |               |                |                     |           |           |             | 3.5            |
| FLAIR zero-shot drone (Laligudi, Espinoza)      |        |       |                  |               |                |                     |           |           |             | 3.5            |
| SharkSpotter (Blumenstein, UTS)                 |        |       |                  |               |                |                     |           |           |             | 3              |
| Turtle drone (Noguchi)                          |        |       |                  |               |                |                     |           |           |             | 3              |
| Hammerhead drone YOLO/TRex (Mansson)            |        |       |                  |               |                |                     |           |           |             | 2.5            |
| KAUST drone rays (Nieuwenhuis)                  |        |       |                  |               |                |                     |           |           |             | 2              |
| SharkEye (McCauley, UCSB)                       |        |       |                  |               |                |                     |           |           |             | 2              |
| Whale shark drone morpho (Zimmermann)           |        |       |                  |               |                |                     |           |           |             | 1              |
| CSUMB pose-ID (Sambold)                         |        |       |                  |               |                |                     |           |           |             | 5.5            |
| Stanford MBWSP (Andrzejczek, Block)             |        |       |                  |               |                |                     |           |           |             | 5              |
| Wildbook / Sharkbook (Holmberg)                 |        |       |                  |               |                |                     |           |           |             | 5              |
| sharkPulse Shark Detector (Jenrette, Ferretti)  |        |       |                  |               |                |                     |           |           |             | 4.5            |
| Social-media CV (Jenrette)                      |        |       |                  |               |                |                     |           |           |             | 3.5            |
| IDtheManta / Manta Catalog (Sobral)             |        |       |                  |               |                |                     |           |           |             | 3              |
| Sevengill photo-ID (OSEA)                       |        |       |                  |               |                |                     |           |           |             | 2              |
| Eagle ray photo-ID comparison (Evans, Esteban)  |        |       |                  |               |                |                     |           |           |             | 1.5            |
| White shark re-ID (Kummann)                     |        |       |                  |               |                |                     |           |           |             | 1.5            |
| Raydar (Morrison, Elston)                       |        |       |                  |               |                |                     |           |           |             | 1              |
| SixgillSharks.org (McGouldrick)                 |        |       |                  |               |                |                     |           |           |             | 1              |
| DRDH porbeagle photo-ID (Zaccagnini)            |        |       |                  |               |                |                     |           |           |             | 0.5            |
| Fuvahmulah tiger photo-ID (Asshauer)            |        |       |                  |               |                |                     |           |           |             | 0.5            |
| Leopard shark photo-ID (Buschmann)              |        |       |                  |               |                |                     |           |           |             | 0.5            |
| Porbeagle lamprey marks via photo-ID (Muller)   |        |       |                  |               |                |                     |           |           |             | 0.5            |
| VIAME (Kitware, NOAA)                           |        |       |                  |               |                |                     |           |           |             | 7.9            |
| ReefVision (Del Valle)                          |        |       |                  |               |                |                     |           |           |             | 5.5            |
| TRex (Walter, Couzin)                           |        |       |                  |               |                |                     |           |           |             | 5.5            |
| FathomNet (Kattija, MBARI)                      |        |       |                  |               |                |                     |           |           |             | 5.5            |
| Fishial.AI (Wye Foundation)                     |        |       |                  |               |                |                     |           |           |             | 5              |
| MegaDetector family (Morris)                    |        |       |                  |               |                |                     |           |           |             | 5              |
| CVision / Ocean Vision AI (MBARI)               |        |       |                  |               |                |                     |           |           |             | 5              |
| BIIGLE (Bielefeld)                              |        |       |                  |               |                |                     |           |           |             | 3.5            |
| Open fish datasets (JCU, KAUST, Alicante)       |        |       |                  |               |                |                     |           |           |             | 2.5            |
| Great Eggcase Hunt verification (Allen, Gordon) |        |       |                  |               |                |                     |           |           |             | 2.5            |
| EcoMorphLab (Gal Or, Levy)                      |        |       |                  |               |                |                     |           |           |             | 2              |
| Beach safety (NSW Government)                   |        |       |                  |               |                |                     |           |           |             | 3              |
| NICMA (Asadi Gharabaghi)                        |        |       |                  |               |                |                     |           |           |             | 2              |
| Kuwait aquarium AI (Alzaidan)                   |        |       |                  |               |                |                     |           |           |             | 2              |
| SE Australia provisioning AI (Pini-Fitzsimmons) |        |       |                  |               |                |                     |           |           |             | 2              |
| Thornback pattern ML (Egloff)                   |        |       |                  |               |                |                     |           |           |             | 1              |
| Blue shark photogrammetry (Lopez Valentin)      |        |       |                  |               |                |                     |           |           |             | 0.5            |
| Azores deep BRUV (Catarino, Fauconnet)          |        |       |                  |               |                |                     |           |           |             | 0              |

Cell: dark = built, light = partial or planned, blank = not in scope. Right column: layers covered (sum of 1 and 0.5 scores), red to green from minimum through the median to maximum. Scores are the compiler's reading of public evidence, 1 and 6 Oct 2026 (rows 44-53 from EEA 2026 abstracts); corrections welcome.

## Where the projects are



## Layers rebuilt in parallel



## Already in the room at EEA 2026

**Photo-ID catalogues matched by hand:** Des Requins et Des Humains porbeagles, Brittany (Zaccagnini, Jung; and Muller's lamprey-mark re-use of the same catalogue); Fuvahmulah tiger sharks (Asshauer and colleagues); Tofo leopard sharks (Buschmann).

**Morphometrics and video:** EcoMorphLab (Gal Or, Levy and colleagues, Shenkar and Tel Aviv University); thornback skate pattern analysis (Egloff, Bangor); whale shark drone photogrammetry (Zimmermann, Dehne, Meekan); NICMA video energetics (Asadi Gharabaghi, Tubner).

**Community verification at scale:** the Great Eggcase Hunt's new Community Verification Platform (Allen, Gordon, Shark Trust).

**Not yet on the list?** If you record, annotate, match, or model shark and ray imagery, tell us: the inventory is public and every row is editable.

Inventory of 53 projects compiled 1 and 6 October 2026 from public sources and the EEA 2026 abstract booklet; scores are the compiler's reading of the evidence. Full inventory, sources, and one-page proposals on the project page.



# Thank You



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Institut de Ciències del Mar (CSIC)

## David Ruiz García

MARBEC (IFREMER)

## To get involved, contact us, & see project here:

[https://github.com/SimonDedman/elasmo\\_analyses](https://github.com/SimonDedman/elasmo_analyses)

Slides: [https://github.com/SimonDedman/elasmo\\_analyses](https://github.com/SimonDedman/elasmo_analyses)

Map: [https://simondedman.github.io/elasmo\\_analyses/network\\_atlas/](https://simondedman.github.io/elasmo_analyses/network_atlas/)

*Collaborators: Alex McInturf, David Shiffman, Chris Mull, Ulrich Bösing, Sophia Pelletier, Deven Guerrero, Ryan McMullen, Lola Riesgo, Emily Warren, Carylanne Maier, Tobi-Dawne Smith, Chiara Gambardella, Nathan Perisic, Andrew Temple, Kate Wing*

*Abstract books: Carylanne Maier, David Green, Brit Finucci, Cat Gordon, Ali Hood*

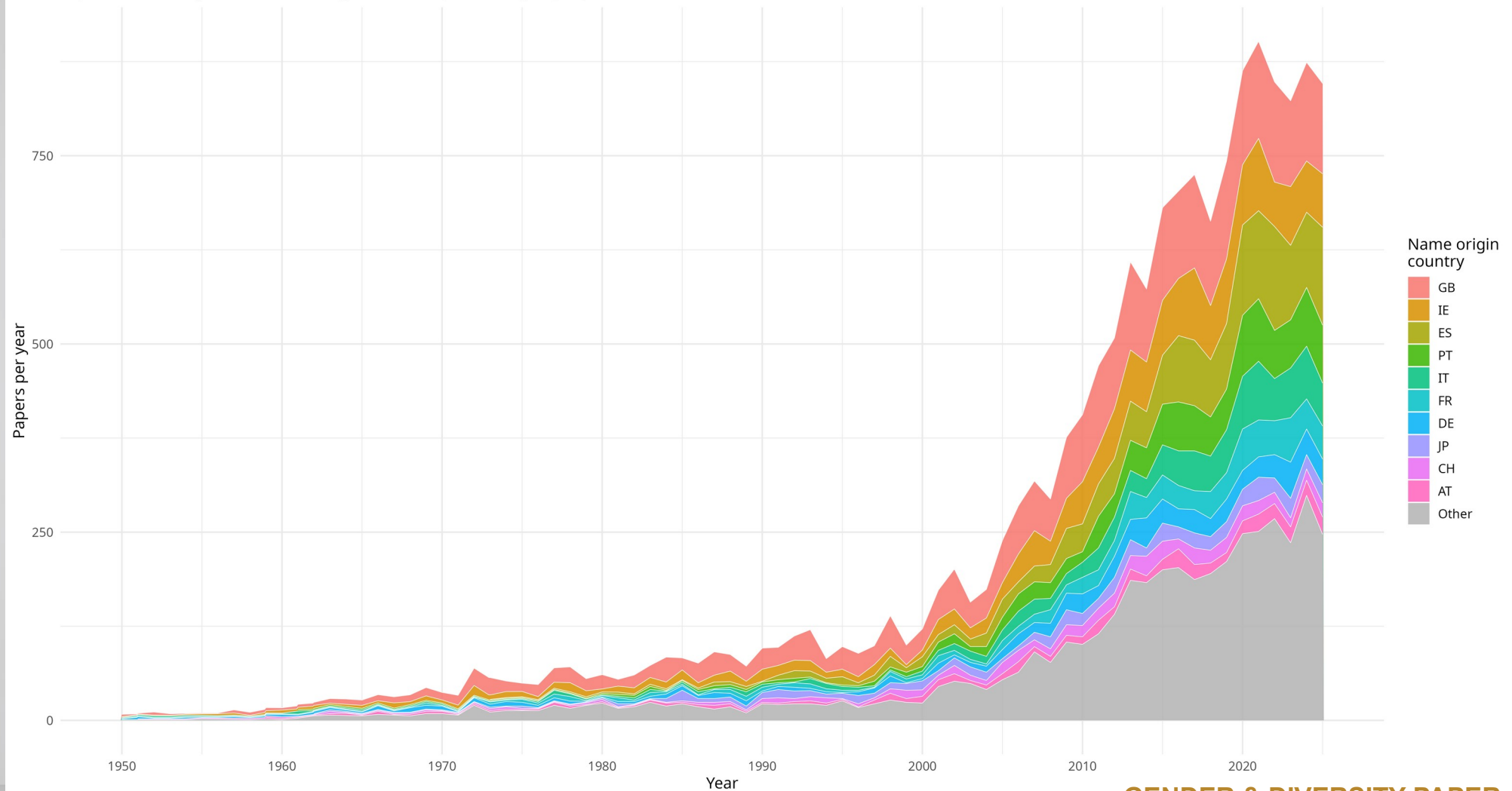
*SharkReferences: Jürgen Pollerspöck & Nicolas Straube*



# Stacked area of first-author name-origin country over time

First author name-origin country over time

Top 10 countries by NamSor name-origin inference (all others grouped)



FIU

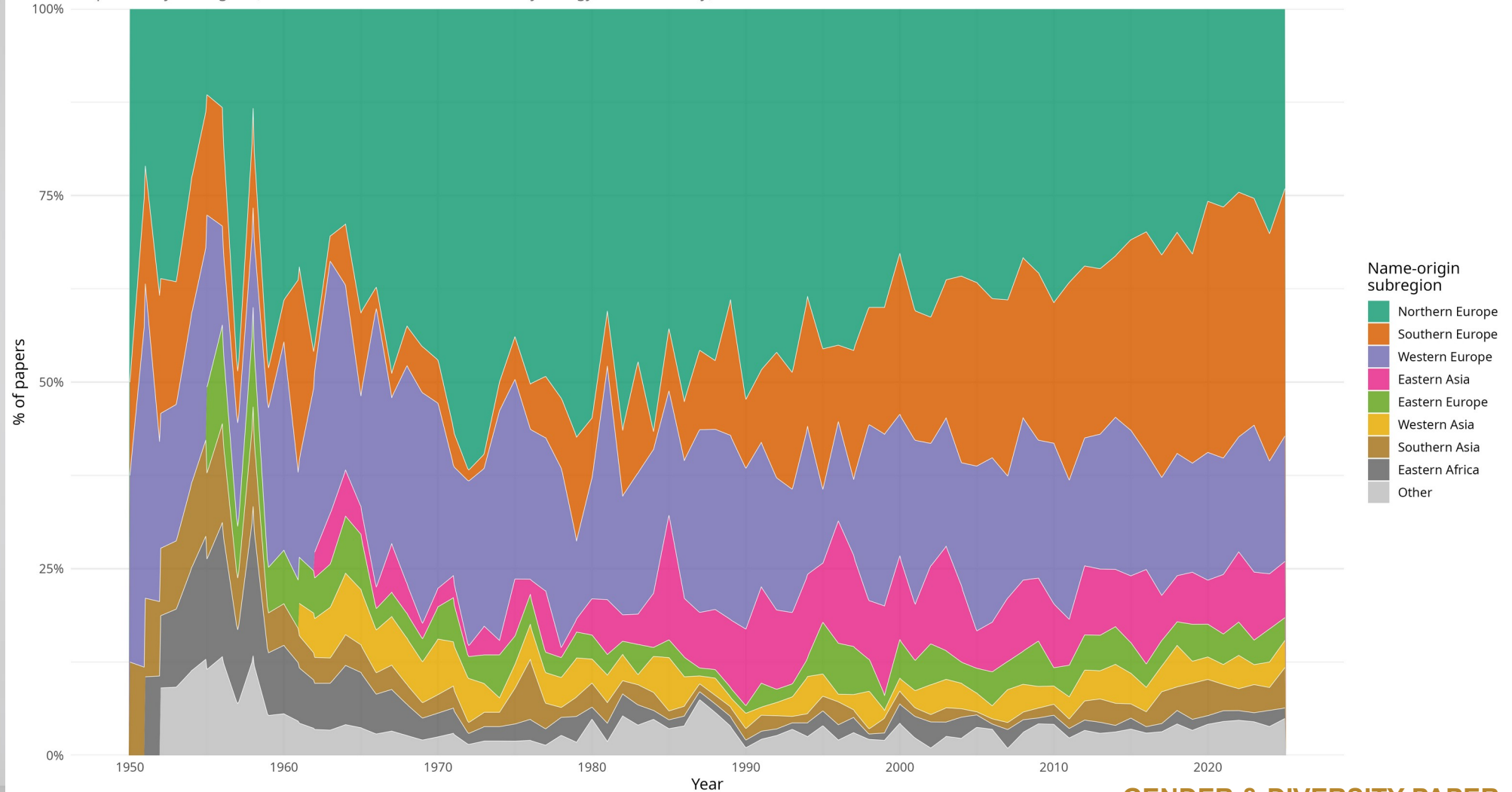
GENDER & DIVERSITY PAPER

REVIEW · N1\_origin\_country\_trends.png

# Proportional stacked area of first-author name-origin subregion over time

**First author name-origin subregion over time**

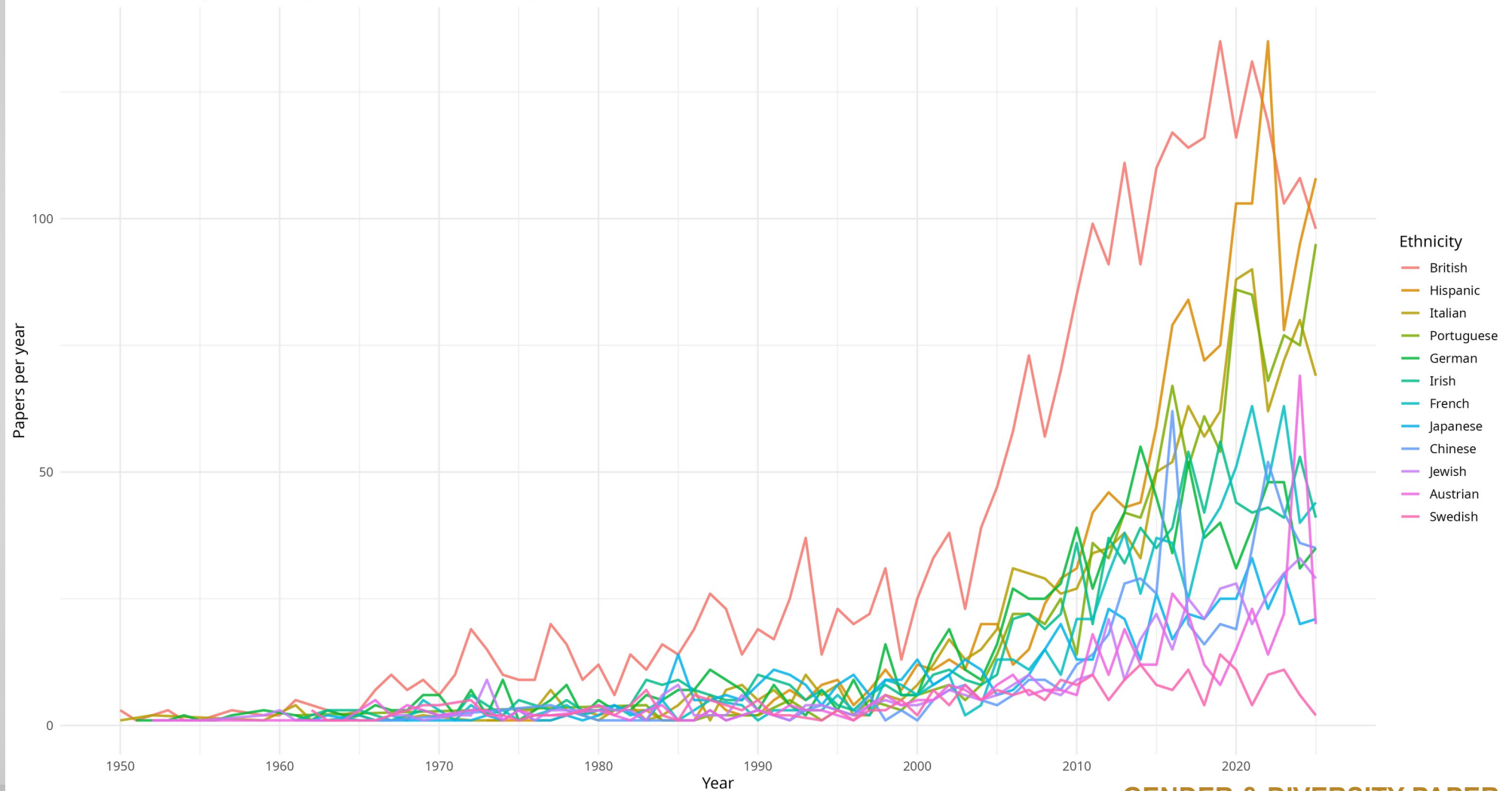
Proportion by subregion (NamSor inference -- reflects name etymology, not nationality)



# First-author ethnicity (NamSor diaspora) papers per year

## First author ethnicity over time

Top 12 ethnicities by NamSor diaspora inference (annual counts)



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REVIEW · N3\_ethnicity\_trends.png

FIU

# % of first authors whose name-origin country differs from institution

## Researcher mobility: name-origin vs institution country

% of first authors whose NamSor name-origin differs from their institution country (5-year bins)



GENDER & DIVERSITY PAPER

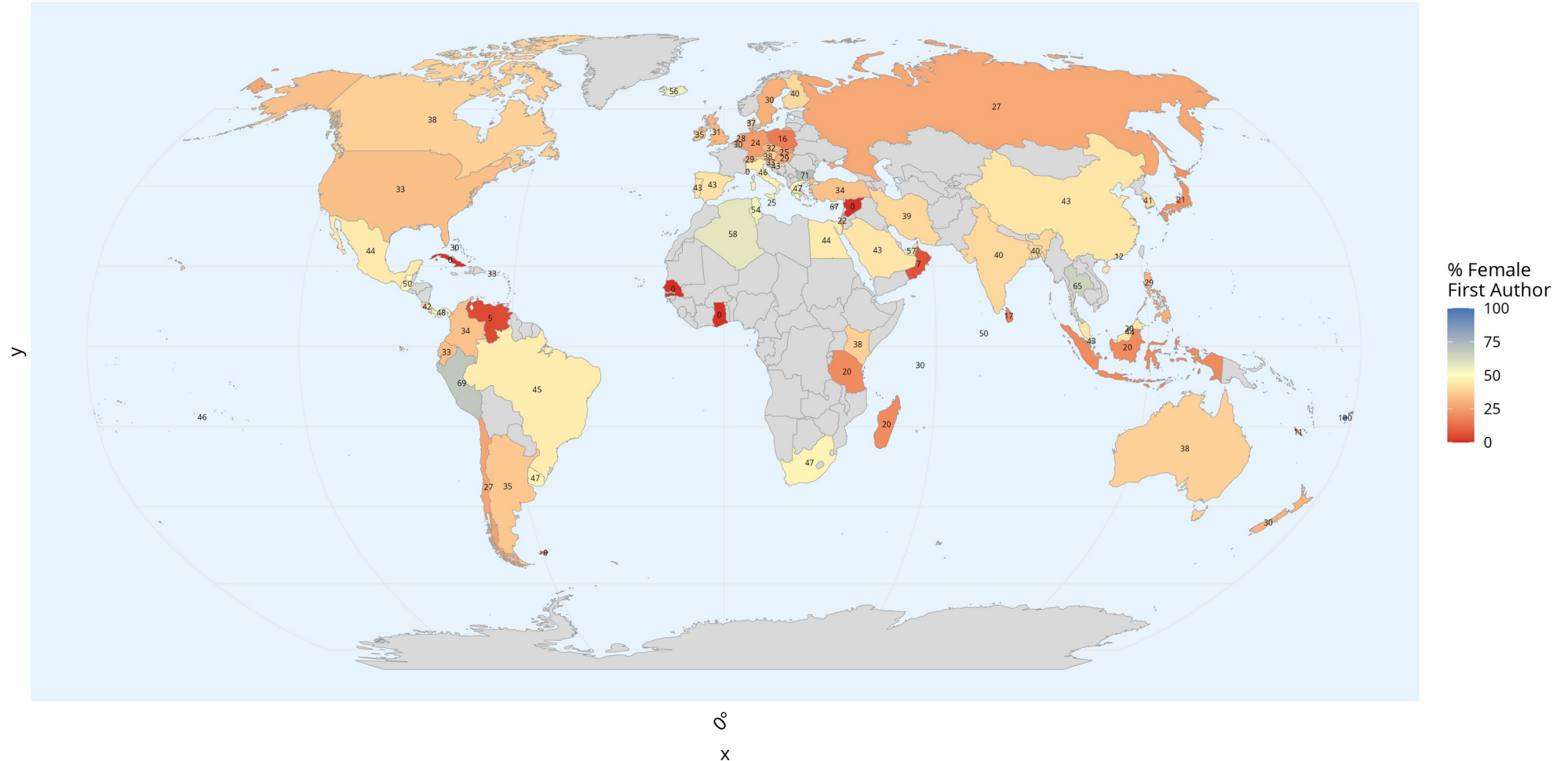
REVIEW · N5\_origin\_institution\_mismatch.png

**FIU**

# World choropleth of % female first authors by author country (NamSor-e)

## Female first authorship rate by country

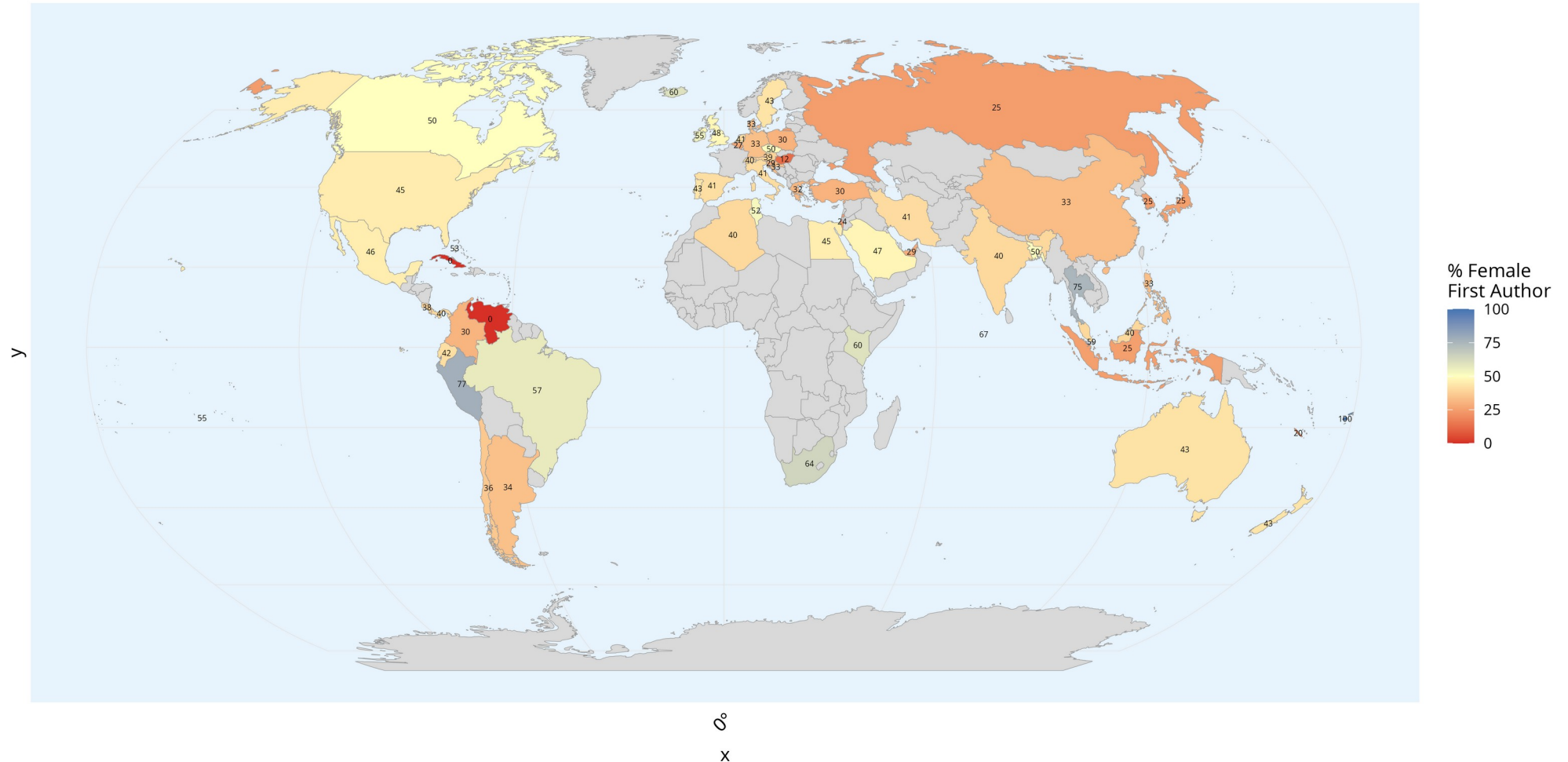
Countries with  $\geq 5$  papers shown; grey = no data (NamSor-enriched). Numbers = % female.



## Same gender choropleth restricted to 2020-2025

### Female first authorship rate by country (2020-2025)

Countries with  $\geq 5$  papers shown; grey = no data. Numbers = % female.

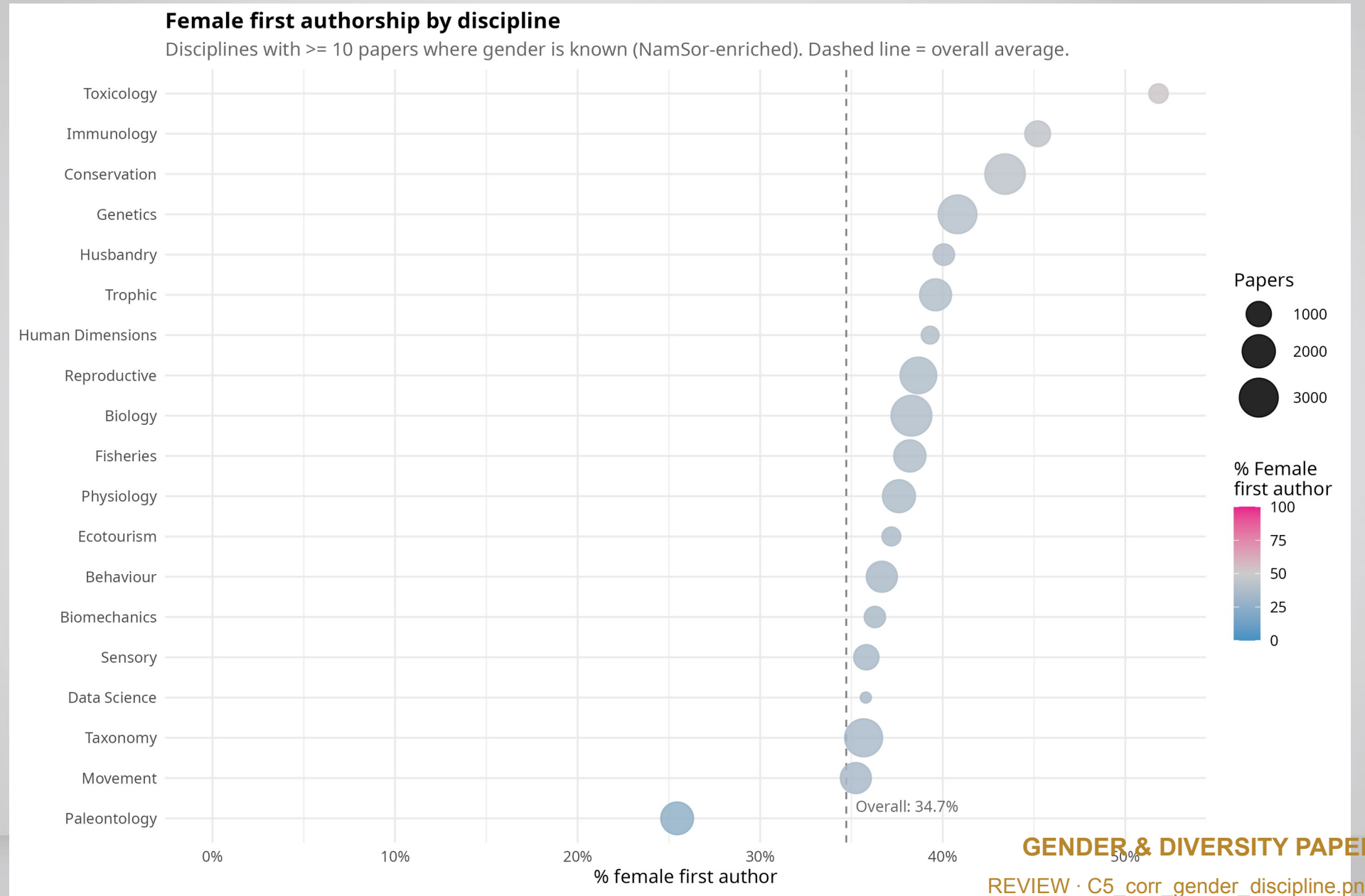


## GENDER & DIVERSITY PAPER

REVIEW · A5b\_spatial\_gender\_map\_recent.png

**FIU**

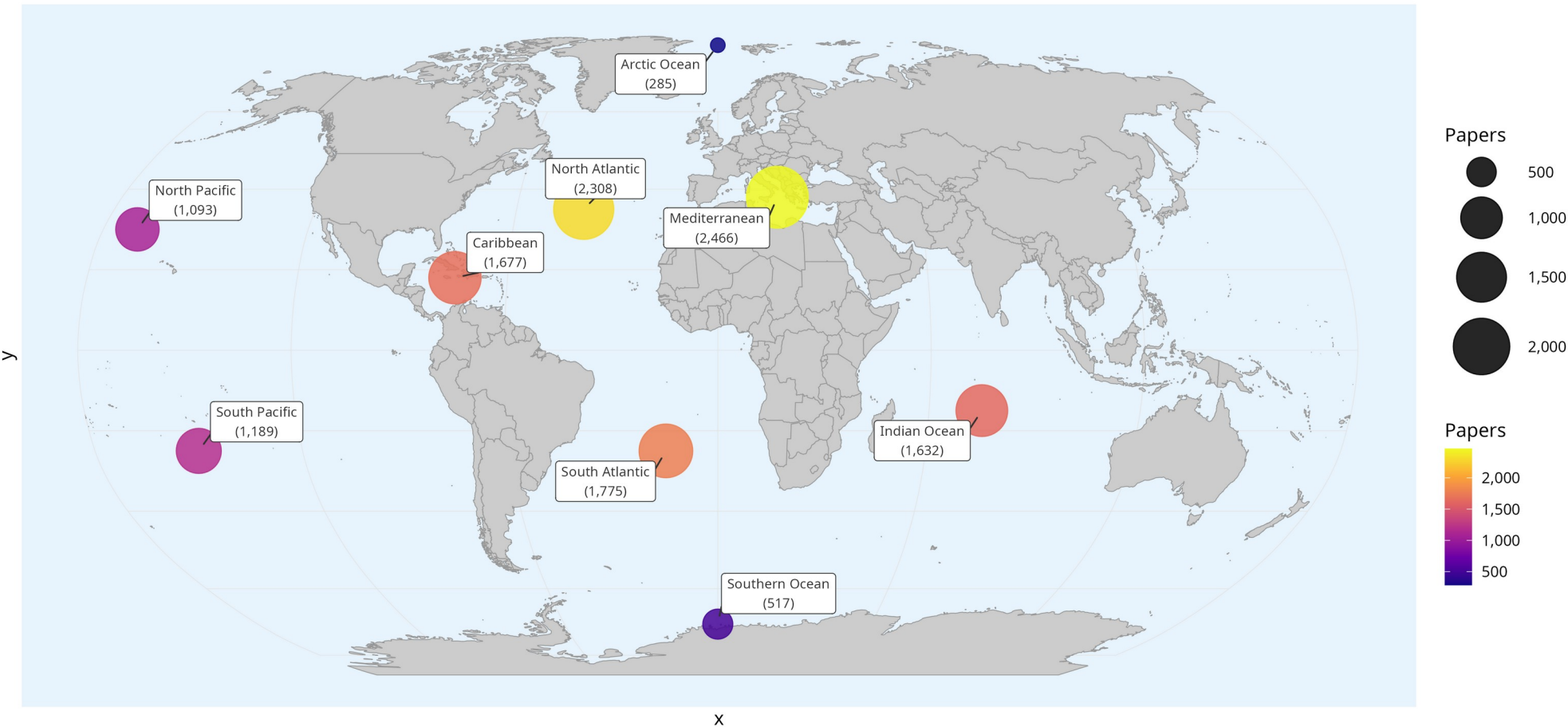
# % female first author by discipline



# Papers by ocean basin (bubbles on world map)

## Elasmobranch Research by Ocean Basin

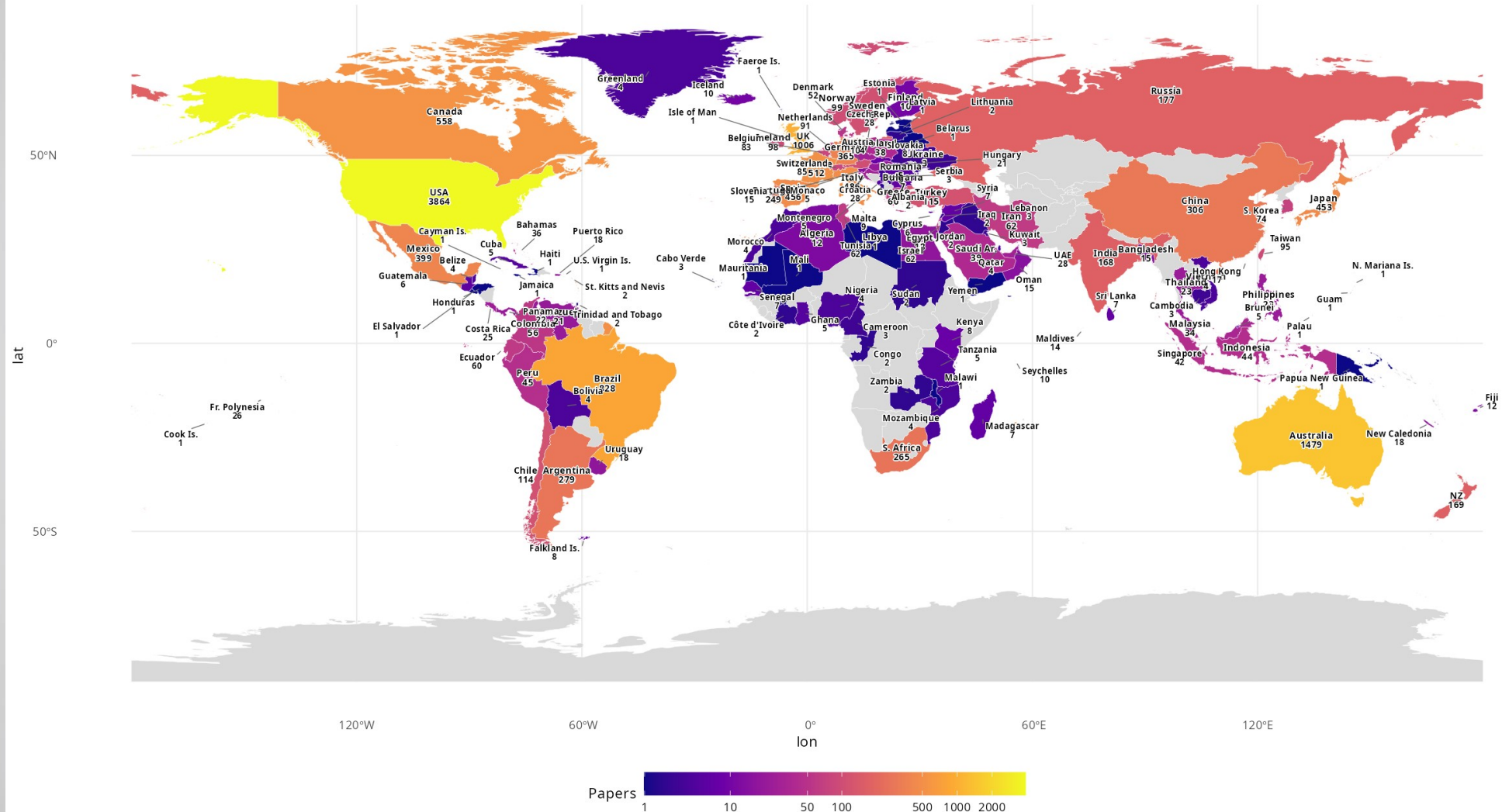
Circle size and colour show total papers per basin (1950-2025)



# Papers by author institution country (1950-2025)

## Shark Research Papers by Author Country (1950-2025)

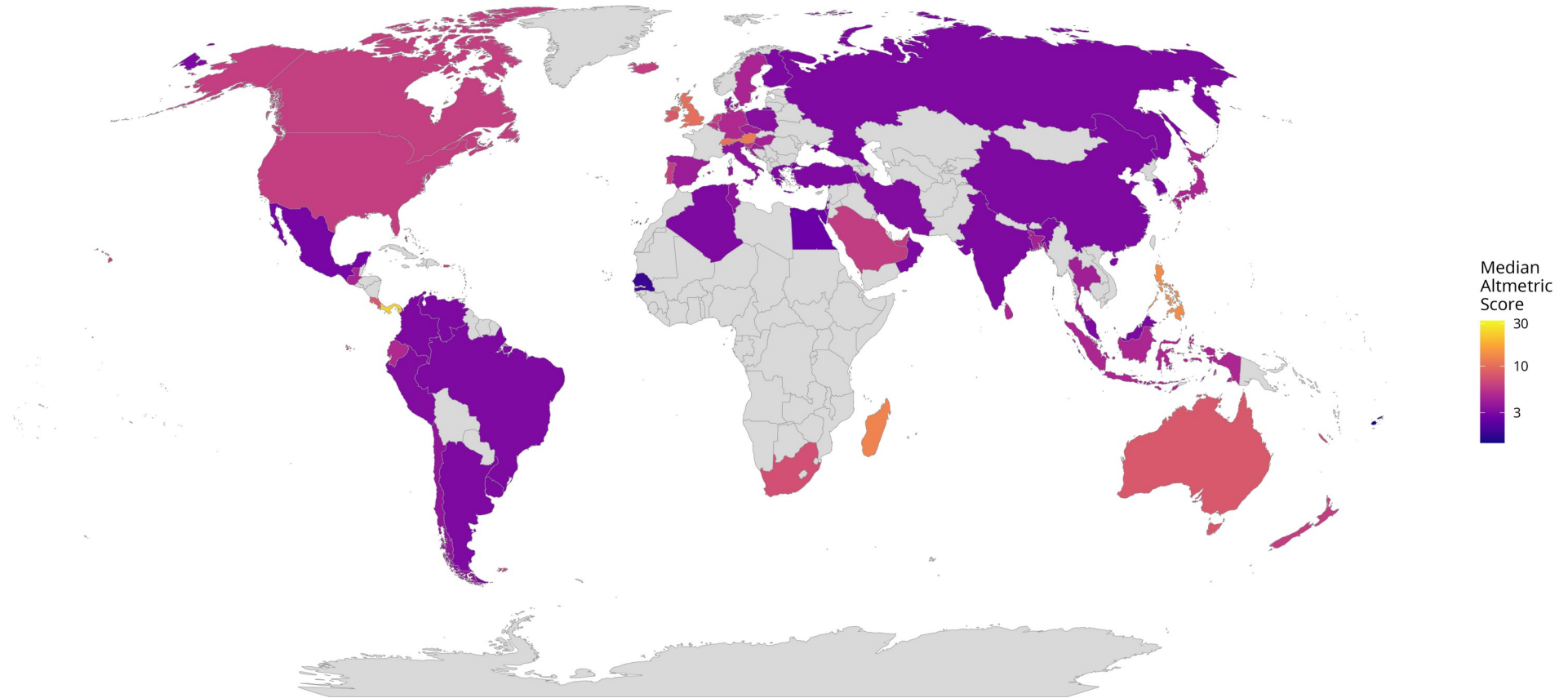
Total: 14,434 papers from 131 countries | Grey = no papers



# World map of median Altmetric score by author country

## Median Altmetric score by author country

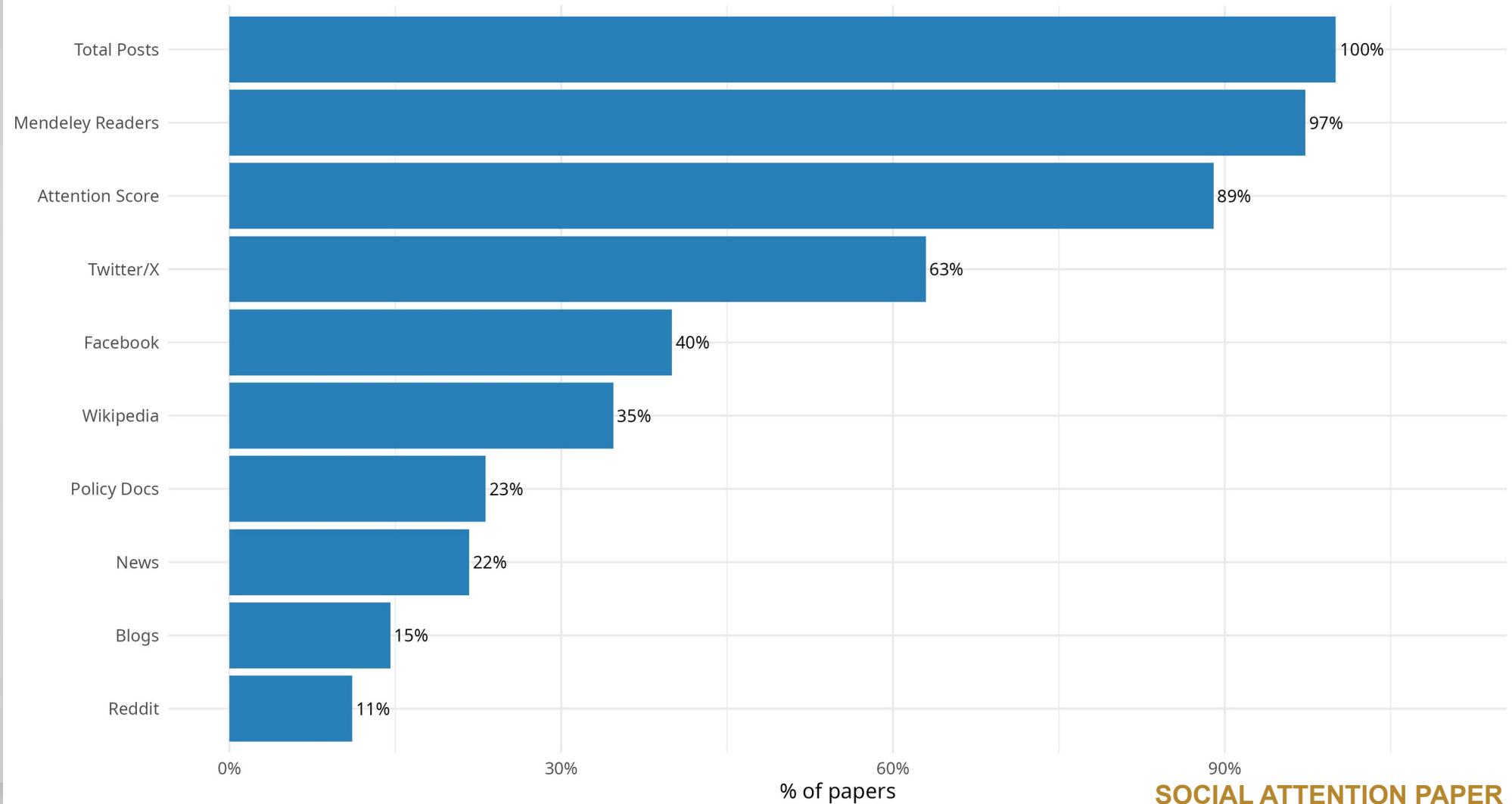
Countries with  $\geq 5$  papers (n = 69 countries shown)



# Altmetric coverage by channel (% papers with at least one mention)

## Altmetric coverage: % of papers with $\geq 1$ mention

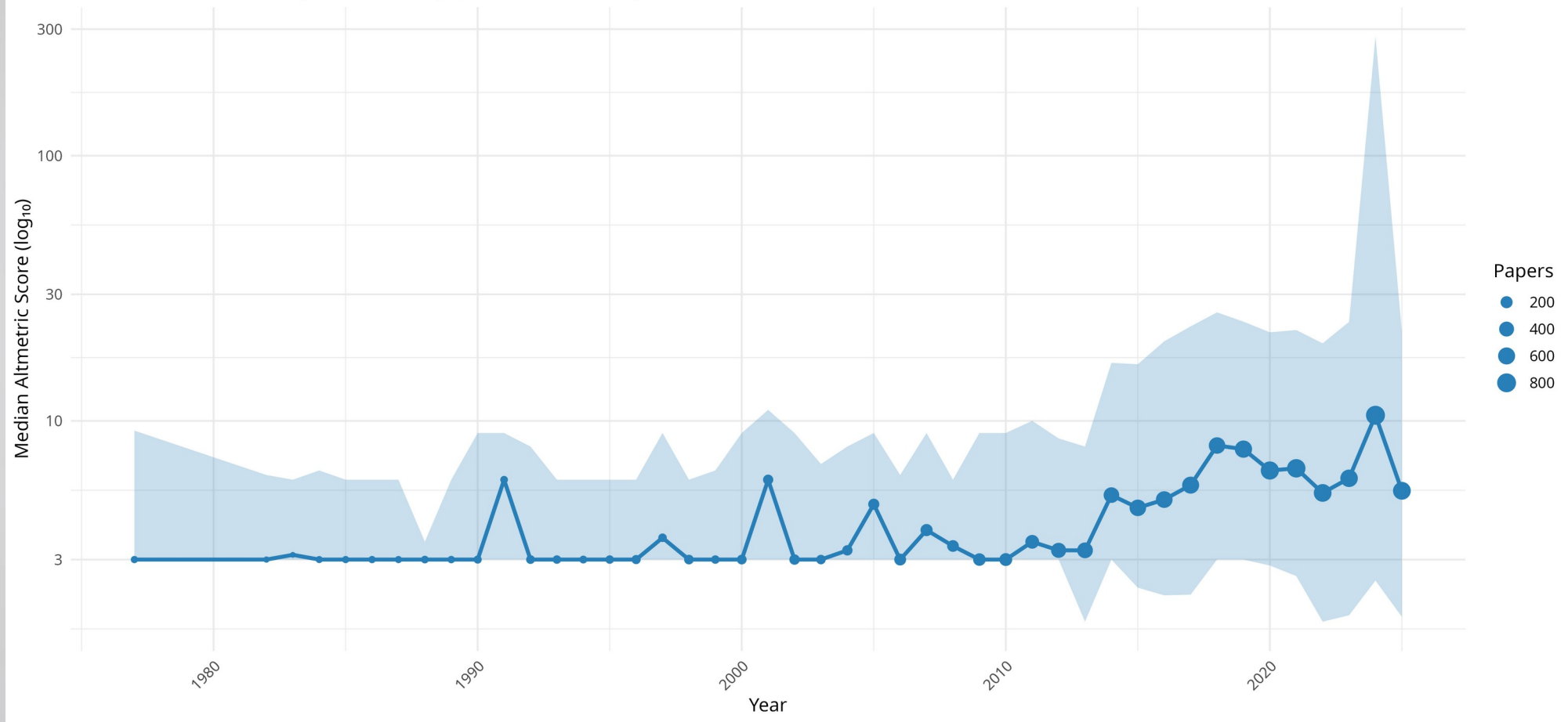
Among 11,633 papers with Altmetric data



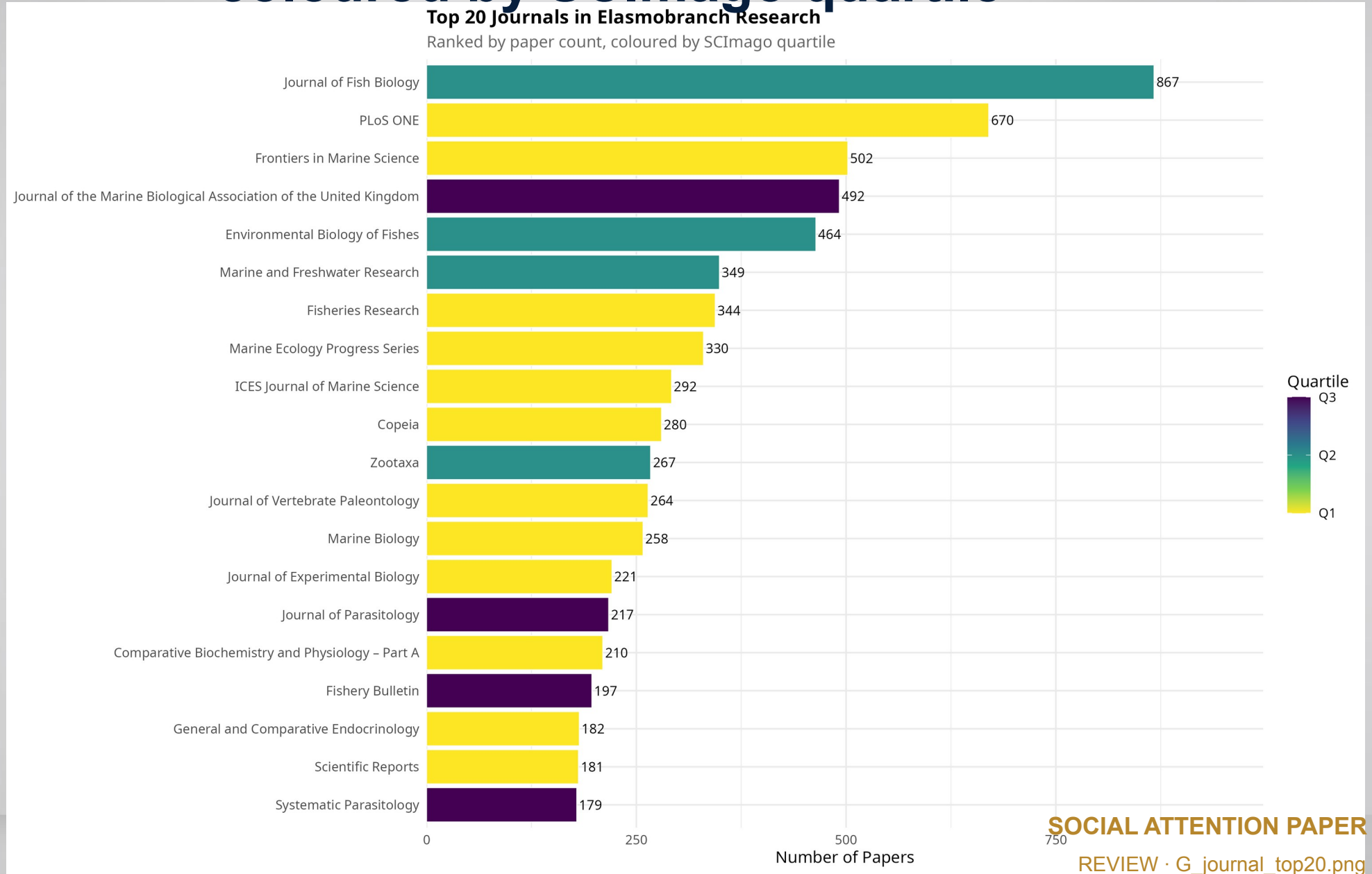
# Median Altmetric score over time with IQR ribbon

## Altmetric Attention Score over time

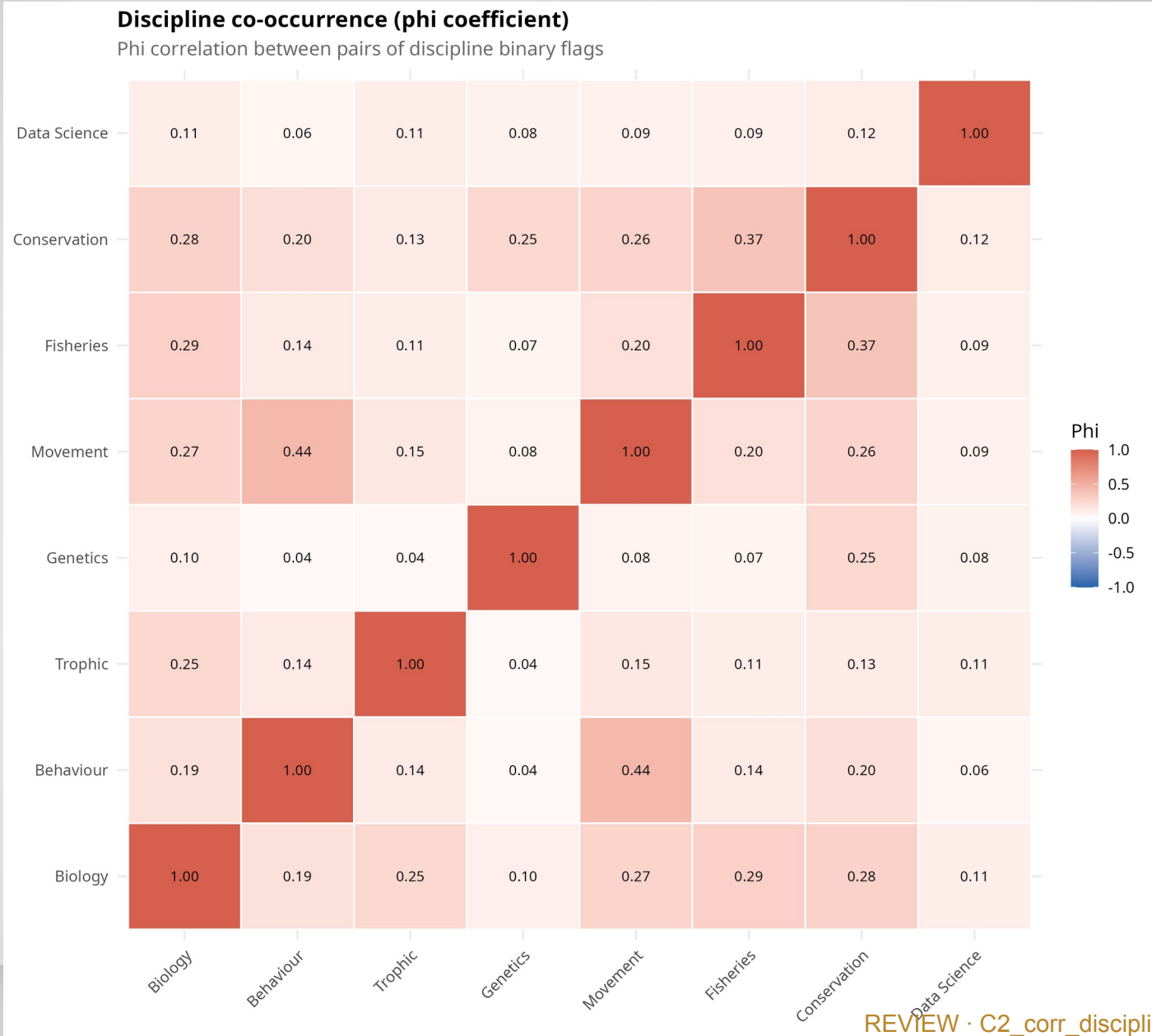
Median (line), IQR ribbon; years with  $\geq 20$  papers with scores only



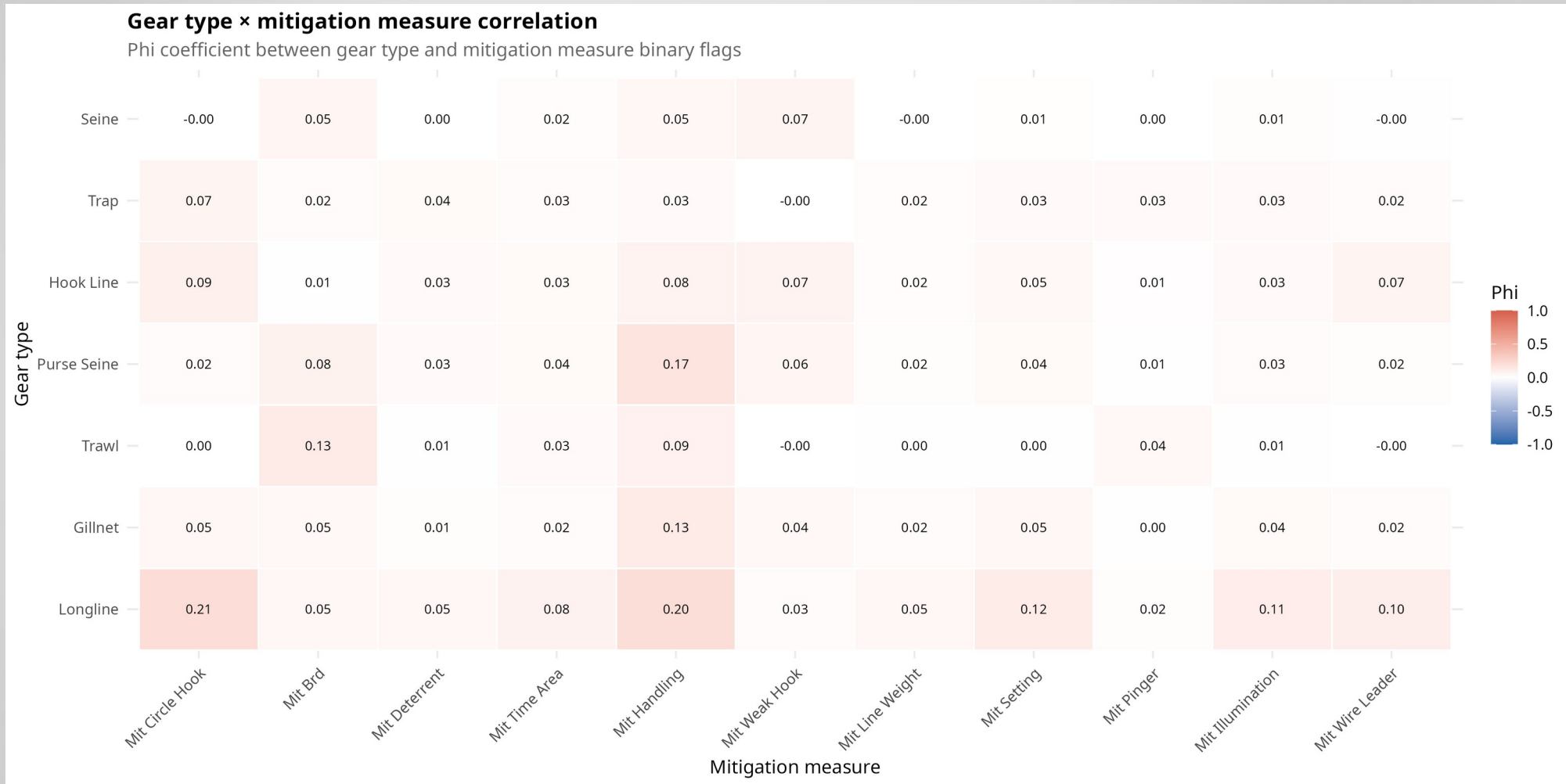
# Top 20 journals by paper count coloured by SCImago quartile



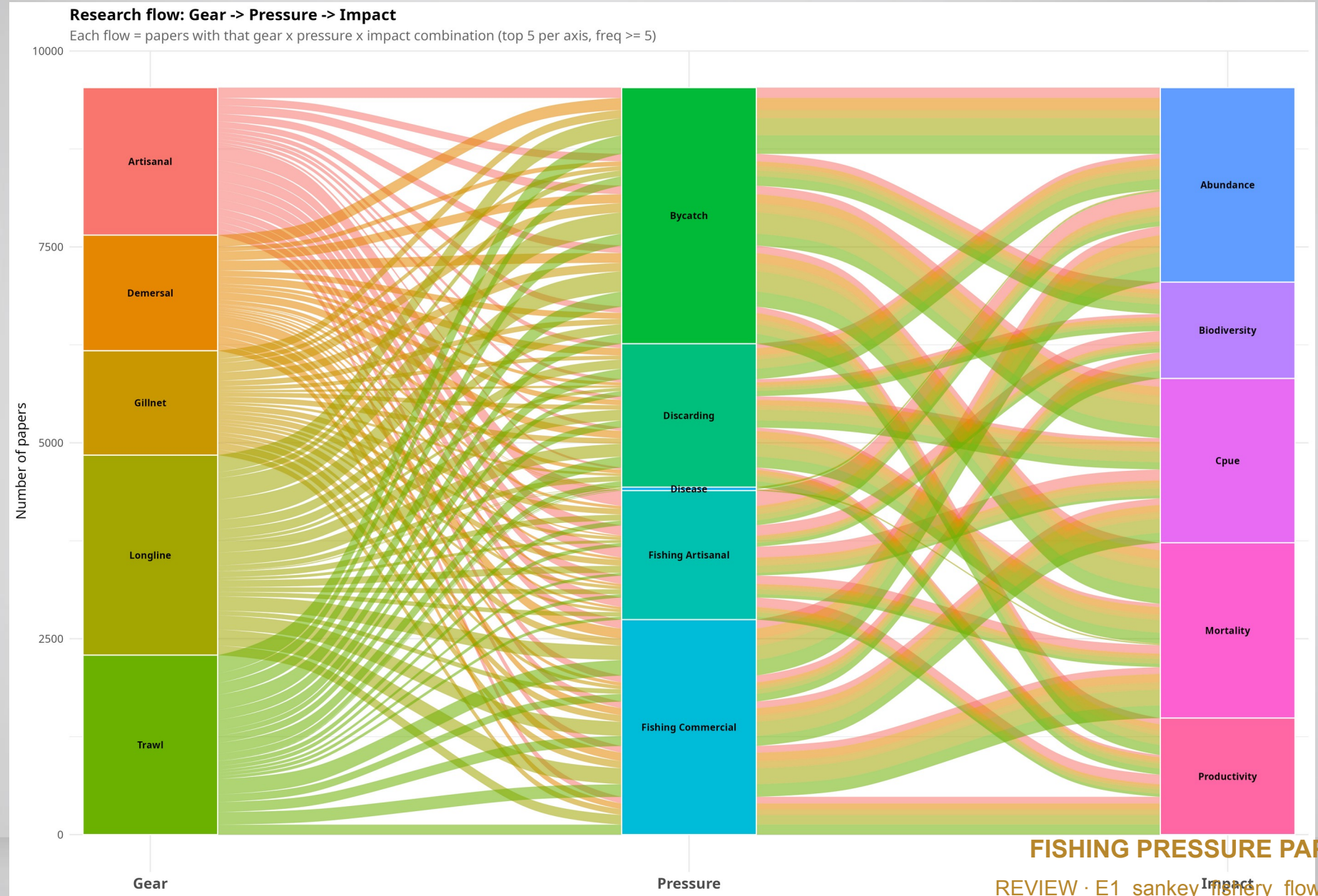
# Heatmap of discipline co-occurrence



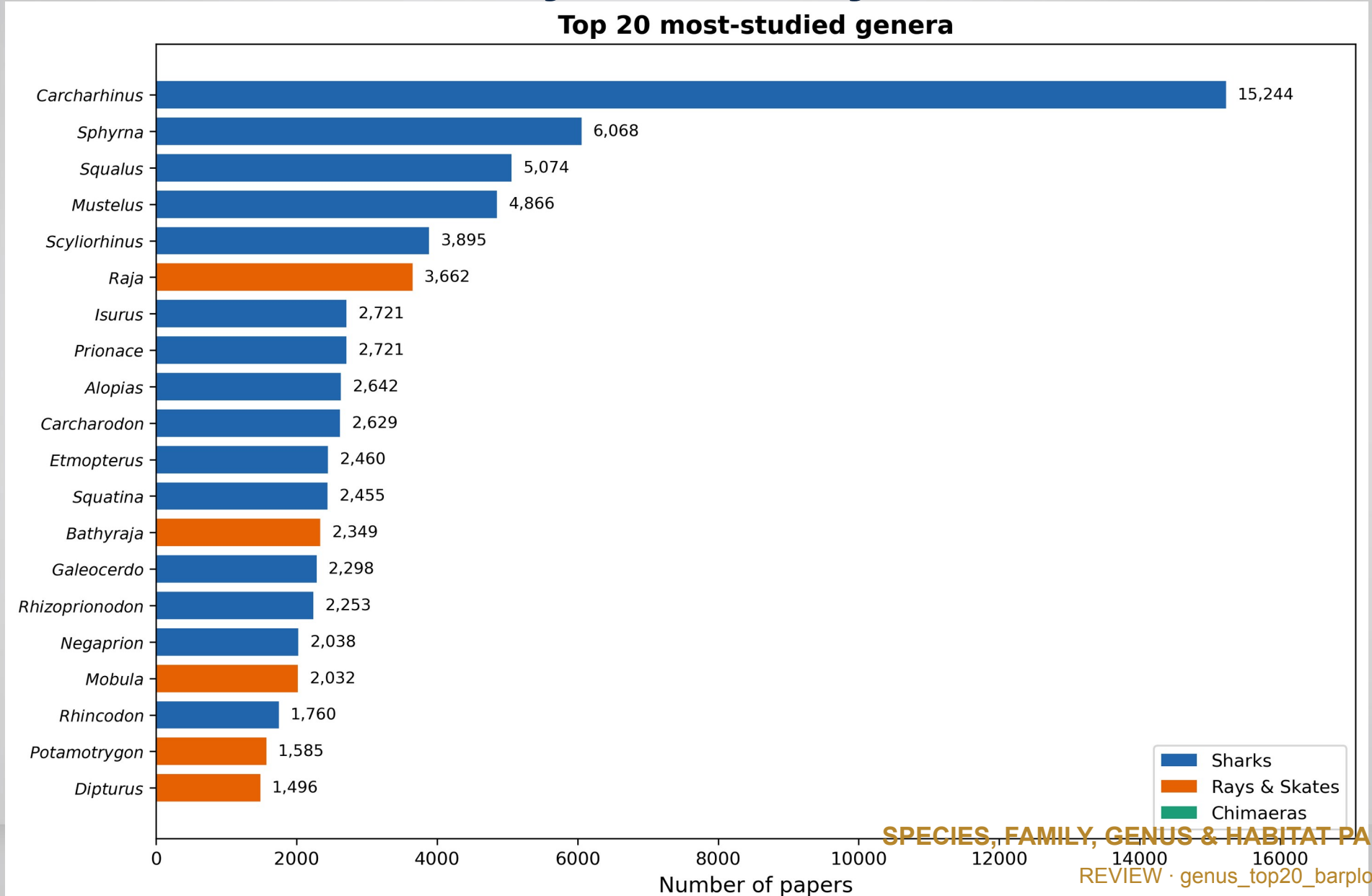
# Heatmap of gear type x mitigation correlation



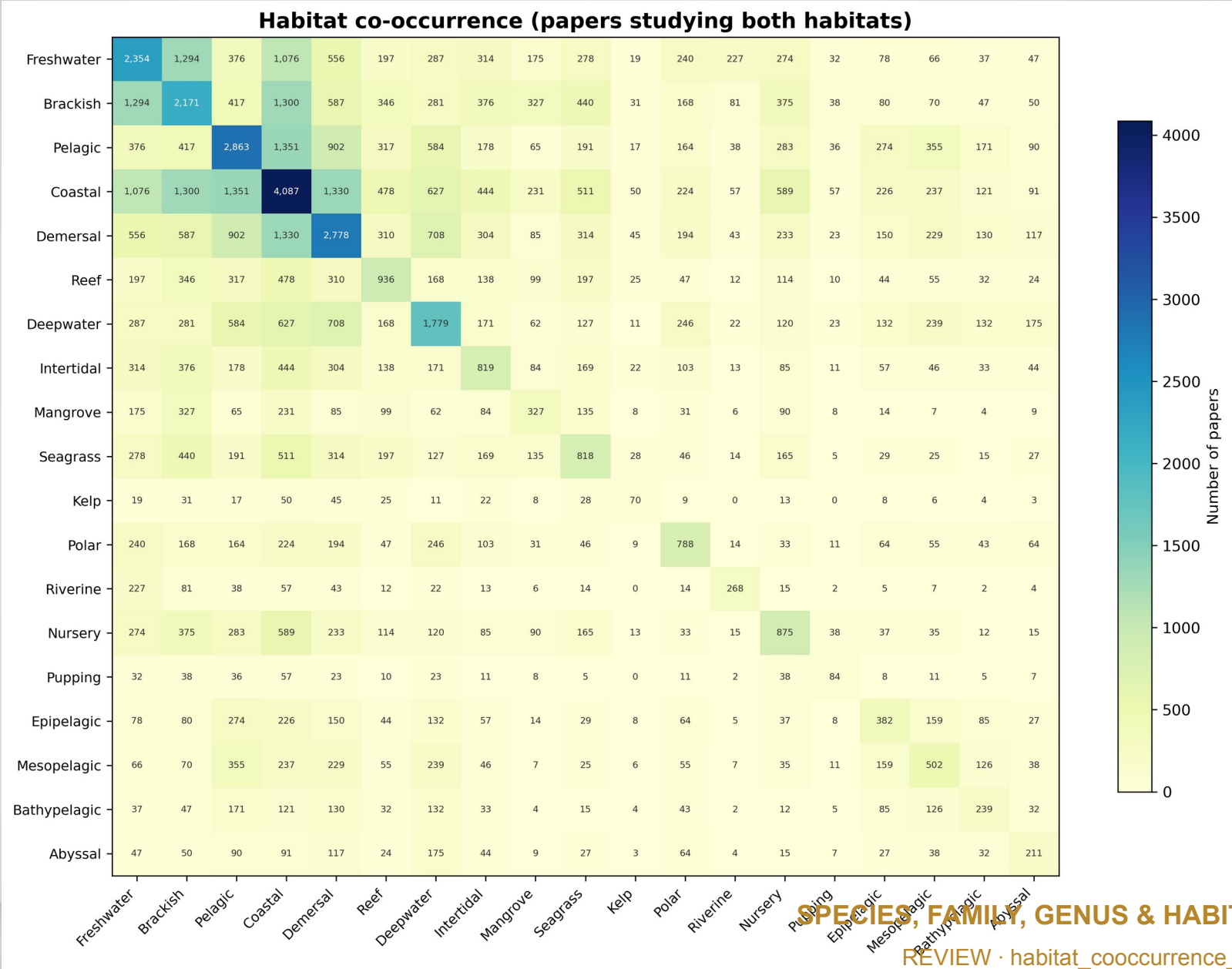
# Sankey of fishery gear -> pressure -> impact flows



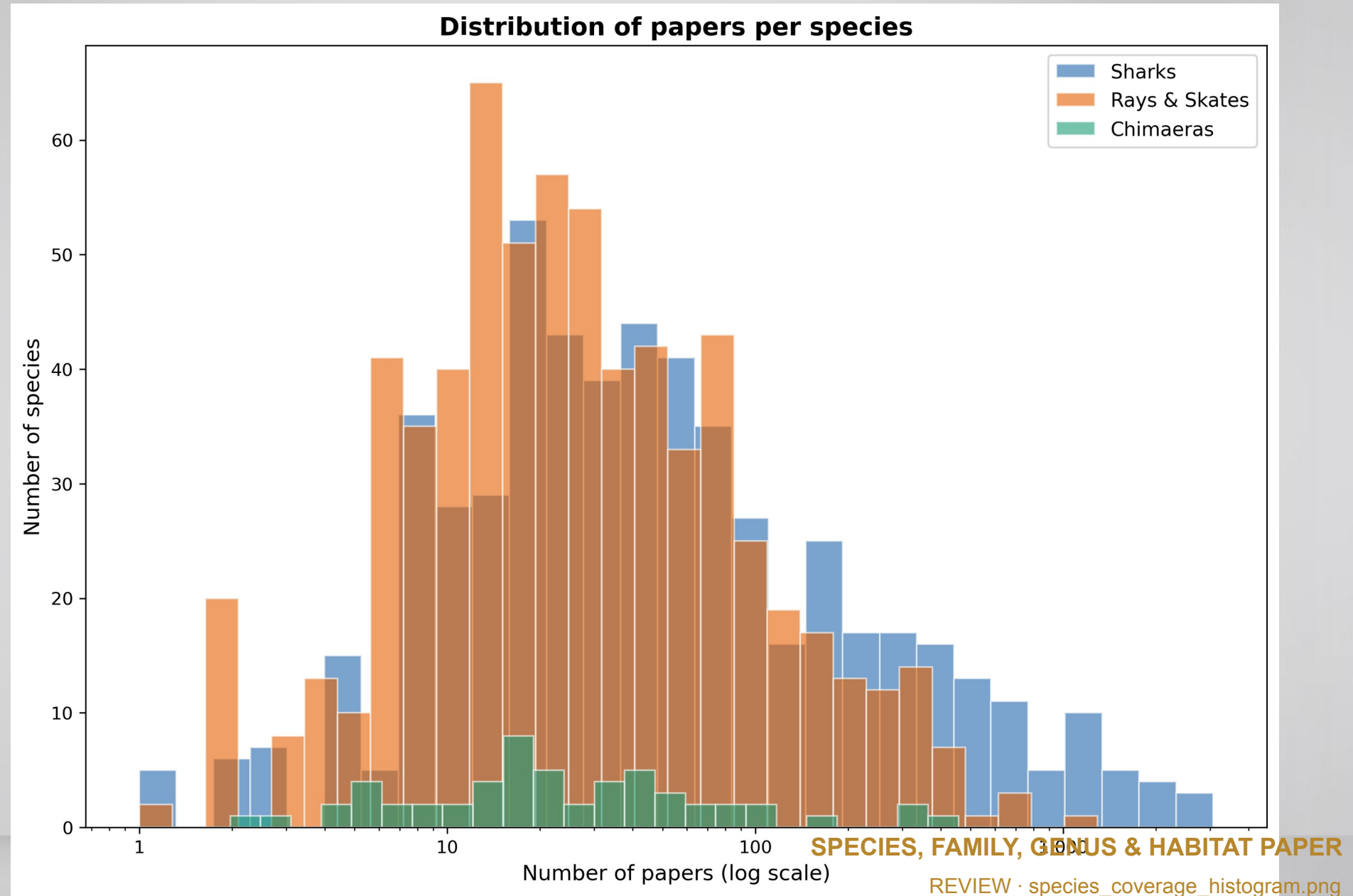
# Top 20 most-studied genera by paper count, coloured by sharks/rays/chimaeras



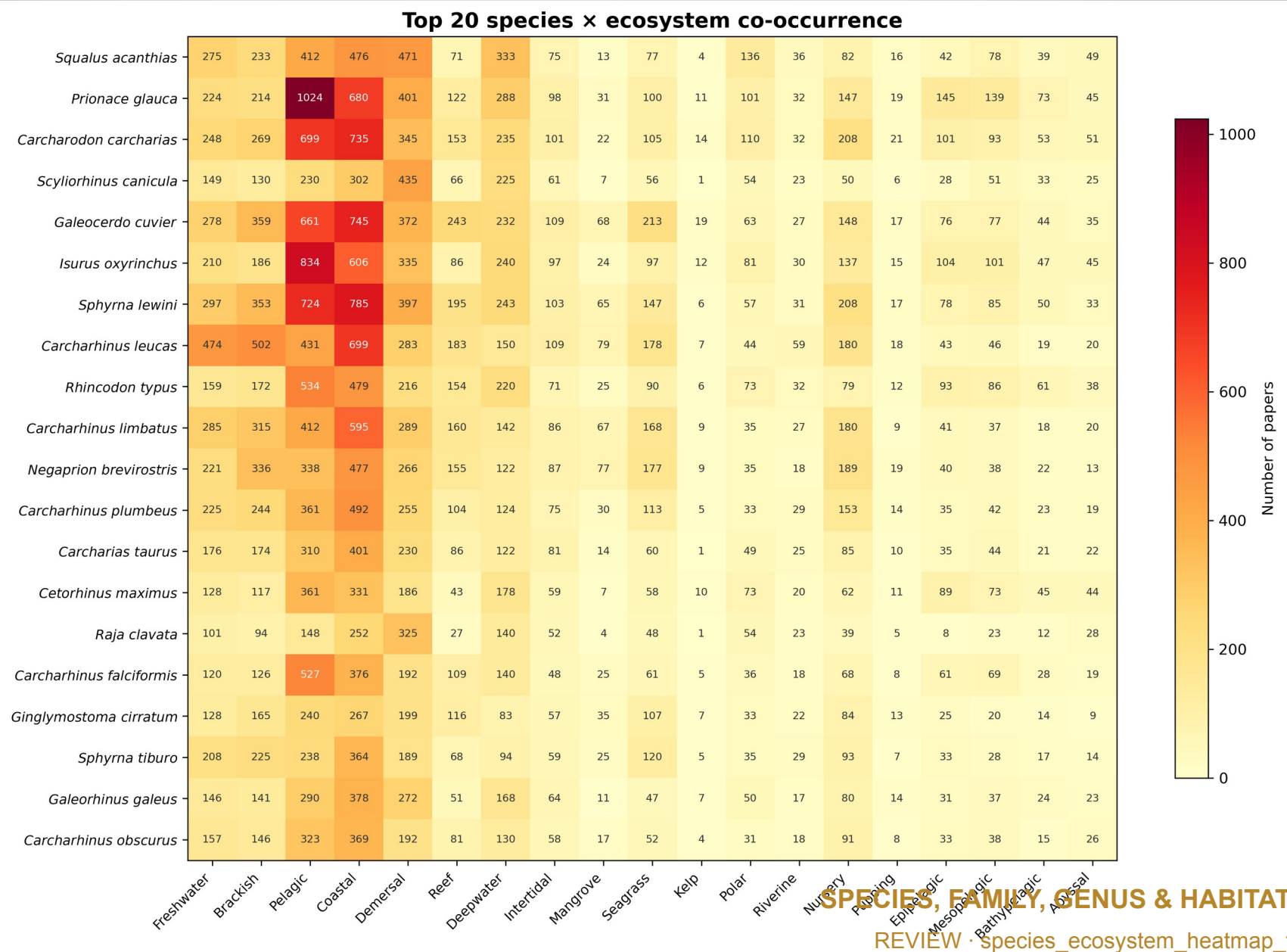
# Heatmap of papers studying pairs of habitats



# Histogram of papers per species (log x) by group

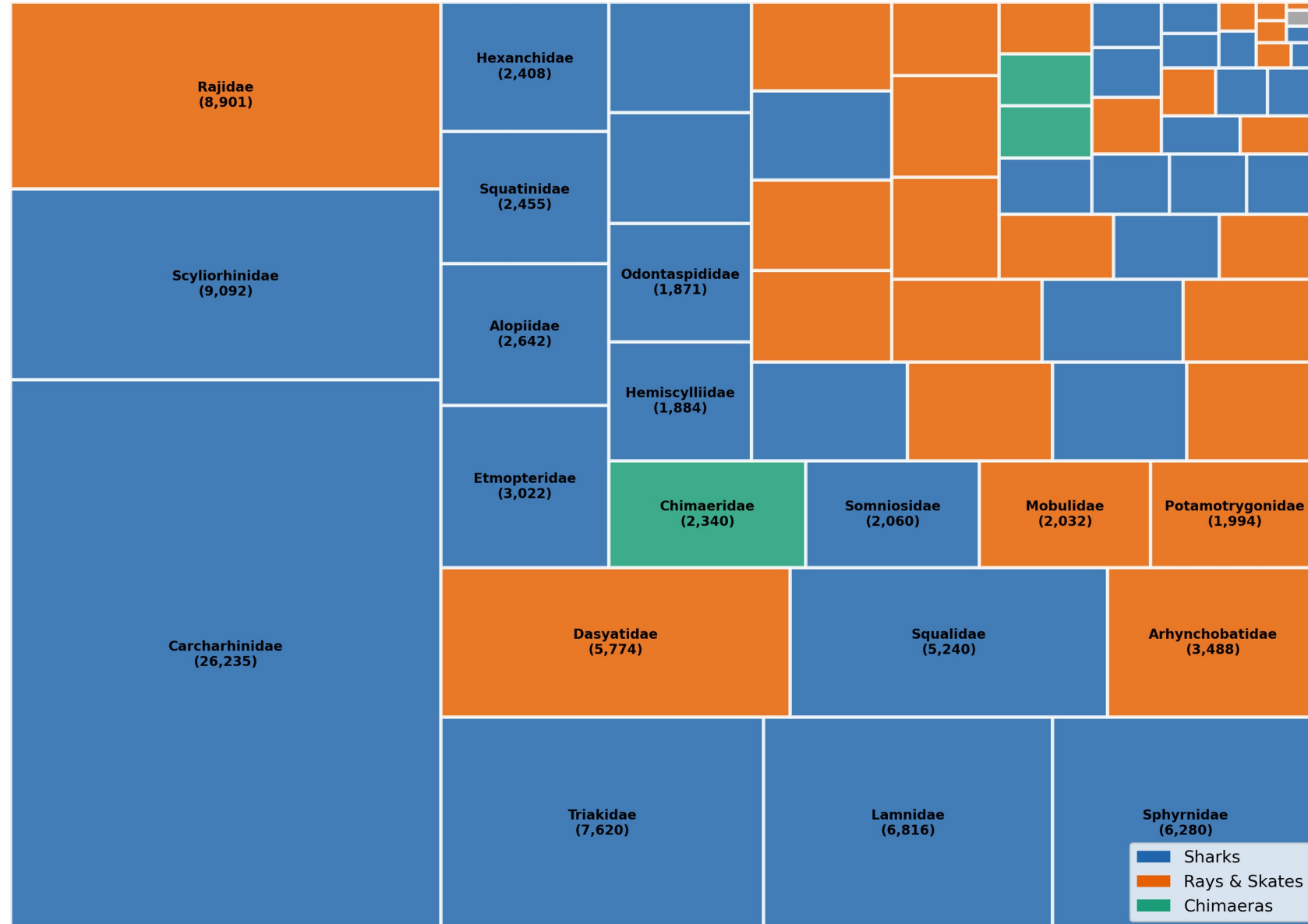


# Top 20 species x ecosystem co-occurrence heatmap



# Treemap of papers per family

Papers per family (size = paper count)



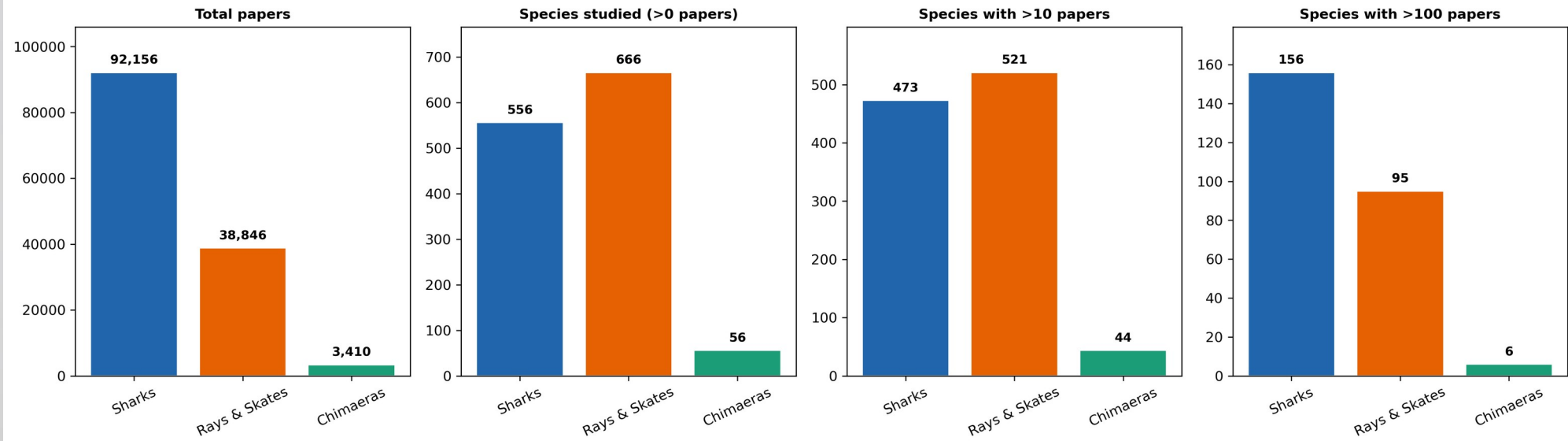
FIU

SPECIES, FAMILY, GENUS & HABITAT PAPER

REVIEW · species\_family\_treemap.png

# Coverage by sharks vs rays/skates vs chimaeras

Research coverage by taxonomic group



# Top 30 most-studied elasmobranch species by papers

